



AHDS MATLAB04 - CURRENT-SENSE CONTROLLED NE555 FREQUENCY GENERATOR

Complete Circuit Study, MATLAB Behavioral Modeling, Test-Point Analysis, R36 Gain Sweep, Derived Frequency Law and
Line-by-Line Code Commentary

MATLAB Study Series - Part 4 | Version 1 -> Version 2.3 -> Version 3.0 | 5 V system | 0-10 A model | 5 mOhm shunt | LM358 +
BC558 + NE555

Document status: This report documents the reference schematic and the MATLAB behavioral model used in Versions 2.3 and 3.0. The Q6 threshold and the frequency-sensitivity coefficient are modeling assumptions, not measured transistor-level constants.

Item	Value
Reference simulations	MATLAB Version 2.3 and R36 Sweep Version 3.0
Time base	0 to 6 ms; dt = 0.5 us
Nominal resistor	R36 = 330 kOhm
First tested threshold-crossing case	R36 = 510 kOhm
Use	Circuit study, design review, parameter selection, and bench-test preparation

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Appendix A. Version 1 - Line-by-Line Annotated MATLAB Source

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1. Purpose and Study Objectives

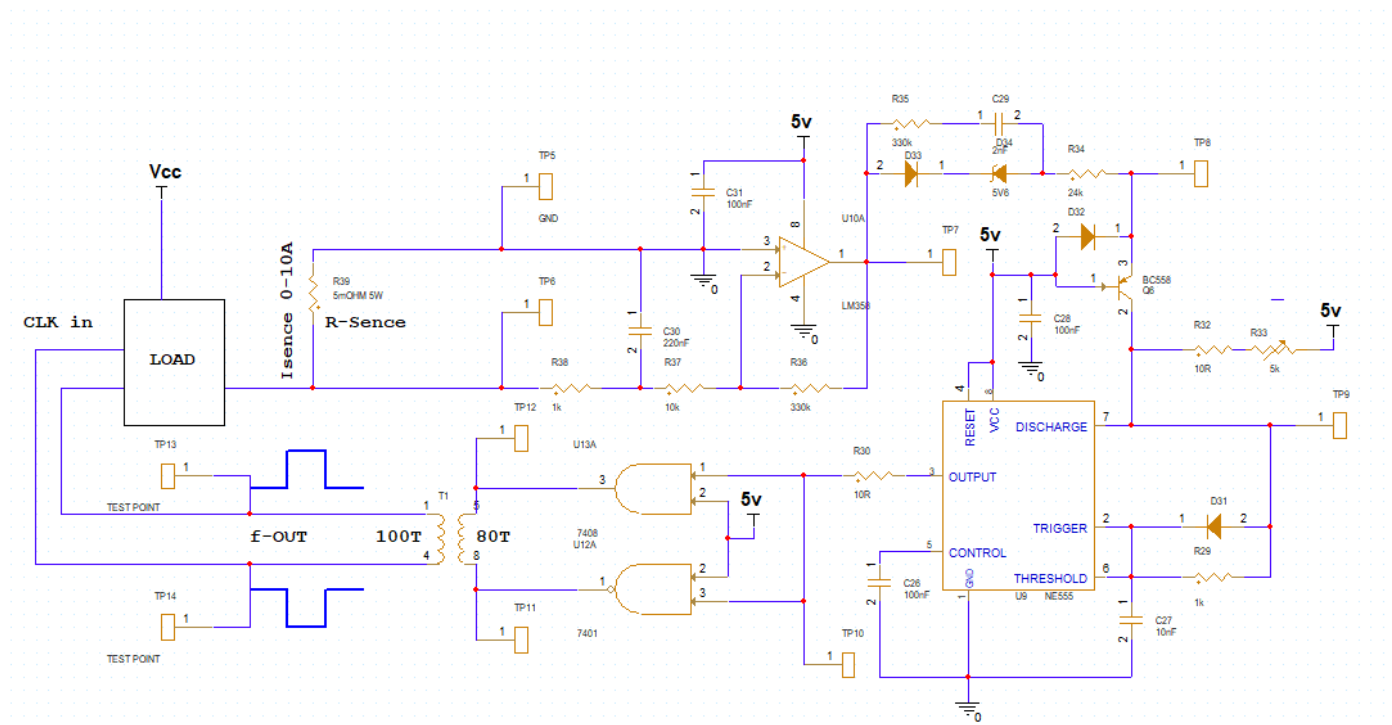
The circuit converts load current into a small shunt voltage, amplifies and filters that signal, converts current transitions into a transient control waveform around TP8, and uses the resulting Q6 control state to influence a NE555 timing model. The timer output drives complementary logic signals and an idealized 80-turn to 100-turn transformer stage. The document is organized so that each physical block, equation, test point, simulation assumption, and result can be reviewed independently.

- Understand how a 5 mOhm current shunt produces TP6 from a 0-10 A load current.
- Understand how the LM358 gain and C30 time constant produce TP7.
- Understand why the R35/C29 path responds mainly to current transitions rather than steady current.
- Understand the behavioral Q6 threshold used in MATLAB and why R36 controls whether the threshold is crossed.
- Understand the nominal NE555 timing calculation and the meaning of TP9/TP10.
- Understand how TP11/TP12 are converted into TP13/TP14 and differential f-OUT in the ideal transformer model.
- Use the R36 sweep to choose the next resistor value for a more detailed model and bench validation.

Key result: With the present behavioral assumptions, R36 = 330, 390, 430, and 470 kOhm do not exceed the 5.65 V Q6 threshold. The first tested crossing occurs at 510 kOhm; 560 kOhm provides a larger modeled margin.

2. Complete Reference Schematic

The schematic below is the reference used to define the MATLAB signal chain and the TP6-TP14 test points. TP5 is the common ground reference and is intentionally not plotted as a separate waveform.



Reference schematic used for the MATLAB model. The schematic includes R39 current shunt, U10A LM358, the D33/D34/C29/Q6 modulation path, U9 NE555, logic gates, transformer T1, and TP5-TP14.



Schematic-specific uncertainty: D32 is visible in the reference schematic but its exact part number was not confirmed. The current MATLAB model therefore does not claim a transistor-accurate D32/Q6 network.

3. Functional Signal Flow

Block	Function
1. Load current	Iload changes between 0 A and 10 A.
2. Current shunt	R39 = 5 mOhm converts current to a small differential voltage. TP6 is modeled with negative polarity for the selected sense direction.
3. LM358 stage	R38/R37/R36 establish the first-order gain. C30 shapes the response. TP7 is the amplified current signal.
4. Transient modulation	R35/C29/R34 form a slow high-pass-like transient path in the behavioral model. TP8 is a biased transient signal.
5. Q6 control	A behavioral threshold of $V_{CC} + 0.65 \text{ V} = 5.65 \text{ V}$ is used. Above threshold, Q6_drive becomes positive.
6. NE555 timing	The nominal oscillator is approximately 18.386 kHz. In Version 3.0, frequency changes are introduced behaviorally only when Q6_drive is positive.
7. Logic drive	TP10 is the 555 output; TP11 and TP12 are complementary logic-drive waveforms.
8. Transformer output	An ideal 80T:100T ratio is applied to produce TP13, TP14, and differential f-OUT = TP13 - TP14.

4. Circuit Values and Model Parameters

Block	Designator	Value	Model role
Supply	VCC	5 V	Model supply
Load current	Iload	0 to 10 A	Step from 1 ms to 4 ms in V2.3
Current shunt	R39 / R-Sense	5 mOhm, 5 W	TP6 sensing element
Op-amp	U10A	LM358	Current-sense amplifier
Sense input	R38 / R37	1 kOhm / 10 kOhm	Effective input path = 11 kOhm
Gain resistor	R36	330 kOhm nominal; 330-560 kOhm sweep	Controls modeled LM358 gain
Sense capacitor	C30	220 nF	First-order current-sense filtering
Modulation network	R35 / R34 / C29	330 kOhm / 24 kOhm / 2 nF	Transient-shaping path
Shaping diode	D33	1N914	Nonlinear path; not explicitly transistor-modeled in V2.3/V3.0
Zener	D34	5.6 V	Nonlinear path; not explicitly transistor-modeled in V2.3/V3.0
Unconfirmed diode	D32	Part not confirmed	Keep as a model uncertainty



Control transistor	Q6	BC558 PNP	Behavioral threshold approximation
Timer IC	U9	NE555	Astable timing block
555 timing	R32 / R33 / R29 / C27	10 Ohm / 5 kOhm / 1 kOhm / 10 nF	Used in nominal timing equation
Timing diode	D31	1N914; Vf approx. 0.60 V	Charge-path correction
555 discharge transistor	VCEsat model	0.10 V	Discharge timing approximation
555 output resistor	R30	10 kOhm	Shown between NE555 output and logic stage
Control capacitor	C26	100 nF	Shown from NE555 control node to ground
Local capacitor	C28	100 nF	Shown near Q6/NE555 network
Logic	U12A / U13A	7401 / 7408	Treated as ideal logic in MATLAB
Transformer	T1	80T drive / 100T output	Ideal turns-ratio model

5. Analytical Calculations

5.1 Current shunt

The model uses a 5 mOhm shunt. For the selected polarity, TP6 is negative when load current is positive.

$$TP6 = -I_{load} \times R_{sense}$$

$$\text{At 10 A: } TP6 = -(10 \text{ A})(0.005 \text{ Ohm}) = -0.050 \text{ V} = -50 \text{ mV}$$

$$P_{shunt} = I^2 R = (10 \text{ A})^2 (0.005 \text{ Ohm}) = 0.50 \text{ W}$$

Iload	TP6	Shunt dissipation
0 A	0 mV	0.000 W
2 A	-10 mV	0.020 W
5 A	-25 mV	0.125 W
8 A	-40 mV	0.320 W
10 A	-50 mV	0.500 W

5.2 LM358 gain and C30 filter

$$\text{Gain magnitude} = R36 / (R38 + R37)$$

$$\text{For } R36 = 330 \text{ kOhm: Gain} = 330 \text{ kOhm} / 11 \text{ kOhm} = 30.0 \text{ V/V}$$

$$TP7 \text{ target approx.} = -\text{Gain} \times TP6$$

At 10 A and $R36 = 330 \text{ kOhm}$, the ideal DC target is approximately +1.50 V. The Version 2.3 script additionally clips the LM358 behavioral output to 0-3.5 V on a 5 V supply.



$$R_{th}(C30) = R38 \parallel R37 = 909.1 \text{ Ohm}$$

$$\tau_{sense} = R_{th} \times C30 = 200.0 \text{ us}$$

$$f_{c_sense} = 1 / (2 \pi \tau) = 795.8 \text{ Hz}$$

5.3 R35/C29 transient path

$$\tau_{mod} = (R35 + R34) \times C29 = 708.0 \text{ us}$$

$$f_{c_mod \text{ approx.}} = 1 / (2 \pi \tau_{mod}) = 224.8 \text{ Hz}$$

Version 2.3 does not solve the diode/Zener/transistor network directly. Instead, TP7 is passed through a low-pass state using τ_{mod} ; the high-pass component $HP_{mod} = TP7 - TP7_{lp_mod}$ is then added to a 4.35 V bias to form TP8. This reproduces the desired transient sensitivity without claiming a physical transistor equation.

5.4 Behavioral Q6 threshold

$$VQ6_{on} = VCC + VBE_{PNP \text{ approx.}} = 5.0 \text{ V} + 0.65 \text{ V} = 5.65 \text{ V}$$

$$Q6_{drive} = \max(TP8 - 5.65 \text{ V}, 0)$$

Interpretation: This threshold is a deliberately simplified switching criterion. Actual BC558 conduction depends on the real emitter, base, collector voltages, device beta, loading, D32, diode/Zener behavior, temperature, and component tolerances.

5.5 NE555 nominal timing

$$VTL = VCC/3 = 1.667 \text{ V}; VTH = 2VCC/3 = 3.333 \text{ V}$$

$$RA = R32 + R33 = 5.01 \text{ kOhm}; RB = R29 = 1.00 \text{ kOhm}$$

$$V_{charge_inf \text{ approx.}} = VCC - V_f(D31) = 4.40 \text{ V}$$

$$t_{charge} = 47.14 \text{ us}$$

$$t_{discharge} = 7.25 \text{ us}$$

$$T = 54.39 \text{ us}; f_{nom} = 18.386 \text{ kHz}; \text{duty approx.} = 86.7\%$$

5.6 Transformer first-order model

$$\text{Turns ratio } n = N_{sec} / N_{pri} = 100 / 80 = 1.25$$

$$V_{sec,diff} = 1.25 \times V_{pri,diff}$$

The MATLAB model represents the secondary as floating around a virtual midpoint: TP13 = +0.5 $V_{sec,diff}$ and TP14 = -0.5 $V_{sec,diff}$. Therefore, $f_{OUT} = TP13 - TP14$. Magnetizing current, leakage, copper resistance, saturation, core loss, load, and parasitic capacitance are not included.

6. Test-Point Map

Point	Node	What to study
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TP5	Ground	Reference only; not plotted
TP6	Current-sense node	0 to -50 mV for 0-10 A with selected polarity
TP7	LM358 output	Amplified and filtered current signal
TP8	Q6 modulation/control node	Transient-biased node compared with 5.65 V behavioral threshold
TP9	NE555 discharge/timing node	5 V during charge; approximately 0.1 V during modeled discharge
TP10	NE555 output	0/5 V oscillator output
TP11	Primary-drive signal A	Complement of TP10 in the ideal logic model
TP12	Primary-drive signal B	Same polarity as TP10 in the ideal logic model
TP13	Secondary end A	Half of the ideal differential secondary voltage about a virtual midpoint
TP14	Secondary end B	Opposite half of the ideal differential secondary voltage
f-OUT	TP13 - TP14	Ideal differential transformer output

7. MATLAB Modeling Method and Assumptions

Element	Implementation
Simulation grid	0-6 ms with $dt = 0.5 \mu s$; 12,001 samples.
Current profile	0 A from 0-1 ms; 10 A from 1-4 ms; 0 A from 4-6 ms.
TP6	Direct shunt equation.
TP7	First-order discrete low-pass toward the ideal amplified target.
TP8	Behavioral high-pass transient around a 4.35 V bias.
Q6	Threshold-only behavior at 5.65 V.
Frequency law	$f_{inst} = f_{nom} + K_f \times Q6_drive$, where $K_f = 5 \text{ kHz/V}$.
Digital phase	Variable-frequency phase accumulator used to generate TP9/TP10.
Logic/transformer	Ideal complementary drive and ideal 1.25 turns-ratio transformer.

Critical model caveat: $K_f = 5 \text{ kHz/V}$ is a behavioral sensitivity chosen for the simulation. Therefore, the calculated 18.928 kHz and 19.619 kHz maxima in the sweep are comparative model outputs, not validated hardware frequencies.

8. Version 1 - Initial 330 kOhm Baseline: Desired Modulation Not Reached

Version 1 was the first complete MATLAB behavioral model of the signal chain. It used $R_{36} = 330 \text{ k}\Omega$ (LM358 gain approximately 30 V/V) and a deliberately slow, piecewise current profile rather than the hard step used later. The run was important because it proved that the MATLAB implementation, current-sense path, NE555 timing, complementary drive, and transformer visualization were functioning, while also showing that the intended frequency-modulation mechanism was not activated.

The desired result was a transient at TP8 large enough to cross the modeled BC558 threshold near 5.65 V. Instead, TP8 remained roughly in the 4.0-4.9 V region. Therefore Q6_drive stayed at zero and the modeled NE555 frequency remained near 18.4 kHz. The timer and transformer output waveforms still existed, but their period stayed constant; what was missing was the desired variable-frequency response, not the basic oscillator output itself.

Version 1 result - why the first attempt did not modulate

$R_{36} = 330 \text{ k}\Omega$; gain approximately 30 V/V; TP7 reaches about 1.5 V at 10 A; TP8 stays below the 5.65 V behavioral Q6 threshold; Q6_drive = 0; instantaneous frequency stays approximately constant at 18.4 kHz. This is a useful failed baseline because it shows exactly which condition was missing.



Figure V1-1. Original Version 1 analog-control-path result. Slow current ramps produce the expected TP6 and TP7 responses, but TP8 remains below the Q6 turn-on threshold.

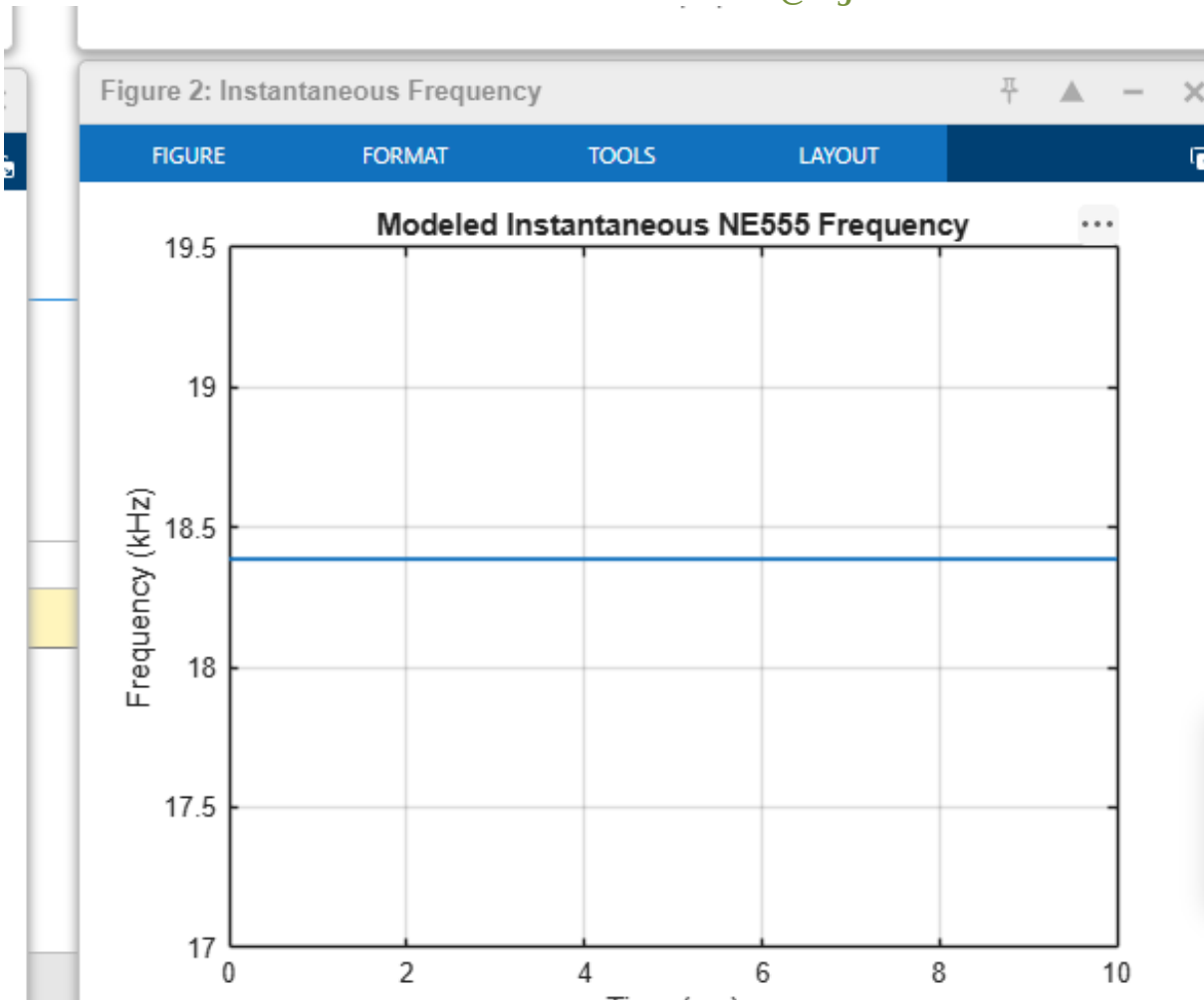


Figure V1-2. Original Version 1 instantaneous-frequency plot. The modeled frequency remains flat near 18.4 kHz because the Q6 modulation condition is not reached.

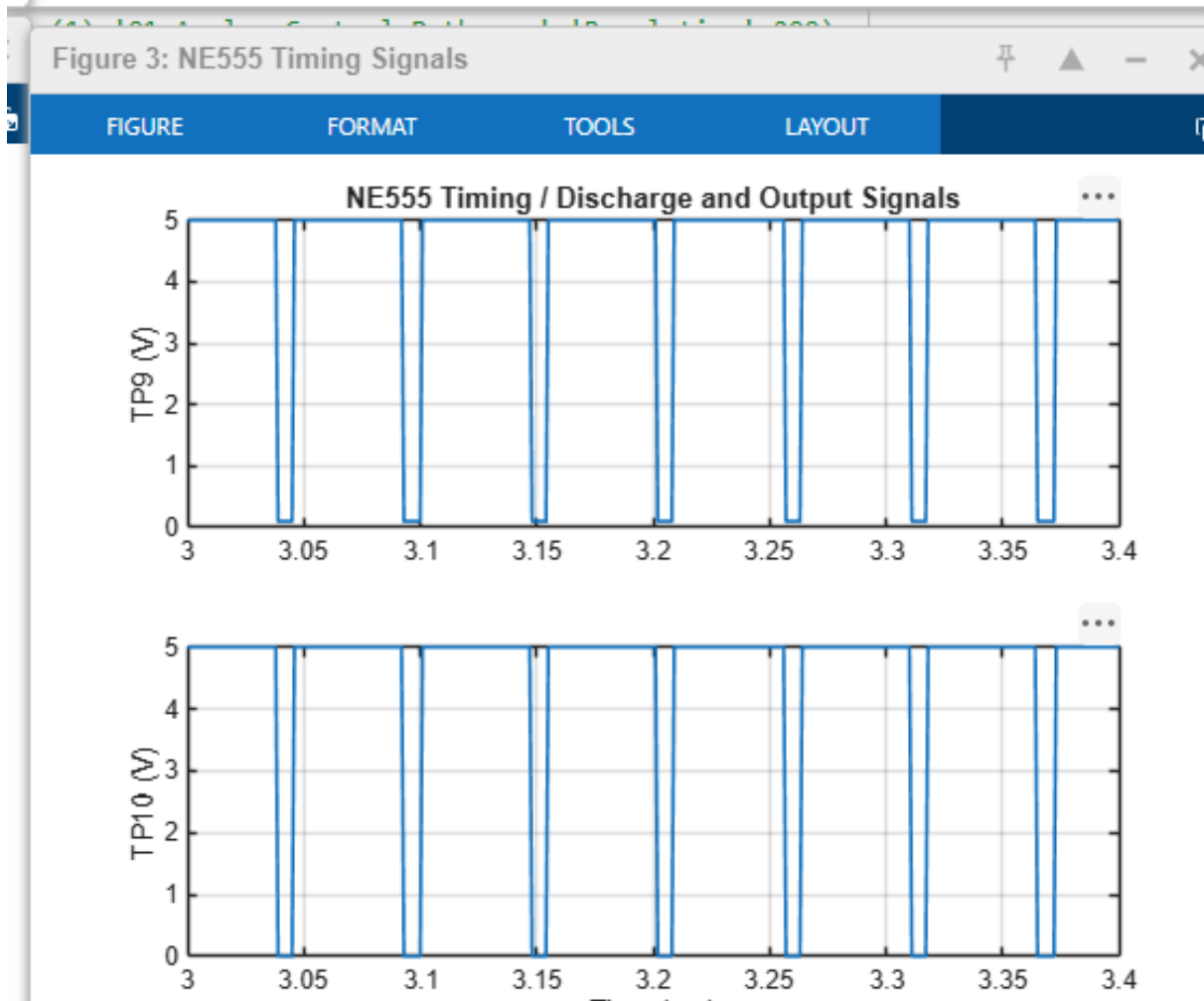


Figure V1-3. Original Version 1 NE555 timing signals. Switching is present even though frequency modulation is absent.



Figure V1-4. Original Version 1 complementary transformer-primary drive.

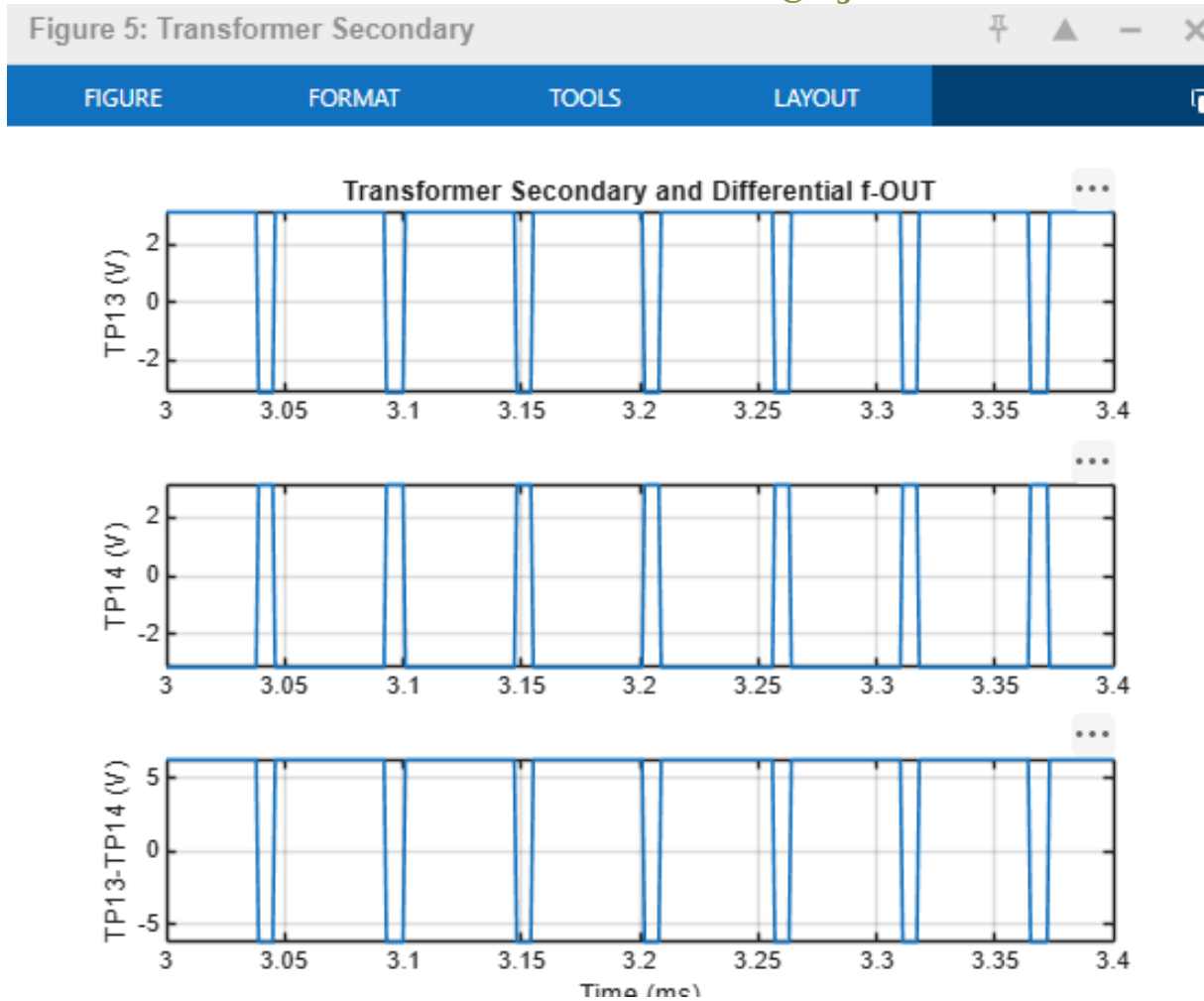


Figure V1-5. Original Version 1 transformer-secondary visualization and differential f-OUT.

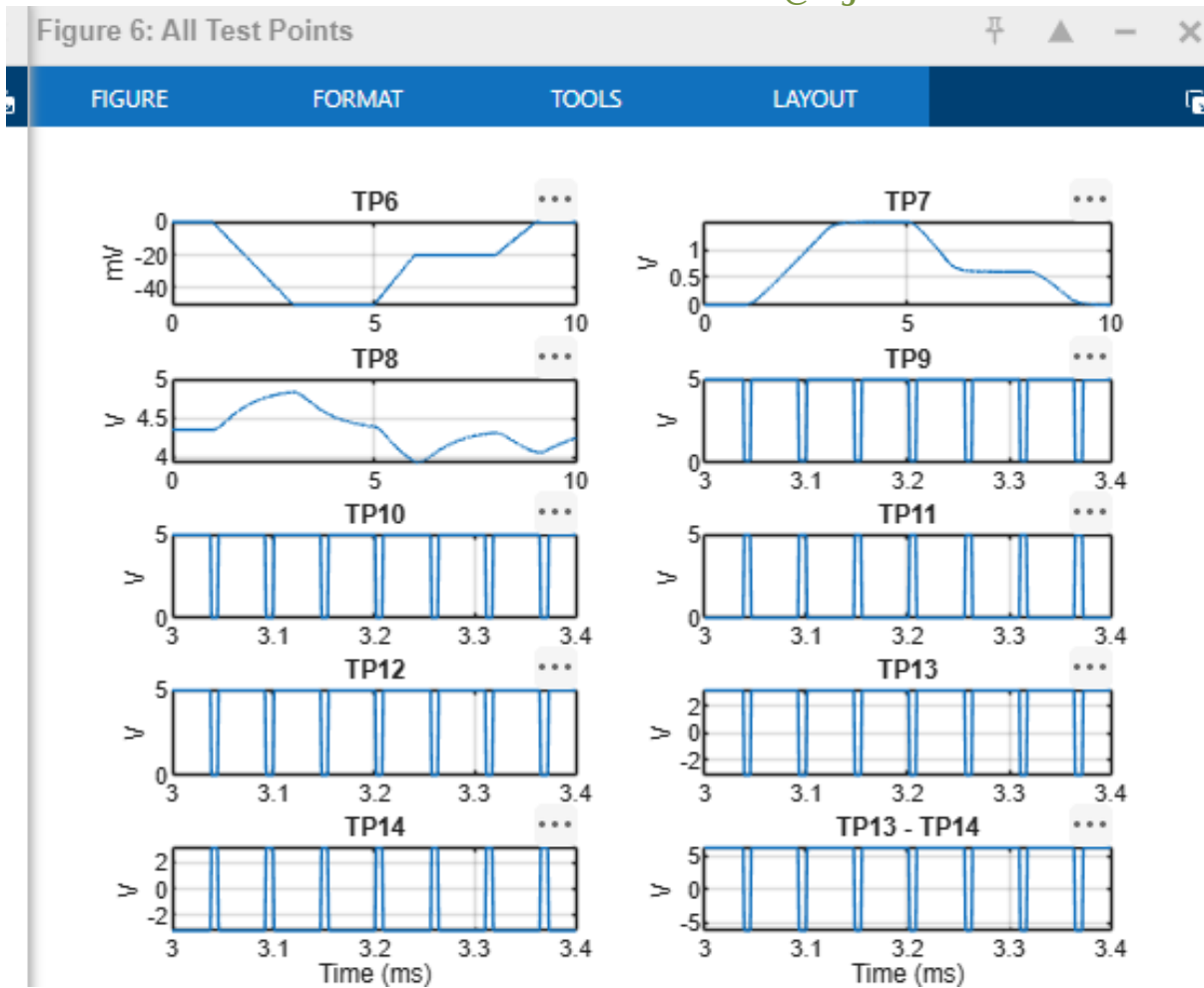


Figure V1-6. Original Version 1 combined diagnostic view of TP6 through TP14 and differential output.

- Lesson 1: a valid MATLAB run can correctly show that the circuit condition is not met; a flat frequency plot is not automatically a software error.
- Lesson 2: the R35/C29 path is transient-sensitive, so edge speed matters. Version 1 used relatively slow millisecond-scale ramps.
- Lesson 3: before changing component values, the next logical experiment was to strengthen the excitation with a fast current step while keeping $R_{36} = 330 \text{ k}\Omega$.

8.1 Simulation Revision History

Several MATLAB files were created during this exercise. Versions 2.1 through 2.3 mainly addressed MATLAB Online plotting/export reliability; they did not change the electrical conclusion that $R_{36} = 330 \text{ k}\Omega$ stays below the modeled Q6 threshold.

Program	Main change	Electrical model change?	Engineering result
Version 1	Slow piecewise 0-10 A current profile; first complete TP6-TP14 model	Baseline	Desired modulation not reached; frequency remains near 18.4 kHz
Version 2.0	Replace slow ramps with a hard 0 A $\rightarrow$ 10 A $\rightarrow$ 0 A step	Stimulus changed only	TP8 rises to about 5.261 V but remains below 5.65 V
Version 2.1	More robust interactive plotting in MATLAB Online	No	Same electrical result as Version 2.0
Version 2.2	Export-only PNG workflow	No	Same electrical result; helped



	to separate model behavior from browser rendering problems		diagnose display/session behavior
Version 2.3	exportgraphics workflow plus CSV output and final baseline plots	No	Stable documented 330 kOhm baseline; no threshold crossing
Version 3.0	Sweep R36 = 330/390/430/470/510/560 kOhm	Yes - gain is intentionally swept	510 kOhm is first tested clear threshold crossing
V3 CSV Plot Test	Read the saved sweep CSV and test simple plotting separately	No	Diagnostic helper for MATLAB Online/session/plot rendering

9. Version 2.3 Results - R36 = 330 kOhm, Fast Current Step

Version 2.3 keeps the original gain of 30 V/V and applies a hard 0 A -> 10 A -> 0 A current pulse. The objective is to test whether a fast current transition alone can move TP8 above the modeled Q6 threshold.

Signal	Minimum	Maximum	Unit
TP6	-50.000	-0.000	mV
TP7	0.000	1.500	V
TP8	3.457	5.261	V
TP9	0.100	5.000	V
TP10	0.000	5.000	V
TP11	0.000	5.000	V
TP12	0.000	5.000	V
TP13	-3.125	3.125	V
TP14	-3.125	3.125	V
f-OUT	-6.250	6.250	V

Time (ms)	Iload (A)	TP6 (mV)	TP7 (V)	TP8 (V)	f (kHz)
0.000	0.0	-0.0	0.000	4.350	18.386
1.000	10.0	-50.0	0.004	4.354	18.386
1.352	10.0	-50.0	1.242	5.261	18.386
2.000	10.0	-50.0	1.490	4.845	18.386
4.000	0.0	-0.0	1.496	4.376	18.386
4.356	0.0	-0.0	0.253	3.457	18.386
5.000	0.0	-0.0	0.010	3.862	18.386
6.000	0.0	-0.0	0.000	4.228	18.386

Version 2.3 conclusion: TP8 reaches 5.261 V at approximately 1.352 ms, which is about 389 mV below the 5.65 V threshold.

Q6_drive therefore remains zero and the modeled frequency stays at 18.386 kHz.

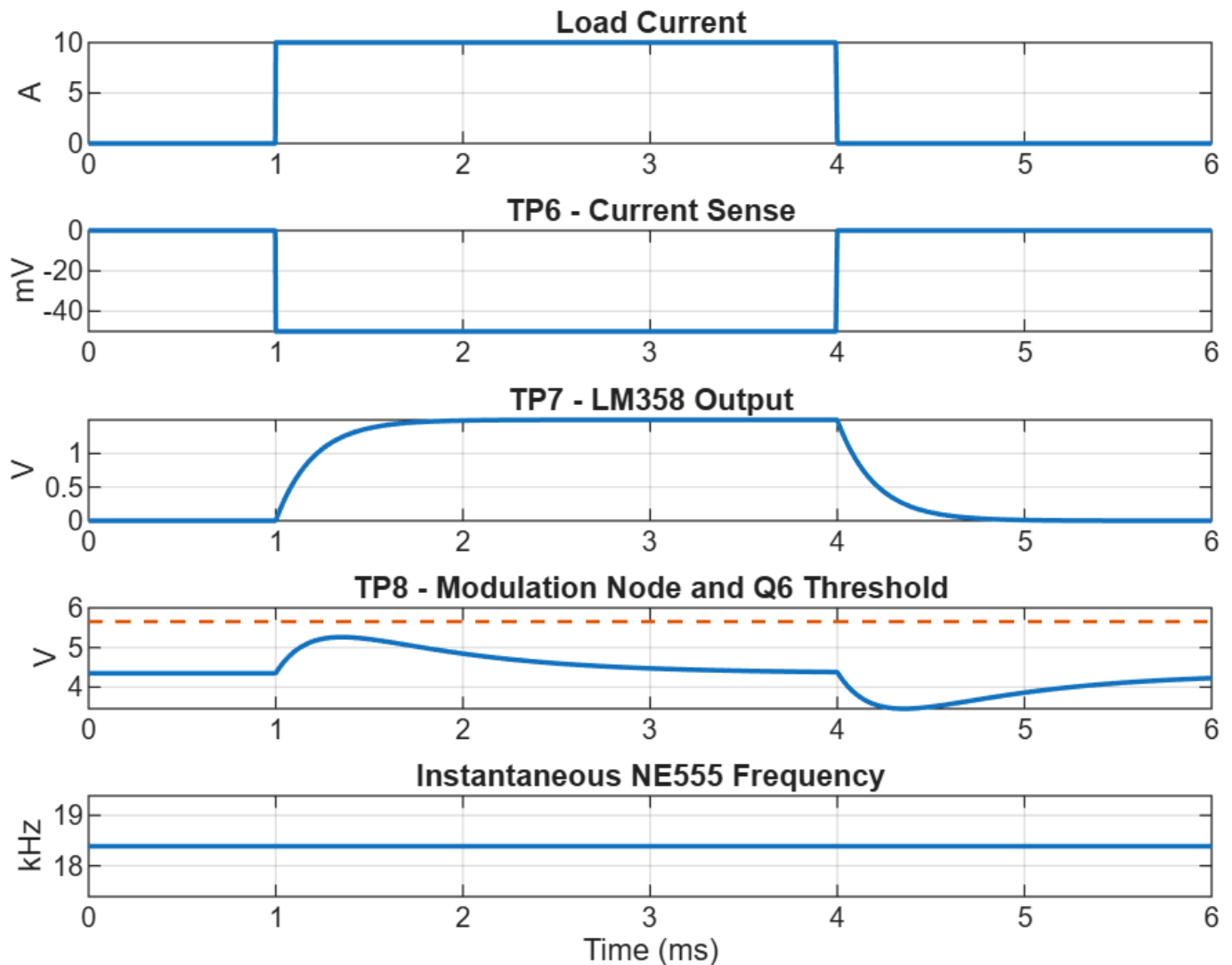


Figure 1. Version 2.3 analog/control path: load current, TP6, TP7, TP8, and instantaneous NE555 frequency.

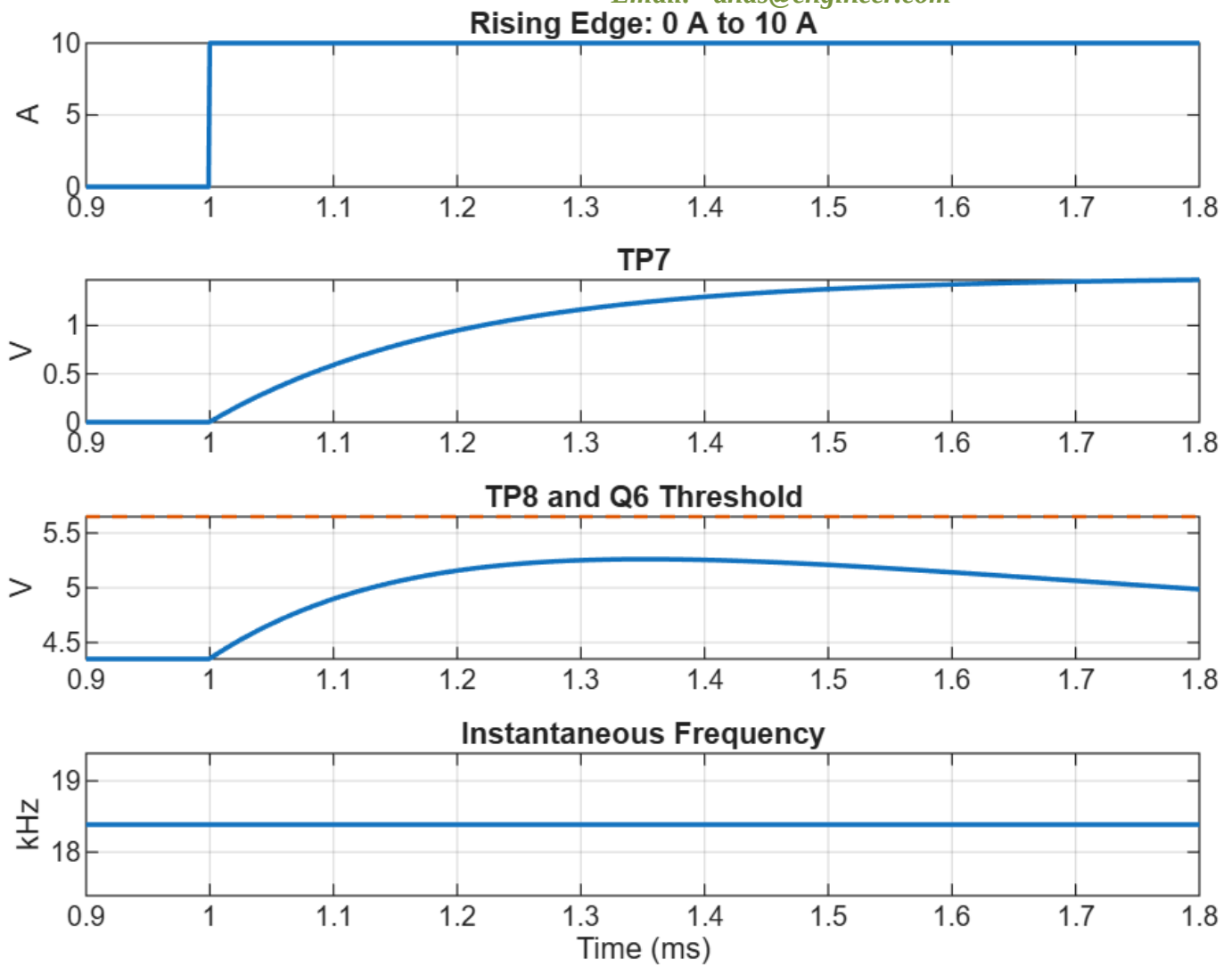


Figure 2. Version 2.3 rising edge, 0 A to 10 A. TP8 peaks below the modeled Q6 threshold with $R_{36} = 330 \text{ k}\Omega$.

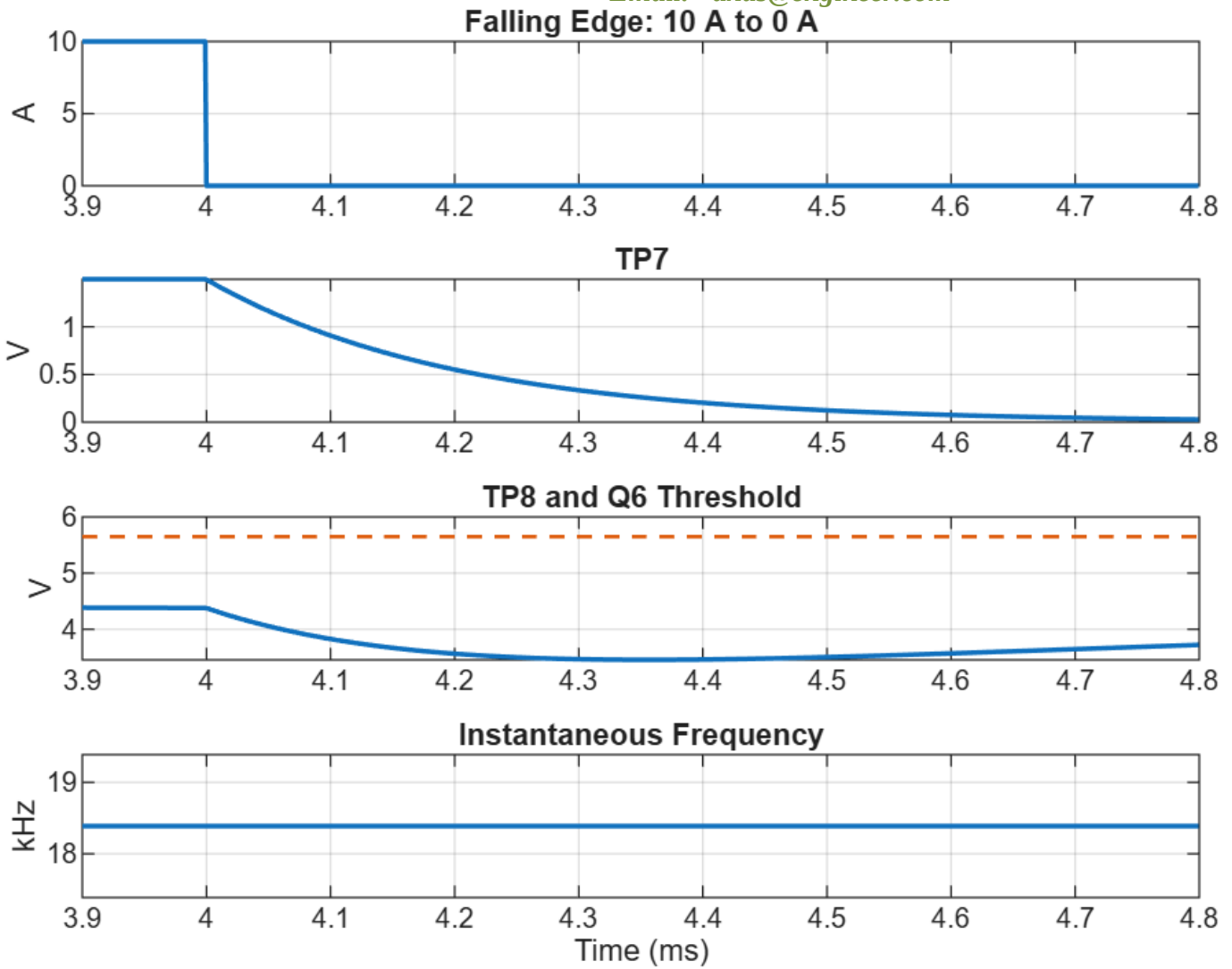


Figure 3. Version 2.3 falling edge, 10 A to 0 A. TP8 moves downward and the modeled frequency remains nominal.

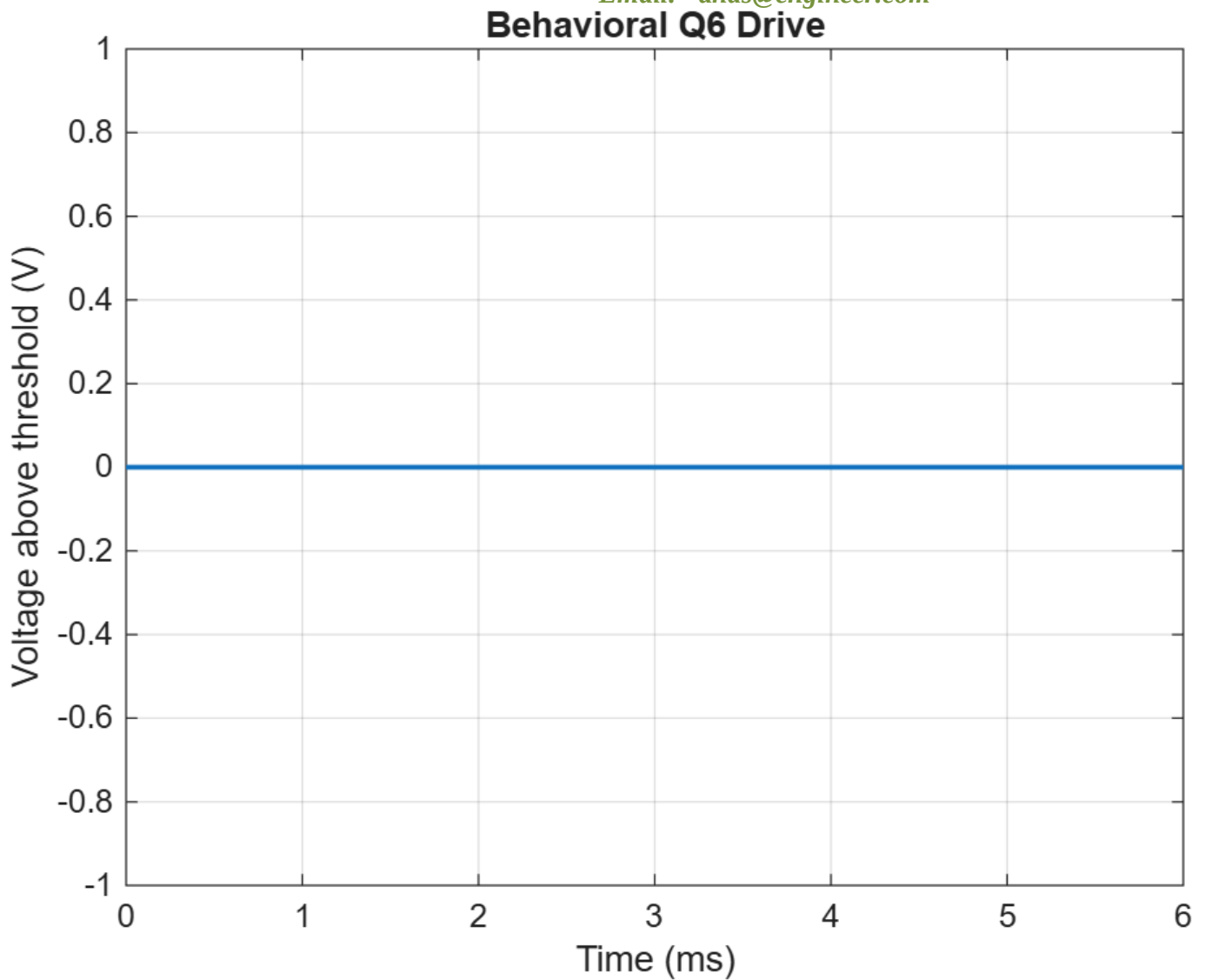


Figure 4. Version 2.3 behavioral Q6 drive. With $R_{36} = 330 \text{ k}\Omega$, TP8 never exceeds the 5.65 V modeled threshold, so Q6_drive remains zero.

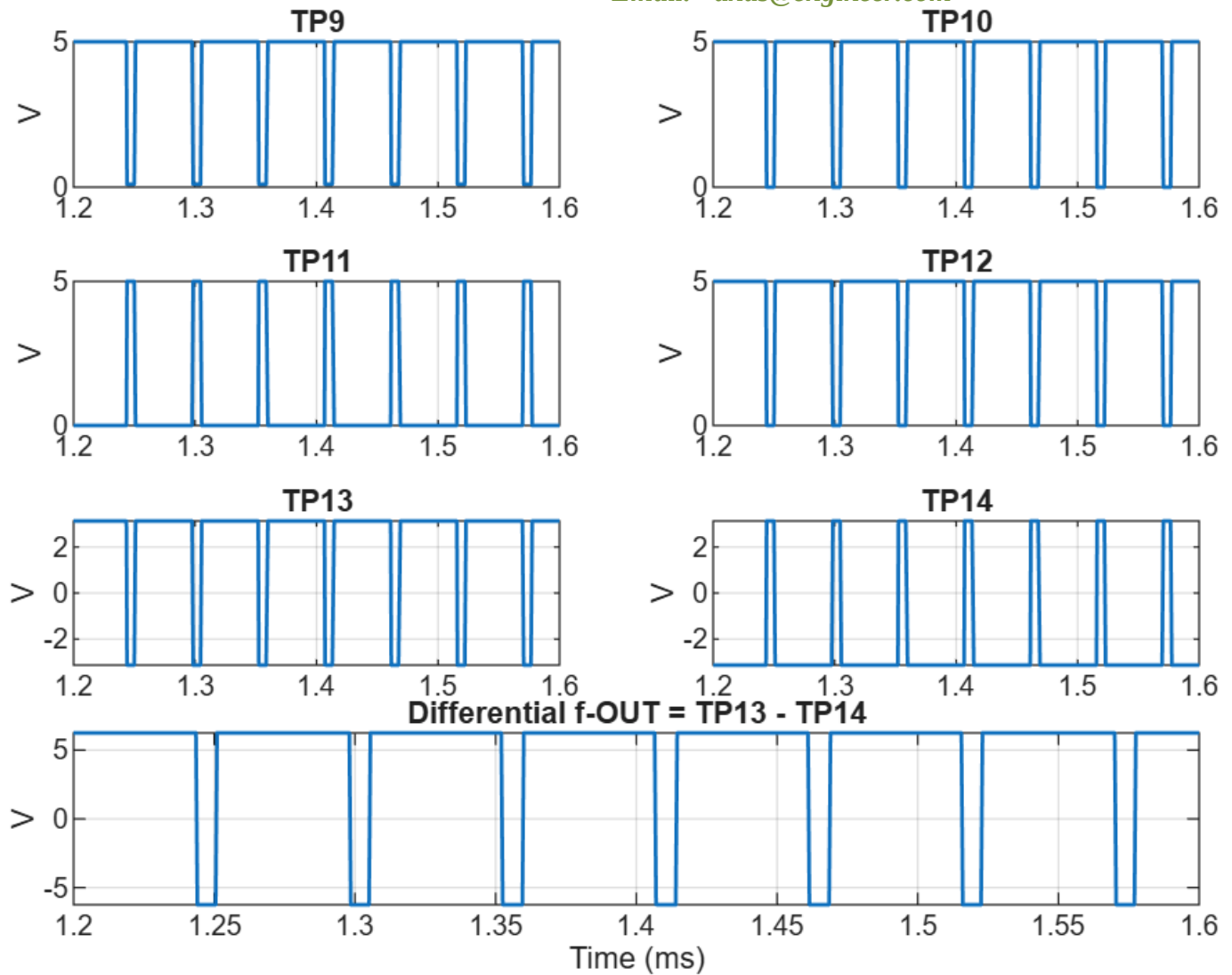


Figure 5. Version 2.3 switching test points TP9 through TP14 and differential f-OUT.

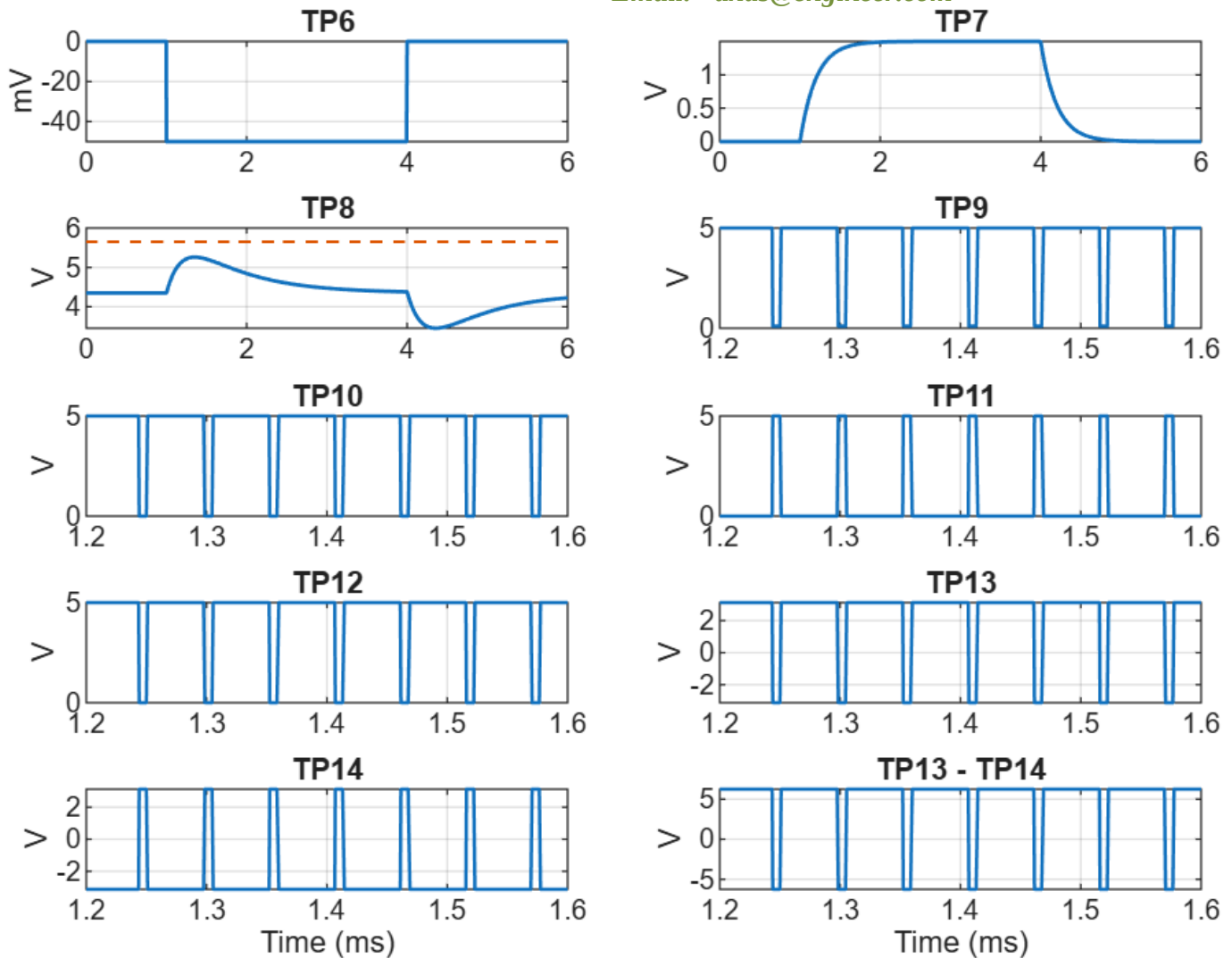


Figure 6. Version 2.3 combined diagnostic view of TP6 through TP14 and differential transformer output.

10. Version 3.0 R36 Gain Sweep

Version 3.0 keeps the same current pulse, time constants, Q6 threshold, NE555 model, and transformer model, but sweeps R36. Because the input path remains 11 kOhm, increasing R36 increases the LM358 gain and therefore increases the transient amplitude applied to TP8.

R36 (kOhm)	Gain	TP7 max (V)	TP8 max (V)	Margin to 5.65 V (mV)	Q6 drive max (V)	f max (kHz)	Cross?
330	30.00	1.500	5.261	-388.7	0.000	18.386	NO
390	35.45	1.773	5.427	-223.0	0.000	18.386	NO
430	39.09	1.955	5.537	-112.5	0.000	18.386	NO
470	42.73	2.136	5.648	-2.0	0.000	18.386	NO
510	46.36	2.318	5.758	+108.4	0.108	18.928	YES
560	50.91	2.545	5.897	+246.5	0.247	19.619	YES

Threshold result: 470 kOhm is only about 2 mV below the modeled threshold and should be regarded as a borderline case. 510 kOhm is the first tested value with positive modeled margin (+108 mV). 560 kOhm provides +247 mV modeled margin.

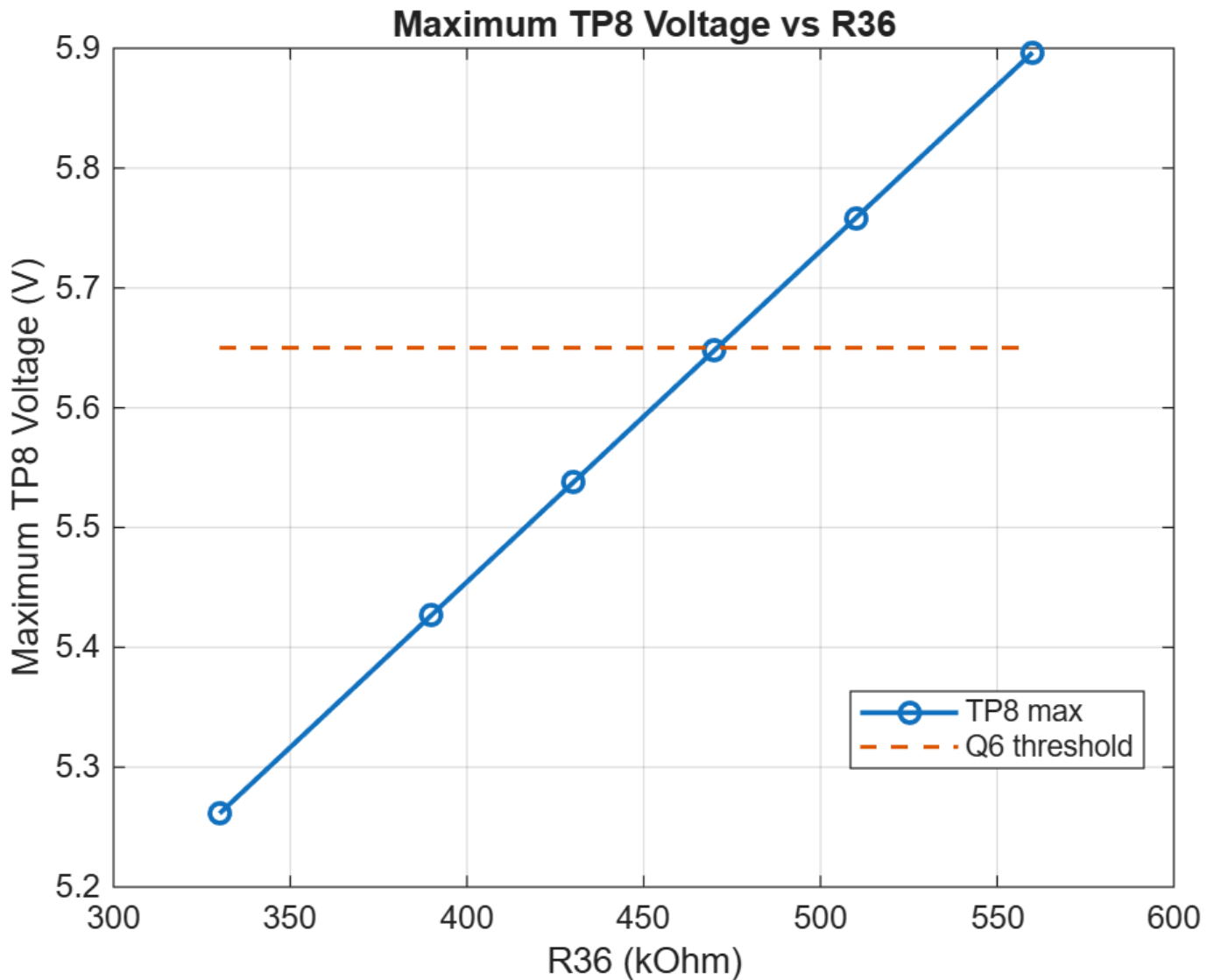


Figure 7. Maximum TP8 versus R36. The threshold crossing begins between 470 kOhm and 510 kOhm for the current behavioral assumptions.

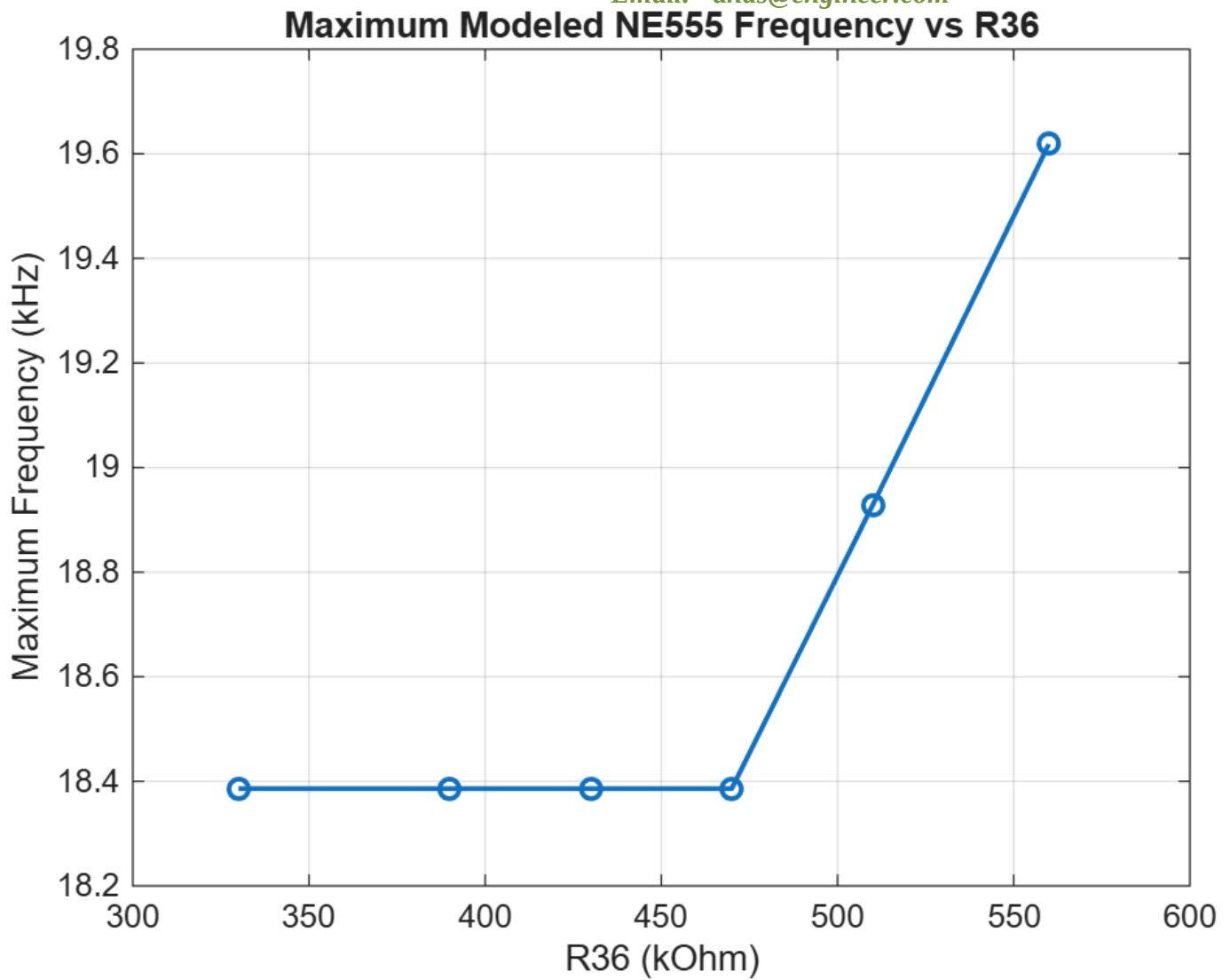


Figure 8. Maximum modeled frequency versus R36. The increase appears only after the behavioral Q6 threshold is crossed.

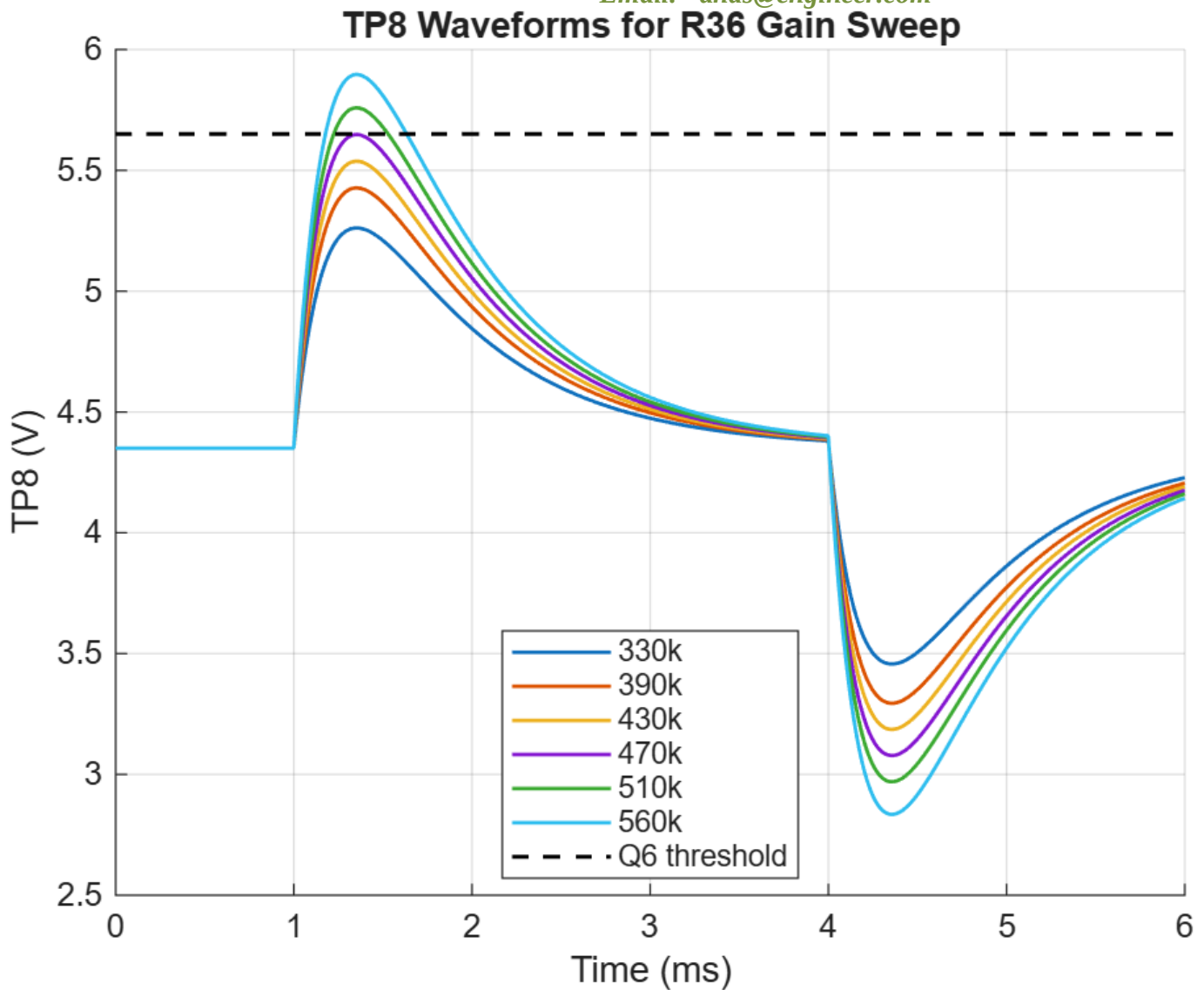


Figure 9. TP8 waveforms for the complete R36 gain sweep.

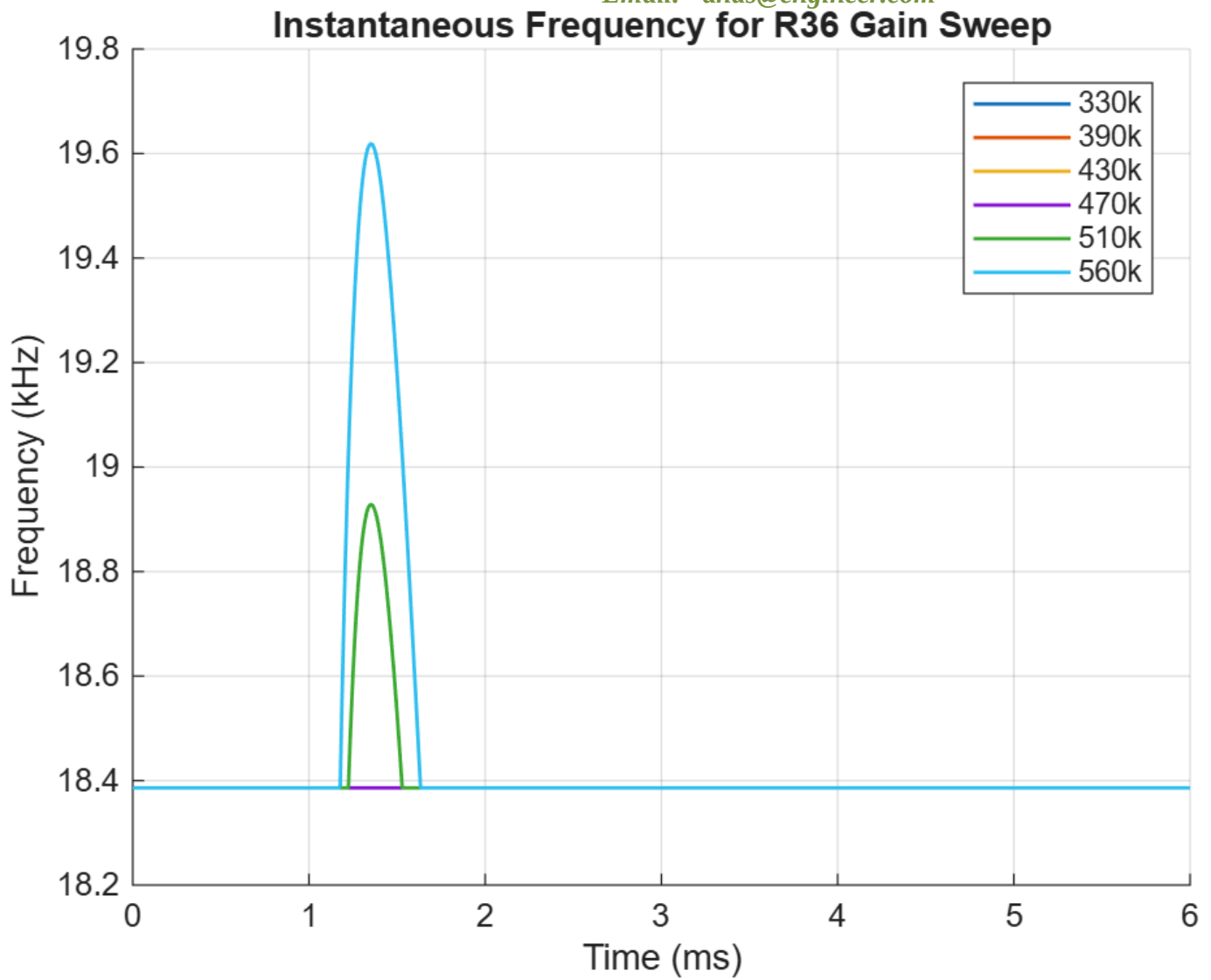


Figure 10. Instantaneous modeled frequency for the complete R36 gain sweep.

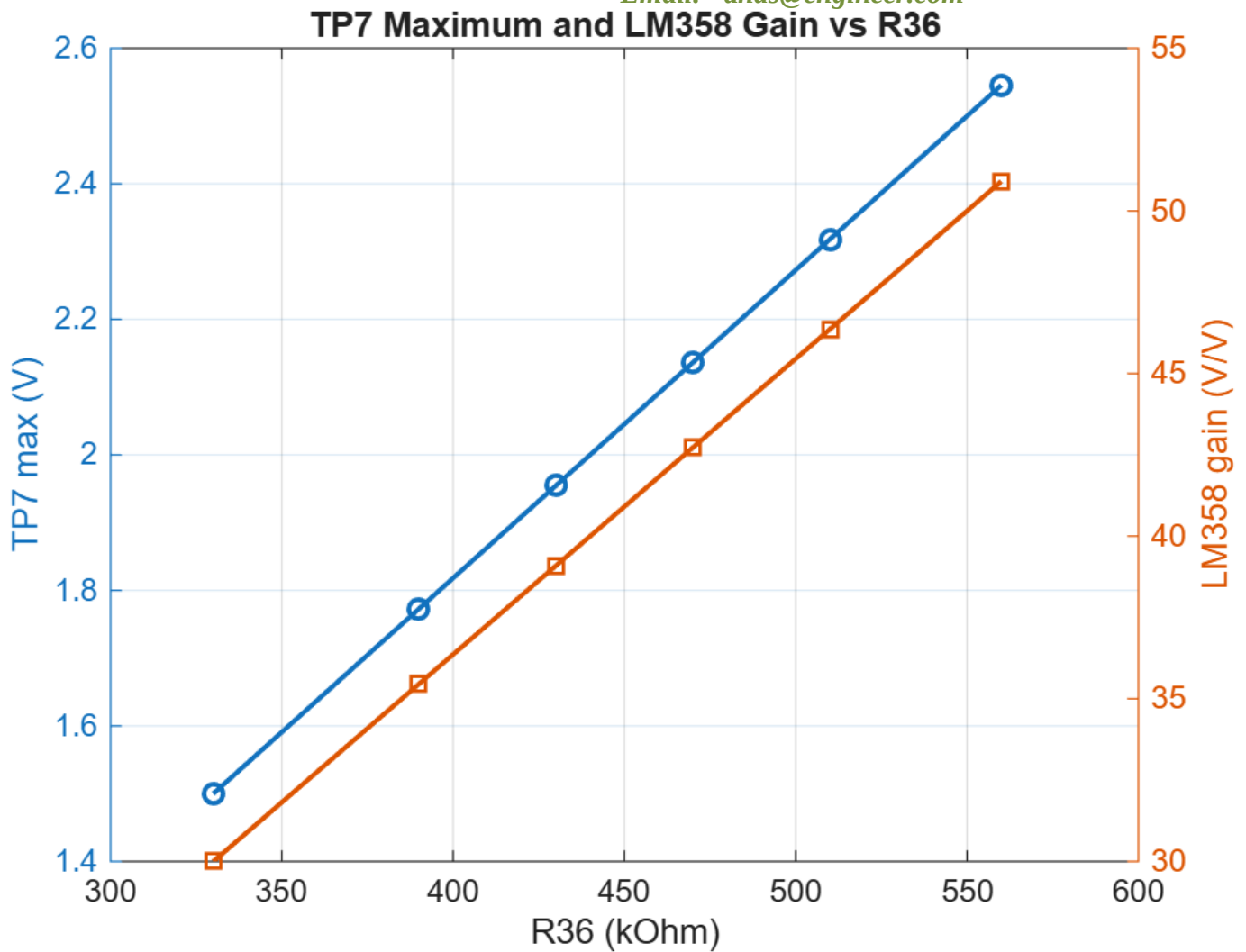


Figure 11. LM358 gain and TP7 maximum versus R36.

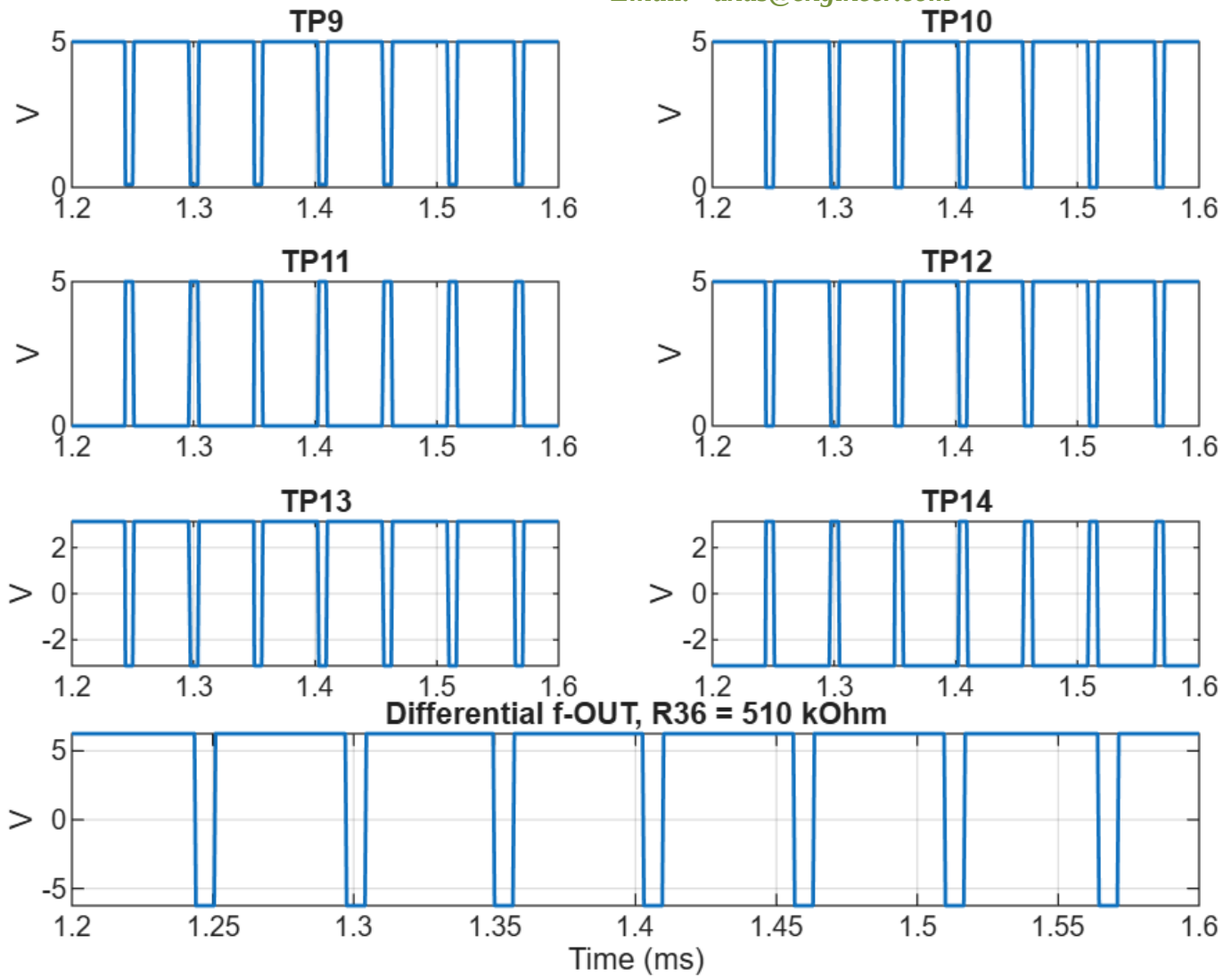


Figure 12. Digital switching waveforms for the first tested threshold-crossing case, $R_{36} = 510 \text{ k}\Omega$.

11. Engineering Interpretation and R36 Selection

The sweep is useful as a design-screening exercise because it separates three regions: clearly below threshold, borderline, and clearly above threshold. It does not yet establish the final production resistor because the transistor and diode network is not modeled physically.

R36	Modeled result	Engineering interpretation
330 k Ω	Gain 30.0; TP8 max 5.261 V	No threshold crossing. Good reference baseline.
390 k Ω	Gain 35.45; TP8 max 5.427 V	Still comfortably below threshold.
430 k Ω	Gain 39.09; TP8 max 5.537 V	Closer, but still 112.5 mV below.
470 k Ω	Gain 42.73; TP8 max 5.648 V	Borderline: only ~2 mV below. Real tolerances could change the result.
510 k Ω	Gain 46.36; TP8 max 5.758 V	First tested crossing; +108 mV model margin. Best next detailed simulation/bench candidate.
560 k Ω	Gain 50.91; TP8 max 5.897 V	Larger model margin; useful comparison point,



		but greater gain must be checked for headroom/noise/tolerance effects.
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Recommended next step: Use R36 = 510 kOhm as the next fixed-value simulation case (V3.1) and for controlled bench correlation. Treat 470 kOhm as a sensitivity/tolerance case and 560 kOhm as a higher-margin comparison, not as an automatic final choice.

12. Derived Frequency Law at R36 = 510 kOhm

Yes - a frequency law can be written for the present behavioral model. The important qualification is that the R35/C29 path behaves like a high-pass/transient path. Therefore the circuit does not produce a simple static frequency-versus-DC-current relationship. For a constant current after the transient has settled, the high-pass contribution returns toward zero and the modeled frequency returns to its nominal value. The most useful closed-form relation is therefore a dynamic law, or a peak-frequency law for a defined current step.

Parameter	Value at R36 = 510 kOhm	Meaning
Selected R36	510 kOhm	First tested value with a useful threshold-crossing margin in Version 3.0
LM358 gain G	$510k / (1k + 10k) = 46.36 \text{ V/V}$	Behavioral gain used by the model
R-sense	5 mOhm	$TP6 = -I_{load} \times R_{sense}$
Current-sense time constant	$\tau_{sense} = 200 \text{ us}$	First-order TP7 low-pass
Transient-path time constant	$\tau_{mod} = 708 \text{ us}$	High-pass shaping around TP8
TP8 bias	4.35 V	Behavioral bias used in Version 2.3 / 3.0
Q6 threshold	5.65 V	Behavioral VCC + VBE approximation
Nominal NE555 frequency	18.386 kHz	Calculated from the timing network
Behavioral frequency coefficient	$K_f = 5 \text{ kHz/V}$	Explicit model parameter; not yet measured

Dynamic transfer from the current-step input to the TP8 transient (continuous-time approximation):

$$\Delta V_8(s) / \Delta I(s) = 0.231818 \times [s \times 708 \text{ us}] / [(1 + s \times 200 \text{ us})(1 + s \times 708 \text{ us})] \quad \text{V/A}$$

For an ideal positive current step of magnitude ΔI , the resulting TP8 transient is:

$$\Delta V_8(t) = 0.231818 \times \Delta I \times [708 / (708 - 200)] \times (\exp(-t/708 \text{ us}) - \exp(-t/200 \text{ us})), \quad t \geq 0$$

The continuous-time peak occurs at approximately $t_{peak} = 352 \text{ us}$. At that instant, the step-response gain is approximately 0.14093 V/A. Therefore:

$$TP8_{peak} \text{ approximately } 4.35 + 0.14093 \times \Delta I \quad [\text{V}], \text{ with } \Delta I \text{ in amperes}$$

With the present 5.65 V behavioral Q6 threshold, the corresponding current-step threshold is approximately 9.22 A. Below this step magnitude, the behavioral model predicts no Q6 drive. Above it, the peak frequency increases according to the explicit K_f model coefficient.

$$f_{peak} [\text{kHz}] \text{ approximately } 18.386 + 5 \times \max(0, 0.14093 \times \Delta I - 1.30)$$

Equivalent form using the R-sense/TP6 step magnitude directly (TP6 in mV):

$$f_{peak} [\text{kHz}] \text{ approximately } 18.386 + 5 \times \max(0, 0.028186 \times |\Delta TP6_{mV}| - 1.30)$$



For the 10 A step used in the simulation, the continuous approximation gives TP8_peak approximately 5.759 V and f_peak approximately 18.933 kHz, which is very close to the discrete Version 3.0 result of about 5.758 V and 18.928 kHz.

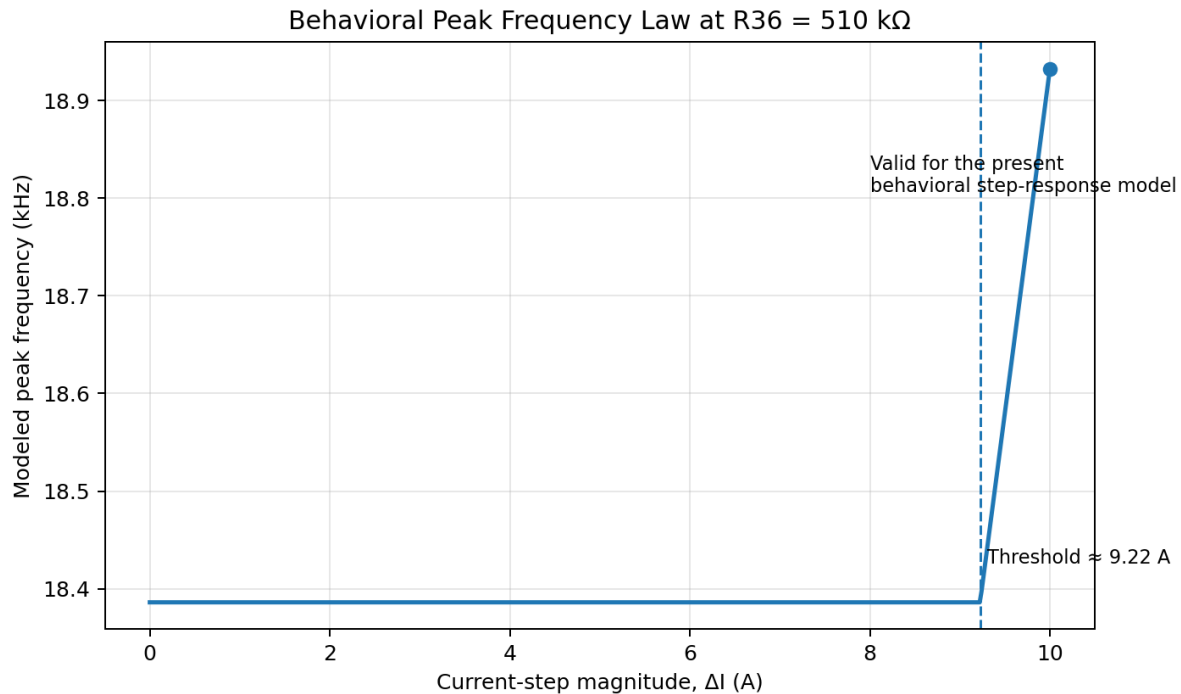


Figure 12-1. Derived peak-frequency law for the present behavioral model at R36 = 510 kOhm. The curve is valid for the defined current-step experiment, not as a measured static transfer function.

Current-step magnitude (A)	Predicted TP8 peak (V)	Behavioral peak frequency (kHz)
8.00	5.477	18.386
9.00	5.618	18.386
9.20	5.647	18.386
9.25	5.654	18.404
9.50	5.689	18.580
9.75	5.724	18.756
10.00	5.759	18.933

How to turn this into a real circuit law later

After bench measurements are available, keep the same model structure but fit $V_{\text{threshold}}$ and K_f from measured data. A simple first fit is $f = f_0 + a \times \max(V_8 - V_{\text{th}}, 0)$. If the measured response is curved, use a piecewise or polynomial fit. Until that calibration is done, the equations above are simulation laws, not guaranteed hardware laws.

13. Model Limitations and What Is Not Yet Proven

Area	Limitation
BC558 threshold	The fixed 5.65 V criterion is a simplified behavioral switch, not a transistor equation.
D32	The diode part is not confirmed, so its real forward drop/dynamic behavior is unknown.
D33/D34	The 1N914 + 5.6 V Zener path is documented in the schematic but not solved as a nonlinear device network in V2.3/V3.0.
LM358	The model uses gain, a first-order time constant, and a simple output clamp. Real input common-mode, bias, slew rate, GBW, output current,



	and saturation behavior are not modeled.
NE555	The nominal timing uses constant diode forward voltage and discharge saturation. Real thresholds, output loading, and control-pin coupling are not included.
Frequency sensitivity	$K_f = 5 \text{ kHz/V}$ is arbitrary/behavioral. The actual circuit transfer from Q6 conduction to NE555 period must be derived or measured.
Logic gates	7408/7401 are treated as ideal logic with no propagation delay or output-drive limitation.
Transformer	Ideal 1.25 turns ratio only. No magnetics, losses, saturation, or load.
Tolerance/temperature	No Monte Carlo, resistor tolerance, diode/Zener tolerance, transistor beta/ V_{BE} spread, or temperature sweep has been applied.
Hardware validation	The plots are MATLAB model outputs, not oscilloscope measurements.

14. Bench-Validation Plan

Step	Check	Purpose
1	Confirm supplies and reference	Verify VCC and the actual ground reference before probing. Confirm the sense polarity at TP6.
2	Calibrate shunt path	Apply known current points (for example 0, 2, 5, 8, 10 A where safe) and verify TP6. Compare with 5 mOhm equation.
3	Verify LM358 gain	Measure TP7 at steady current for $R_{36} = 330 \text{ kOhm}$ first. Then compare 470/510/560 kOhm only after confirming op-amp headroom.
4	Capture current edges	Use the fastest safe current transition available and capture TP6, TP7, and TP8 on the same time base.
5	Measure Q6 pins	Measure emitter, base, and collector to determine the actual conduction condition rather than assuming $V_{BE} = 0.65 \text{ V}$.
6	Identify D32	Confirm the D32 part number/orientation and include it in the next physical model.
7	Measure NE555 timing	Capture TP9 and TP10. Measure actual period, charge/discharge intervals, and any period change during the TP8 transient.
8	Check primary drive	Compare TP11 and TP12 timing and levels. Confirm dead-time/overlap behavior if relevant to the real hardware.
9	Measure secondary safely	Use an appropriate differential/isolated measurement method for TP13, TP14, and f-OUT. Do not assume oscilloscope ground can be tied to a floating node.
10	Correlate model	Overlay measured and modeled TP6/TP7/TP8 first. Only after that refine the Q6-to-frequency law and transformer model.

15. Study Checklist and Review Questions

1. Why is TP6 only 50 mV in magnitude at 10 A?



2. What determines the LM358 gain in this simplified model?
3. Why does C30 make TP7 slower than an ideal instantaneous amplifier?
4. Why is the R35/C29 branch more sensitive to current edges than to a constant 10 A plateau?
5. Why can TP8 peak after the current step instead of simply following current amplitude?
6. What does the 5.65 V Q6 threshold represent, and why is it only an approximation?
7. Why does Version 2.3 show a flat frequency even though TP8 clearly changes?
8. Why does increasing R36 eventually create a modeled frequency transient?
9. Why is 470 kOhm a poor choice for declaring guaranteed operation from this model alone?
10. Why is 510 kOhm a useful next test value?
11. Why can the 19.619 kHz result at 560 kOhm not yet be claimed as a physical hardware prediction?
12. What do TP9 and TP10 represent?
13. Why are TP11 and TP12 complementary in the present model?
14. Why are TP13 and TP14 shown as positive/negative around a virtual midpoint?
15. What additional transformer parameters are required for a realistic magnetic model?
16. Which measurements would most directly validate the Q6 threshold assumption?
17. What could temperature do to VBE, diode drops, Zener voltage, and the threshold margin?
18. How would resistor tolerance affect the 470 kOhm borderline case?
19. What is the difference between validating the signal chain and validating the absolute frequency law?
20. What should be changed in the model before using it for final component release?



Appendix A. Version 1 - Line-by-Line Annotated MATLAB Source

Every source line is reproduced exactly in the left column. The right column explains the purpose in plain English so a reader with little or no MATLAB experience can follow the program from top to bottom.

Line	MATLAB source line	English explanation
1	% NE555 Current-Sense / Frequency-Modulation Behavioral Model	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
2	% Project: Current-sense controlled NE555 / transformer signal generator	Source-code comment: Project: Current-sense controlled NE555 / transformer signal generator
3	% Purpose: First-pass MATLAB model for test points TP6...TP14	Source-code comment: Purpose: First-pass MATLAB model for test points TP6...TP14
4	%	Source-code comment explaining the model assumption or the purpose of the following lines.
5	% IMPORTANT:	Source-code comment: IMPORTANT:
6	% This is a BEHAVIORAL / ENGINEERING APPROXIMATION based on the current	Source-code comment: This is a BEHAVIORAL / ENGINEERING APPROXIMATION based on the current
7	% schematic and the preliminary calculations. It is intended to visualize	Source-code comment: schematic and the preliminary calculations. It is intended to visualize
8	% signal relationships and timing before a more detailed transistor/diode	Source-code comment: signal relationships and timing before a more detailed transistor/diode
9	% model is added.	Source-code comment: model is added.
10	%	Source-code comment explaining the model assumption or the purpose of the following lines.
11	% Assumptions used in this first version:	Source-code comment: Assumptions used in this first version:
12	% VCC = 5 V	Source-code comment: VCC = 5 V
13	% R-sense = 5 mOhm	Source-code comment: R-sense = 5 mOhm
14	% Load current = 0...10 A	Source-code comment: Load current = 0...10 A
15	% LM358 DC gain ~= 30	Source-code comment: LM358 DC gain ~= 30
16	% C30 current-sense LPF tau ~= 200 us	Source-code comment: C30 current-sense LPF tau ~= 200 us
17	% NE555 nominal freq ~= 18.4 kHz (including ~0.6 V on D31)	Source-code comment: NE555 nominal freq ~= 18.4 kHz (including ~0.6 V on D31)
18	% D33 = 1N914	Source-code comment: D33 = 1N914
19	% D34 = 5.6 V Zener	Source-code comment: D34 = 5.6 V Zener
20	% R35/C29 modulation tau ~= 708 us	Source-code comment: R35/C29 modulation tau ~= 708 us
21	% TP8 is modeled as a transient control node around a DC bias.	Source-code comment: TP8 is modeled as a transient control node around a DC bias.
22	% Q6/555 modulation sensitivity is represented by a tunable coefficient.	Source-code comment: Q6/555 modulation sensitivity is represented by a tunable coefficient.
23	% TP11/TP12 are treated as complementary 5-V primary-drive signals.	Source-code comment: TP11/TP12 are treated as complementary 5-V primary-drive signals.
24	% Transformer ratio is approximated as 100T / 80T = 1.25.	Source-code comment: Transformer ratio is approximated as 100T / 80T = 1.25.
25	%	Source-code comment explaining the model assumption or the purpose of the following lines.
26	% All test-point waveforms are referenced to GND except the isolated	Source-code comment: All test-point waveforms are referenced to GND except the isolated
27	% transformer secondary, where TP13/TP14 are shown using an assumed	Source-code comment: transformer secondary, where TP13/TP14 are shown using an assumed
28	% midpoint reference for visualization. The differential TP13-TP14 is the	Source-code comment: midpoint reference for visualization. The differential TP13-TP14 is the
29	% physically meaningful output.	Source-code comment: physically meaningful output.
30	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
31	clear;	Removes variables from the MATLAB workspace so values from an earlier run cannot contaminate this simulation.
32	clc;	Clears the MATLAB Command Window so the new run starts with readable console output.
33	close all;	Closes previously open figure windows so old plots are not confused with the current simulation results.
34	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
35	% 1. Global simulation settings	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
36	VCC = 5.0; % V	Assigns or calculates the 5 V supply used by the behavioral model from the expression on the right-hand side.
37	dt = 1e-6; % 1 us time step	Assigns or calculates the simulation time step from the expression on the right-hand side.
38	Tsim = 10e-3; % 10 ms simulation	Assigns or calculates the total simulated time from the expression on the right-hand side.



Line	MATLAB source line	English explanation
39	t = 0:dt:Tsim;	Assigns or calculates the simulation time vector from the expression on the right-hand side.
40	N = numel(t);	Calculates the number of time samples by counting the number of elements in the referenced vector.
41	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
42	%% 2. Circuit parameters	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
43	Rsense = 5e-3; % 5 mOhm	Assigns or calculates the current-shunt resistance from the expression on the right-hand side.
44	R38 = 1e3;	Assigns or calculates the 1 kOhm LM358 input-path resistor from the expression on the right-hand side.
45	R37 = 10e3;	Assigns or calculates the 10 kOhm LM358 input-path resistor from the expression on the right-hand side.
46	R36 = 330e3;	Assigns or calculates the LM358 gain-setting resistor from the expression on the right-hand side.
47	C30 = 220e-9;	Assigns or calculates the current-sense filtering capacitor from the expression on the right-hand side.
48	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
49	R35 = 330e3;	Assigns or calculates the main resistor in the transient-modulation path from the expression on the right-hand side.
50	R34 = 24e3;	Assigns or calculates the 24 kOhm modulation-path resistor from the expression on the right-hand side.
51	C29 = 2e-9;	Assigns or calculates the transient-modulation capacitor from the expression on the right-hand side.
52	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
53	R32 = 10;	Assigns or calculates the small series resistor in the NE555 charge path from the expression on the right-hand side.
54	R33 = 5e3;	Assigns or calculates the main NE555 charge resistor from the expression on the right-hand side.
55	R29 = 1e3;	Assigns or calculates the NE555 discharge resistor from the expression on the right-hand side.
56	C27 = 10e-9;	Assigns or calculates the NE555 timing capacitor from the expression on the right-hand side.
57	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
58	Vf_D31 = 0.60; % approximate 1N914 forward drop	Assigns or calculates the assumed D31 forward drop from the expression on the right-hand side.
59	Vf_D33 = 0.60; % approximate 1N914 forward drop	Assigns or calculates the assumed D33 forward drop from the expression on the right-hand side.
60	Vz_D34 = 5.60; % 5.6-V Zener	Assigns or calculates the assumed D34 Zener voltage from the expression on the right-hand side.
61	VCEsat = 0.10; % approximate 555 discharge saturation voltage	Assigns or calculates the assumed saturation voltage of the NE555 discharge transistor from the expression on the right-hand side.
62	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
63	Npri = 80; % transformer primary turns	Assigns or calculates the transformer primary-turn count from the expression on the right-hand side.
64	Nsec = 100; % transformer secondary turns	Assigns or calculates the transformer secondary-turn count from the expression on the right-hand side.
65	turns_ratio = Nsec / Npri;	Assigns or calculates the ideal transformer turns ratio from the expression on the right-hand side.
66	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
67	%% 3. Derived values	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
68	Rin_eff = R38 + R37;	Assigns or calculates the effective LM358 input-path resistance from the expression on the right-hand side.
69	A_LM358 = R36 / Rin_eff; % ~= 30	Assigns or calculates the modeled LM358 closed-loop gain magnitude from the expression on the right-hand side.



Line	MATLAB source line	English explanation
70	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
71	Rth_C30 = 1 / (1/R38 + 1/R37);	Calculates the Thevenin resistance seen by C30 from the stated reciprocal relationship.
72	tau_sense = Rth_C30 * C30;	Assigns or calculates the current-sense first-order time constant from the expression on the right-hand side.
73	fc_sense = 1 / (2*pi*tau_sense);	Calculates the corresponding current-sense cutoff frequency from the stated reciprocal relationship.
74	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
75	tau_mod = (R35 + R34) * C29;	Assigns or calculates the transient-path time constant from the expression on the right-hand side.
76	fc_mod = 1 / (2*pi*tau_mod);	Calculates the corresponding transient-path cutoff frequency from the stated reciprocal relationship.
77	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
78	RA = R32 + R33;	Assigns or calculates the NE555 effective charge resistance from the expression on the right-hand side.
79	RB = R29;	Assigns or calculates the NE555 discharge resistance from the expression on the right-hand side.
80	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
81	% NE555 thresholds	Source-code comment: NE555 thresholds
82	VTL = VCC/3;	Assigns or calculates the lower NE555 timing threshold (one-third VCC) from the expression on the right-hand side.
83	VTH = 2*VCC/3;	Assigns or calculates the upper NE555 timing threshold (two-thirds VCC) from the expression on the right-hand side.
84	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
85	% First-pass frequency estimate including D31 forward drop.	Source-code comment: First-pass frequency estimate including D31 forward drop.
86	Vcharge_inf = VCC - Vf_D31;	Assigns or calculates the asymptotic timing-capacitor charge voltage after the diode drop from the expression on the right-hand side.
87	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
88	t_charge = RA*C27 * log((Vcharge_inf - VTL) / (Vcharge_inf - VTH));	Calculates the calculated NE555 charge interval using the analytical exponential charge/discharge timing equation.
89	t_discharge = RB*C27 * log((VTH - VCEsat) / (VTL - VCEsat));	Calculates the calculated NE555 discharge interval using the analytical exponential charge/discharge timing equation.
90	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
91	T555_nom = t_charge + t_discharge;	Assigns or calculates the nominal NE555 oscillation period from the expression on the right-hand side.
92	f555_nom = 1 / T555_nom;	Assigns or calculates the nominal NE555 frequency from the expression on the right-hand side.
93	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
94	fprintf('LM358 approximate DC gain : %.2f V/V\n', A_LM358);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
95	fprintf('Current-sense LPF tau : %.1f us\n', tau_sense*1e6);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
96	fprintf('Current-sense LPF fc : %.1f Hz\n', fc_sense);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
97	fprintf('Modulation path tau : %.1f us\n', tau_mod*1e6);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
98	fprintf('Modulation path fc : %.1f Hz\n', fc_mod);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
99	fprintf('Nominal NE555 frequency : %.1f kHz\n', f555_nom/1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
100	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
		effect.
101	%% 4. Load-current profile: 0...10 A	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
102	% This profile is intentionally dynamic so that TP7/TP8 response can be seen.	Source-code comment: This profile is intentionally dynamic so that TP7/TP8 response can be seen.
103	Iload = zeros(size(t));	Initializes the simulated load-current waveform with zeros, giving it the required array size before later samples are calculated.
104	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
105	% 0 to 1 ms: 0 A	Source-code comment: 0 to 1 ms: 0 A
106	idx = t >= 1e-3 & t < 3e-3;	Builds a logical time mask selecting the interval where the following load-current assignment applies.
107	Iload(idx) = 10 * (t(idx)-1e-3)/(2e-3); % ramp 0 -> 10 A	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
108	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
109	idx = t >= 3e-3 & t < 5e-3;	Builds a logical time mask selecting the interval where the following load-current assignment applies.
110	Iload(idx) = 10; % hold at 10 A	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
111	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
112	idx = t >= 5e-3 & t < 6e-3;	Builds a logical time mask selecting the interval where the following load-current assignment applies.
113	Iload(idx) = 10 - 6*(t(idx)-5e-3)/(1e-3); % ramp 10 -> 4 A	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
114	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
115	idx = t >= 6e-3 & t < 8e-3;	Builds a logical time mask selecting the interval where the following load-current assignment applies.
116	Iload(idx) = 4; % hold at 4 A	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
117	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
118	idx = t >= 8e-3 & t < 9e-3;	Builds a logical time mask selecting the interval where the following load-current assignment applies.
119	Iload(idx) = 4 - 4*(t(idx)-8e-3)/(1e-3); % ramp 4 -> 0 A	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
120	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
121	Iload(t >= 9e-3) = 0;	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
122	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
123	%% 5. TP6 - R-sense signal	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
124	% Polarity chosen so the inverting LM358 stage generates a positive output.	Source-code comment: Polarity chosen so the inverting LM358 stage generates a positive output.
125	TP6 = -Iload * Rsense; % V, 0 to -50 mV	Converts load current into the shunt voltage at TP6. The negative sign represents the polarity chosen for the inverting LM358 model.
126	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
127	%% 6. TP7 - LM358 amplified + filtered current-sense output	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
128	% Ideal target output from the inverting stage:	Source-code comment: Ideal target output from the inverting stage:
129	TP7_target = -A_LM358 * TP6;	Calculates the ideal amplified current-sense voltage before the first-order output/filter response is applied.
130	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
		effect.
131	% Simple single-pole low-pass approximation from C30 network.	Source-code comment: Simple single-pole low-pass approximation from C30 network.
132	TP7 = zeros(size(t));	Initializes the filtered behavioral LM358 output with zeros, giving it the required array size before later samples are calculated.
133	alpha_sense = dt / (tau_sense + dt);	Assigns or calculates the discrete first-order filter coefficient for TP7 from the expression on the right-hand side.
134	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
135	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
136	TP7(k) = TP7(k-1) + alpha_sense*(TP7_target(k)-TP7(k-1));	Updates TP7 with a first-order discrete low-pass equation, moving the previous output toward the instantaneous target.
137	end	Ends the current for-loop or if/else block.
138	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
139	% Approximate LM358 output swing limitation on 5-V supply.	Source-code comment: Approximate LM358 output swing limitation on 5-V supply.
140	TP7 = min(max(TP7, 0), 3.5);	Clamps the behavioral LM358 output to the assumed usable output range on a 5 V supply.
141	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
142	%% 7. TP8 - transient modulation/control node	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
143	% The R35/C29 path is modeled as a high-pass response to TP7.	Source-code comment: The R35/C29 path is modeled as a high-pass response to TP7.
144	% A bias near the D32-clamped region is used so the transient can approach	Source-code comment: A bias near the D32-clamped region is used so the transient can approach
145	% the BC558 conduction threshold.	Source-code comment: the BC558 conduction threshold.
146	Vbias_TP8 = 4.35; % approximate bias/clamp level for first-pass model	Assigns or calculates the assumed TP8 bias/clamp level from the expression on the right-hand side.
147	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
148	TP7_lp_mod = zeros(size(t));	Initializes the low-pass state used to form the high-pass transient with zeros, giving it the required array size before later samples are calculated.
149	alpha_mod = dt / (tau_mod + dt);	Assigns or calculates the discrete transient-path filter coefficient from the expression on the right-hand side.
150	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
151	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
152	TP7_lp_mod(k) = TP7_lp_mod(k-1) + alpha_mod*(TP7(k)-TP7_lp_mod(k-1));	Updates the low-pass state used to subtract the slow component of TP7 and isolate the transient/high-pass component.
153	end	Ends the current for-loop or if/else block.
154	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
155	HP_mod = TP7 - TP7_lp_mod;	Subtracts the slow low-pass component from TP7 to obtain the transient/high-pass component that drives TP8.
156	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
157	% Tunable shaping gain. Start with 1.0 for transparent interpretation.	Source-code comment: Tunable shaping gain. Start with 1.0 for transparent interpretation.
158	Kmod_node = 1.0;	Assigns or calculates the behavioral TP8 shaping gain from the expression on the right-hand side.
159	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
160	TP8 = Vbias_TP8 + Kmod_node*HP_mod;	Adds the transient component to the assumed TP8 DC bias, forming the behavioral control-node voltage.
161	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
162	% Keep visualization within a reasonable transient range.	Source-code comment: Keep visualization within a reasonable transient range.
163	TP8 = min(max(TP8, 0), 6.5);	Clamps the behavioral TP8 voltage to a bounded plotting range so extreme numerical values cannot dominate the visualization.
164	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
		effect.
165	%% 8. Convert TP8 transient into instantaneous 555 frequency	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
166	% Approximate BC558 turn-on threshold around VCC + 0.65 V.	Source-code comment: Approximate BC558 turn-on threshold around VCC + 0.65 V.
167	VBE_PNP = 0.65;	Assigns or calculates the assumed BC558 base-emitter magnitude from the expression on the right-hand side.
168	VQ6_on = VCC + VBE_PNP;	Assigns or calculates the modeled Q6 turn-on threshold from the expression on the right-hand side.
169	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
170	% Only the amount above threshold modulates frequency in this first model.	Source-code comment: Only the amount above threshold modulates frequency in this first model.
171	Q6_drive = max(TP8 - VQ6_on, 0);	Implements the behavioral Q6 threshold: values below threshold produce zero drive; only voltage above threshold contributes to modulation.
172	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
173	% Frequency sensitivity is not yet known from a full transistor-level	Source-code comment: Frequency sensitivity is not yet known from a full transistor-level
174	% derivation. Keep it explicit and tunable.	Source-code comment: derivation. Keep it explicit and tunable.
175	Kf = 5e3; % Hz/V, behavioral tuning coefficient	Assigns or calculates the behavioral frequency sensitivity coefficient from the expression on the right-hand side.
176	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
177	f_inst = f555_nom + Kf*Q6_drive;	Creates the behavioral frequency law by adding Kf times Q6 threshold overdrive to the nominal NE555 frequency.
178	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
179	% Keep frequency inside a practical plotting range.	Source-code comment: Keep frequency inside a practical plotting range.
180	f_inst = min(max(f_inst, 5e3), 40e3);	Clamps the behavioral frequency to a practical plotting range; this is a numerical safeguard rather than a measured circuit limit.
181	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
182	%% 9. Generate variable-frequency NE555 output phase	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
183	phase = zeros(size(t));	Initializes the accumulated oscillator phase with zeros, giving it the required array size before later samples are calculated.
184	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
185	phase(k) = phase(k-1) + 2*pi*f_inst(k)*dt;	Numerically integrates instantaneous frequency into oscillator phase for this time sample.
186	end	Ends the current for-loop or if/else block.
187	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
188	% Approximate charge/discharge duty cycle from calculated timing.	Source-code comment: Approximate charge/discharge duty cycle from calculated timing.
189	duty_nom = t_charge / T555_nom;	Assigns or calculates the nominal high-state duty fraction from the expression on the right-hand side.
190	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
191	phase01 = mod(phase, 2*pi)/(2*pi);	Wraps accumulated phase into a normalized 0-to-1 cycle position used to synthesize the square-wave output.
192	TP10 = VCC * (phase01 < duty_nom); % 555 OUTPUT after R30	Converts normalized oscillator phase and duty cycle into a 0 V / VCC square-wave approximation of the NE555 output.
193	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
194	%% 10. TP9 - 555 DISCHARGE node	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
195	% Approximate:	Source-code comment: Approximate:
196	% charge interval -> node near VCC	Source-code comment: charge interval -> node near VCC
197	% discharge interval -> node near VCEsat	Source-code comment: discharge interval -> node near VCEsat
198	TP9 = VCC * ones(size(t));	Initializes the modeled NE555 discharge-node voltage with ones of the required array size;



Line	MATLAB source line	English explanation
		later lines modify selected samples.
199	TP9(phase01 >= duty_nom) = VCEsat;	For time samples in the discharge phase, forces TP9 down to the assumed discharge-transistor saturation voltage.
200	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
201	%% 11. TP11 and TP12 - complementary transformer-drive signals	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
202	% First-pass digital model of the two logic branches.	Source-code comment: First-pass digital model of the two logic branches.
203	TP12 = TP10;	Uses TP10 directly as the second complementary primary-drive waveform.
204	TP11 = VCC - TP10;	Creates the complementary primary-drive waveform by logically inverting TP10 around the 5 V supply.
205	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
206	%% 12. TP13 and TP14 - transformer secondary visualization	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
207	% Primary differential drive:	Source-code comment: Primary differential drive:
208	Vpri_diff = TP12 - TP11; % +/- 5 V in this idealized model	Calculates the differential voltage applied across the idealized transformer primary.
209	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
210	% Ideal transformer differential output:	Source-code comment: Ideal transformer differential output:
211	Vsec_diff = turns_ratio * Vpri_diff;	Applies the ideal turns ratio to the primary differential voltage to estimate the transformer secondary differential voltage.
212	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
213	% The secondary is floating in the schematic.	Source-code comment: The secondary is floating in the schematic.
214	% For plotting only, assume a virtual midpoint reference:	Source-code comment: For plotting only, assume a virtual midpoint reference:
215	TP13 = +0.5 * Vsec_diff;	Assigns one secondary terminal to half of the differential secondary voltage about a virtual midpoint used only for plotting.
216	TP14 = -0.5 * Vsec_diff;	Assigns the opposite secondary terminal to the negative half of the differential secondary voltage about the virtual plotting midpoint.
217	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
218	f_OUT = TP13 - TP14; % differential output	Calculates the differential output voltage; this is the meaningful transformer-secondary signal because the secondary is floating.
219	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
220	%% 13. Basic numerical summaries	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
221	fprintf('\n--- Signal ranges ---\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
222	fprintf('Iload : %.2f to %.2f A\n', min(Iload), max(Iload));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
223	fprintf('TP6 : %.1f to %.1f mV\n', min(TP6)*1e3, max(TP6)*1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
224	fprintf('TP7 : %.2f to %.2f V\n', min(TP7), max(TP7));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
225	fprintf('TP8 : %.2f to %.2f V\n', min(TP8), max(TP8));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
226	fprintf('f_inst : %.2f to %.2f kHz\n', min(f_inst)/1e3, max(f_inst)/1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
227	fprintf('f_OUT differential peak: %.2f V\n', max(abs(f_OUT)));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
228	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
229	%% 14. FIGURE 1 - Current-sense and analog control path	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
230	figure('Name', 'Analog Control Path', 'Color', 'w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be



Line	MATLAB source line	English explanation
		exported directly without interactive display.
231	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
232	subplot(4,1,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
233	plot(t*1e3, Iload, 'LineWidth', 1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
234	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
235	ylabel('I {LOAD} (A)');	Sets the y-axis label and therefore documents the plotted engineering unit.
236	title('Load Current and Analog Test Points');	Sets the plot title so the test point or analysis purpose is explicit.
237	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
238	subplot(4,1,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
239	plot(t*1e3, TP6*1e3, 'LineWidth', 1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
240	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
241	ylabel('TP6 (mV)');	Sets the y-axis label and therefore documents the plotted engineering unit.
242	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
243	subplot(4,1,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
244	plot(t*1e3, TP7, 'LineWidth', 1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
245	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
246	ylabel('TP7 (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
247	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
248	subplot(4,1,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
249	plot(t*1e3, TP8, 'LineWidth', 1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
250	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
251	ylabel('TP8 (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
252	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
253	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
254	%% 15. FIGURE 2 - Instantaneous frequency	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
255	figure('Name','Instantaneous Frequency','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
256	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
257	plot(t*1e3, f_inst/1e3, 'LineWidth', 1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
258	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
259	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
260	ylabel('Frequency (kHz)');	Sets the y-axis label and therefore documents the plotted engineering unit.
261	title('Modeled Instantaneous NE555 Frequency');	Sets the plot title so the test point or analysis purpose is explicit.
262	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
263	%% 16. FIGURE 3 - NE555 timing signals, zoomed view	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
264	zoom_start = 3.0e-3;	Assigns or calculates the start time of the digital zoom window from the expression on the right-hand side.
265	zoom_stop = 3.4e-3;	Assigns or calculates the stop time of the digital zoom window from the expression on the right-hand side.



Line	MATLAB source line	English explanation
266	<code>z = t >= zoom_start & t <= zoom_stop;</code>	Builds a logical time mask so only the requested zoom interval is plotted.
267	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
268	<code>figure('Name','NE555 Timing Signals','Color','w');</code>	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
269	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
270	<code>subplot(2,1,1);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
271	<code>plot(t(z)*1e3, TP9(z), 'LineWidth', 1.2);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
272	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
273	<code>ylabel('TP9 (V)');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
274	<code>title('NE555 Timing / Discharge and Output Signals');</code>	Sets the plot title so the test point or analysis purpose is explicit.
275	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
276	<code>subplot(2,1,2);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
277	<code>plot(t(z)*1e3, TP10(z), 'LineWidth', 1.2);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
278	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
279	<code>ylabel('TP10 (V)');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
280	<code>xlabel('Time (ms)');</code>	Sets the x-axis label for the current plot.
281	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
282	<code>% 17. FIGURE 4 - Transformer primary drive, zoomed view</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
283	<code>figure('Name','Transformer Primary Drive','Color','w');</code>	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
284	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
285	<code>subplot(2,1,1);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
286	<code>plot(t(z)*1e3, TP11(z), 'LineWidth', 1.2);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
287	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
288	<code>ylabel('TP11 (V)');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
289	<code>title('Transformer Primary Drive Signals');</code>	Sets the plot title so the test point or analysis purpose is explicit.
290	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
291	<code>subplot(2,1,2);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
292	<code>plot(t(z)*1e3, TP12(z), 'LineWidth', 1.2);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
293	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
294	<code>ylabel('TP12 (V)');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
295	<code>xlabel('Time (ms)');</code>	Sets the x-axis label for the current plot.
296	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
297	<code>% 18. FIGURE 5 - Transformer secondary output, zoomed view</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
298	<code>figure('Name','Transformer Secondary','Color','w');</code>	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
299	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
300	<code>subplot(3,1,1);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.



Line	MATLAB source line	English explanation
301	<code>plot(t(z)*1e3, TP13(z), 'LineWidth', 1.2);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
302	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
303	<code>ylabel('TP13 (V)');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
304	<code>title('Transformer Secondary and Differential f-OUT');</code>	Sets the plot title so the test point or analysis purpose is explicit.
305	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
306	<code>subplot(3,1,2);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
307	<code>plot(t(z)*1e3, TP14(z), 'LineWidth', 1.2);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
308	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
309	<code>ylabel('TP14 (V)');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
310	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
311	<code>subplot(3,1,3);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
312	<code>plot(t(z)*1e3, f_OUT(z), 'LineWidth', 1.2);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
313	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
314	<code>ylabel('TP13-TP14 (V)');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
315	<code>xlabel('Time (ms)');</code>	Sets the x-axis label for the current plot.
316	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
317	<code>%% 19. FIGURE 6 - All test points on one diagnostic figure</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
318	<code>% This is intentionally dense and intended as a virtual oscilloscope view.</code>	Source-code comment: This is intentionally dense and intended as a virtual oscilloscope view.
319	<code>figure('Name','All Test Points','Color','w');</code>	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
320	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
321	<code>subplot(5,2,1);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
322	<code>plot(t*1e3, TP6*1e3); grid on;</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
323	<code>title('TP6'); ylabel('mV');</code>	Sets the plot title so the test point or analysis purpose is explicit.
324	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
325	<code>subplot(5,2,2);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
326	<code>plot(t*1e3, TP7); grid on;</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
327	<code>title('TP7'); ylabel('V');</code>	Sets the plot title so the test point or analysis purpose is explicit.
328	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
329	<code>subplot(5,2,3);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
330	<code>plot(t*1e3, TP8); grid on;</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
331	<code>title('TP8'); ylabel('V');</code>	Sets the plot title so the test point or analysis purpose is explicit.
332	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
333	<code>subplot(5,2,4);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
334	<code>plot(t(z)*1e3, TP9(z)); grid on;</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
335	<code>title('TP9'); ylabel('V');</code>	Sets the plot title so the test point or analysis purpose is explicit.



Line	MATLAB source line	English explanation
336	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
337	subplot(5,2,5);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
338	plot(t(z)*1e3, TP10(z)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
339	title('TP10'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
340	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
341	subplot(5,2,6);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
342	plot(t(z)*1e3, TP11(z)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
343	title('TP11'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
344	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
345	subplot(5,2,7);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
346	plot(t(z)*1e3, TP12(z)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
347	title('TP12'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
348	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
349	subplot(5,2,8);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
350	plot(t(z)*1e3, TP13(z)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
351	title('TP13'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
352	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
353	subplot(5,2,9);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
354	plot(t(z)*1e3, TP14(z)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
355	title('TP14'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
356	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
357	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
358	subplot(5,2,10);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
359	plot(t(z)*1e3, f_OUT(z)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
360	title('TP13 - TP14'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
361	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
362	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
363	%% 20. Optional automatic PNG export	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
364	% Uncomment these lines after checking the figures in MATLAB Online:	Source-code comment: Uncomment these lines after checking the figures in MATLAB Online:
365	%	Source-code comment explaining the model assumption or the purpose of the following lines.
366	% exportgraphics(figure(1),'01_Analog_Control_Path.png','Resolution',200);	Source-code comment: exportgraphics(figure(1),'01_Analog_Control_Path.png','Resolution',200);
367	%	Source-code comment: exportgraphics(figure(2),'02_Instantaneous_Frequency.png','Resolution',200);
368	% exportgraphics(figure(3),'03_NE555_Timing.png','Resolution',200);	Source-code comment: exportgraphics(figure(3),'03_NE555_Timing.png','Resolution',200);
369	% exportgraphics(figure(4),'04_Primary_Drive.png','Resolution',200);	Source-code comment: exportgraphics(figure(4),'04_Primary_Drive.png','Resolution',200);
370	% exportgraphics(figure(5),'05_Secondary_Output.png','Resolution',200);	Source-code comment: exportgraphics(figure(5),'05_Secondary_Output.png','Resolution',200);



Line	MATLAB source line	English explanation
371	% exportgraphics(figure(6),'06_All_Test_Points.png','Resolution',200);	Source-code comment: exportgraphics(figure(6),'06_All_Test_Points.png','Resolution',200);
372	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
373	%% 21. Notes for the next model revision	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
374	% 1. Replace behavioral Q6/Kf model with explicit BC558 transistor equations.	Source-code comment: 1. Replace behavioral Q6/Kf model with explicit BC558 transistor equations.
375	% 2. Add an explicit model for D32 once its exact part number is confirmed.	Source-code comment: 2. Add an explicit model for D32 once its exact part number is confirmed.
376	% 3. Replace constant diode drops with Shockley-diode models if desired.	Source-code comment: 3. Replace constant diode drops with Shockley-diode models if desired.
377	% 4. Derive the exact frequency-control law from the 555 timing network.	Source-code comment: 4. Derive the exact frequency-control law from the 555 timing network.
378	% 5. Add transformer leakage inductance, winding resistance and load.	Source-code comment: 5. Add transformer leakage inductance, winding resistance and load.
379	% 6. Compare MATLAB results with oscilloscope measurements at TP6...TP14.	Source-code comment: 6. Compare MATLAB results with oscilloscope measurements at TP6...TP14.



Appendix B. Version 2.0 - Line-by-Line Annotated MATLAB Source

Every source line is reproduced exactly in the left column. The right column explains the purpose in plain English so a reader with little or no MATLAB experience can follow the program from top to bottom.

Line	MATLAB source line	English explanation
1	%% NE555 Current-Sense / Frequency-Modulation Behavioral Model - VERSION 2	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
2	% Purpose:	Source-code comment: Purpose:
3	% Test the existing circuit with a FAST 0 -> 10 A current step.	Source-code comment: Test the existing circuit with a FAST 0 -> 10 A current step.
4	%	Source-code comment explaining the model assumption or the purpose of the following lines.
5	% IMPORTANT:	Source-code comment: IMPORTANT:
6	% Component values, LM358 gain and transistor threshold assumptions are	Source-code comment: Component values, LM358 gain and transistor threshold assumptions
7	% intentionally kept the same as Version 1.	Source-code comment: intentionally kept the same as Version 1.
8	%	Source-code comment explaining the model assumption or the purpose of the following lines.
9	% The only major change is the load-current waveform:	Source-code comment: The only major change is the load-current waveform:
10	% 0 A -> 10 A STEP -> hold -> 0 A STEP	Source-code comment: 0 A -> 10 A STEP -> hold -> 0 A STEP
11	%	Source-code comment explaining the model assumption or the purpose of the following lines.
12	% This lets us test whether the R35/C29 transient path can raise TP8 high	Source-code comment: This lets us test whether the R35/C29 transient path can raise TP8 high
13	% enough to turn on Q6 (BC558) and therefore change the NE555 frequency.	Source-code comment: enough to turn on Q6 (BC558) and therefore change the NE555 frequency.
14	%	Source-code comment explaining the model assumption or the purpose of the following lines.
15	% If TP8 still does not exceed the approximate Q6 threshold, the next	Source-code comment: If TP8 still does not exceed the approximate Q6 threshold, the next
16	% step will be to revisit the gain / bias / transistor model rather than	Source-code comment: step will be to revisit the gain / bias / transistor model rather than
17	% artificially forcing modulation.	Source-code comment: artificially forcing modulation.
18	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
19	clear;	Removes variables from the MATLAB workspace so values from an earlier run cannot contaminate this simulation.
20	clc;	Clears the MATLAB Command Window so the new run starts with readable console output.
21	close all;	Closes previously open figure windows so old plots are not confused with the current simulation results.
22	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
23	%% 1. Global simulation settings	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
24	VCC = 5.0;	Assigns or calculates the 5 V supply used by the behavioral model from the expression on the right-hand side.
25	dt = 0.5e-6; % 0.5 us for better edge resolution	Assigns or calculates the simulation time step from the expression on the right-hand side.
26	Tsim = 6e-3; % 6 ms	Assigns or calculates the total simulated time from the expression on the right-hand side.
27	t = 0:dt:Tsim;	Assigns or calculates the simulation time vector from the expression on the right-hand side.
28	N = numel(t);	Calculates the number of time samples by counting the number of elements in the referenced vector.
29	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
30	%% 2. Circuit parameters	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
31	Rsense = 5e-3; % 5 mOhm	Assigns or calculates the current-shunt resistance from the expression on the right-hand side.
32	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
33	R38 = 1e3;	Assigns or calculates the 1 kOhm LM358 input-path resistor from the expression on the right-hand side.
34	R37 = 10e3;	Assigns or calculates the 10 kOhm LM358 input-path resistor from the expression on the



Line	MATLAB source line	English explanation
		right-hand side.
35	R36 = 330e3;	Assigns or calculates the LM358 gain-setting resistor from the expression on the right-hand side.
36	C30 = 220e-9;	Assigns or calculates the current-sense filtering capacitor from the expression on the right-hand side.
37	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
38	R35 = 330e3;	Assigns or calculates the main resistor in the transient-modulation path from the expression on the right-hand side.
39	R34 = 24e3;	Assigns or calculates the 24 kOhm modulation-path resistor from the expression on the right-hand side.
40	C29 = 2e-9;	Assigns or calculates the transient-modulation capacitor from the expression on the right-hand side.
41	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
42	R32 = 10;	Assigns or calculates the small series resistor in the NE555 charge path from the expression on the right-hand side.
43	R33 = 5e3;	Assigns or calculates the main NE555 charge resistor from the expression on the right-hand side.
44	R29 = 1e3;	Assigns or calculates the NE555 discharge resistor from the expression on the right-hand side.
45	C27 = 10e-9;	Assigns or calculates the NE555 timing capacitor from the expression on the right-hand side.
46	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
47	Vf_D31 = 0.60; % 1N914 approximation	Assigns or calculates the assumed D31 forward drop from the expression on the right-hand side.
48	Vf_D33 = 0.60; % 1N914 approximation	Assigns or calculates the assumed D33 forward drop from the expression on the right-hand side.
49	Vz_D34 = 5.60; % Zener	Assigns or calculates the assumed D34 Zener voltage from the expression on the right-hand side.
50	VCEsat = 0.10;	Assigns or calculates the assumed saturation voltage of the NE555 discharge transistor from the expression on the right-hand side.
51	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
52	Npri = 80;	Assigns or calculates the transformer primary-turn count from the expression on the right-hand side.
53	Nsec = 100;	Assigns or calculates the transformer secondary-turn count from the expression on the right-hand side.
54	turns_ratio = Nsec/Npri;	Assigns or calculates the ideal transformer turns ratio from the expression on the right-hand side.
55	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
56	%% 3. Derived values	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
57	Rin_eff = R38 + R37;	Assigns or calculates the effective LM358 input-path resistance from the expression on the right-hand side.
58	A_LM358 = R36 / Rin_eff;	Assigns or calculates the modeled LM358 closed-loop gain magnitude from the expression on the right-hand side.
59	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
60	Rth_C30 = 1/(1/R38 + 1/R37);	Calculates the Thevenin resistance seen by C30 from the stated reciprocal relationship.
61	tau_sense = Rth_C30*C30;	Assigns or calculates the current-sense first-order time constant from the expression on the right-hand side.
62	fc_sense = 1/(2*pi*tau_sense);	Calculates the corresponding current-sense cutoff frequency from the stated reciprocal relationship.
63	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
64	tau_mod = (R35 + R34)*C29;	Assigns or calculates the transient-path time constant from the expression on the right-hand



Line	MATLAB source line	English explanation
		side.
65	<code>fc_mod = 1/(2*pi*tau_mod);</code>	Calculates the corresponding transient-path cutoff frequency from the stated reciprocal relationship.
66	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
67	<code>RA = R32 + R33;</code>	Assigns or calculates the NE555 effective charge resistance from the expression on the right-hand side.
68	<code>RB = R29;</code>	Assigns or calculates the NE555 discharge resistance from the expression on the right-hand side.
69	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
70	<code>VTL = VCC/3;</code>	Assigns or calculates the lower NE555 timing threshold (one-third VCC) from the expression on the right-hand side.
71	<code>VTH = 2*VCC/3;</code>	Assigns or calculates the upper NE555 timing threshold (two-thirds VCC) from the expression on the right-hand side.
72	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
73	<code>Vcharge_inf = VCC - Vf_D31;</code>	Assigns or calculates the asymptotic timing-capacitor charge voltage after the diode drop from the expression on the right-hand side.
74	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
75	<code>t_charge = RA*C27 * log((Vcharge_inf - VTL)/(Vcharge_inf - VTH));</code>	Calculates the calculated NE555 charge interval using the analytical exponential charge/discharge timing equation.
76	<code>t_discharge = RB*C27 * log((VTH - VCEsat)/(VTL - VCEsat));</code>	Calculates the calculated NE555 discharge interval using the analytical exponential charge/discharge timing equation.
77	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
78	<code>T555_nom = t_charge + t_discharge;</code>	Assigns or calculates the nominal NE555 oscillation period from the expression on the right-hand side.
79	<code>f555_nom = 1/T555_nom;</code>	Assigns or calculates the nominal NE555 frequency from the expression on the right-hand side.
80	<code>duty_nom = t_charge/T555_nom;</code>	Assigns or calculates the nominal high-state duty fraction from the expression on the right-hand side.
81	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
82	<code>fprintf('VERSION 2 - FAST CURRENT STEP TEST\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
83	<code>fprintf('LM358 gain : %.2f V/V\n', A_LM358);</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
84	<code>fprintf('Current-sense tau : %.1f us\n', tau_sense*1e6);</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
85	<code>fprintf('Current-sense fc : %.1f Hz\n', fc_sense);</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
86	<code>fprintf('Modulation-path tau : %.1f us\n', tau_mod*1e6);</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
87	<code>fprintf('Modulation-path fc : %.1f Hz\n', fc_mod);</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
88	<code>fprintf('Nominal NE555 frequency : %.2f kHz\n', f555_nom/1e3);</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
89	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
90	<code>%% 4. FAST load-current profile</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
91	<code>% 0 A from 0 to 1 ms</code>	Source-code comment: 0 A from 0 to 1 ms
92	<code>% 10 A step at 1 ms</code>	Source-code comment: 10 A step at 1 ms
93	<code>% hold 10 A until 4 ms</code>	Source-code comment: hold 10 A until 4 ms
94	<code>% 0 A step at 4 ms</code>	Source-code comment: 0 A step at 4 ms



Line	MATLAB source line	English explanation
95	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
96	Iload = zeros(size(t));	Initializes the simulated load-current waveform with zeros, giving it the required array size before later samples are calculated.
97	Iload(t >= 1e-3 & t < 4e-3) = 10;	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
98	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
99	%% 5. TP6 - R-sense voltage	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
100	% Polarity selected for the inverting LM358 model.	Source-code comment: Polarity selected for the inverting LM358 model.
101	TP6 = -Iload * Rsense;	Converts load current into the shunt voltage at TP6. The negative sign represents the polarity chosen for the inverting LM358 model.
102	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
103	%% 6. TP7 - LM358 output	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
104	TP7_target = -A_LM358 * TP6;	Calculates the ideal amplified current-sense voltage before the first-order output/filter response is applied.
105	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
106	TP7 = zeros(size(t));	Initializes the filtered behavioral LM358 output with zeros, giving it the required array size before later samples are calculated.
107	alpha_sense = dt/(tau_sense + dt);	Assigns or calculates the discrete first-order filter coefficient for TP7 from the expression on the right-hand side.
108	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
109	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
110	TP7(k) = TP7(k-1) + alpha_sense*(TP7_target(k)-TP7(k-1));	Updates TP7 with a first-order discrete low-pass equation, moving the previous output toward the instantaneous target.
111	end	Ends the current for-loop or if/else block.
112	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
113	% Approximate LM358 output range on 5-V supply	Source-code comment: Approximate LM358 output range on 5-V supply
114	TP7 = min(max(TP7,0),3.5);	Clamps the behavioral LM358 output to the assumed usable output range on a 5 V supply.
115	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
116	%% 7. TP8 - AC-coupled modulation node	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
117	% First-pass model:	Source-code comment: First-pass model:
118	% R35/C29 acts as a high-pass / transient path.	Source-code comment: R35/C29 acts as a high-pass / transient path.
119	% TP8 has an assumed DC bias around 4.35 V.	Source-code comment: TP8 has an assumed DC bias around 4.35 V.
120	%	Source-code comment explaining the model assumption or the purpose of the following lines.
121	% NOTE:	Source-code comment: NOTE:
122	% This bias is still a behavioral approximation and will be refined later.	Source-code comment: This bias is still a behavioral approximation and will be refined later.
123	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
124	Vbias_TP8 = 4.35;	Assigns or calculates the assumed TP8 bias/clamp level from the expression on the right-hand side.
125	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
126	TP7_lp_mod = zeros(size(t));	Initializes the low-pass state used to form the high-pass transient with zeros, giving it the required array size before later samples are calculated.
127	alpha_mod = dt/(tau_mod + dt);	Assigns or calculates the discrete transient-path filter coefficient from the expression on the right-hand side.
128	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
		effect.
129	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
130	TP7_lp_mod(k) = TP7_lp_mod(k-1) + alpha_mod*(TP7(k)-TP7_lp_mod(k-1));	Updates the low-pass state used to subtract the slow component of TP7 and isolate the transient/high-pass component.
131	end	Ends the current for-loop or if/else block.
132	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
133	HP_mod = TP7 - TP7_lp_mod;	Subtracts the slow low-pass component from TP7 to obtain the transient/high-pass component that drives TP8.
134	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
135	Kmod_node = 1.0; % do NOT artificially increase in Version 2	Assigns or calculates the behavioral TP8 shaping gain from the expression on the right-hand side.
136	TP8 = Vbias_TP8 + Kmod_node*HP_mod;	Adds the transient component to the assumed TP8 DC bias, forming the behavioral control-node voltage.
137	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
138	TP8 = min(max(TP8,0),6.5);	Clamps the behavioral TP8 voltage to a bounded plotting range so extreme numerical values cannot dominate the visualization.
139	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
140	%% 8. Q6 threshold and drive	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
141	% BC558 is PNP.	Source-code comment: BC558 is PNP.
142	% With base approximately at +5 V, emitter must rise above base by ~0.65 V.	Source-code comment: With base approximately at +5 V, emitter must rise above base by ~0.65 V.
143	VBE_PNP = 0.65;	Assigns or calculates the assumed BC558 base-emitter magnitude from the expression on the right-hand side.
144	VQ6_on = VCC + VBE_PNP; % ~5.65 V	Assigns or calculates the modeled Q6 turn-on threshold from the expression on the right-hand side.
145	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
146	Q6_drive = max(TP8 - VQ6_on,0);	Implements the behavioral Q6 threshold: values below threshold produce zero drive; only voltage above threshold contributes to modulation.
147	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
148	%% 9. Frequency modulation model	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
149	% Keep the same behavioral sensitivity as Version 1.	Source-code comment: Keep the same behavioral sensitivity as Version 1.
150	% If Q6_drive remains zero, frequency MUST remain nominal.	Source-code comment: If Q6_drive remains zero, frequency MUST remain nominal.
151	Kf = 5e3; % Hz/V	Assigns or calculates the behavioral frequency sensitivity coefficient from the expression on the right-hand side.
152	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
153	f_inst = f555_nom + Kf*Q6_drive;	Creates the behavioral frequency law by adding Kf times Q6 threshold overdrive to the nominal NE555 frequency.
154	f_inst = min(max(f_inst,5e3),40e3);	Clamps the behavioral frequency to a practical plotting range; this is a numerical safeguard rather than a measured circuit limit.
155	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
156	%% 10. Variable-frequency phase generator	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
157	phase = zeros(size(t));	Initializes the accumulated oscillator phase with zeros, giving it the required array size before later samples are calculated.
158	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
159	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.



Line	MATLAB source line	English explanation
160	phase(k) = phase(k-1) + 2*pi*f_inst(k)*dt;	Numerically integrates instantaneous frequency into oscillator phase for this time sample.
161	end	Ends the current for-loop or if/else block.
162	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
163	phase01 = mod(phase,2*pi)/(2*pi);	Wraps accumulated phase into a normalized 0-to-1 cycle position used to synthesize the square-wave output.
164	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
165	%% 11. TP10 - NE555 output	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
166	TP10 = VCC*(phase01 < duty_nom);	Converts normalized oscillator phase and duty cycle into a 0 V / VCC square-wave approximation of the NE555 output.
167	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
168	%% 12. TP9 - NE555 discharge node	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
169	TP9 = VCC*ones(size(t));	Initializes the modeled NE555 discharge-node voltage with ones of the required array size; later lines modify selected samples.
170	TP9(phase01 >= duty_nom) = VCEsat;	For time samples in the discharge phase, forces TP9 down to the assumed discharge-transistor saturation voltage.
171	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
172	%% 13. TP11 / TP12 - transformer primary drive	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
173	TP12 = TP10;	Uses TP10 directly as the second complementary primary-drive waveform.
174	TP11 = VCC - TP10;	Creates the complementary primary-drive waveform by logically inverting TP10 around the 5 V supply.
175	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
176	%% 14. TP13 / TP14 - transformer secondary	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
177	Vpri_diff = TP12 - TP11;	Calculates the differential voltage applied across the idealized transformer primary.
178	Vsec_diff = turns_ratio*Vpri_diff;	Applies the ideal turns ratio to the primary differential voltage to estimate the transformer secondary differential voltage.
179	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
180	% Floating secondary represented with a virtual midpoint for plotting only	Source-code comment: Floating secondary represented with a virtual midpoint for plotting only
181	TP13 = +0.5*Vsec_diff;	Assigns one secondary terminal to half of the differential secondary voltage about a virtual midpoint used only for plotting.
182	TP14 = -0.5*Vsec_diff;	Assigns the opposite secondary terminal to the negative half of the differential secondary voltage about the virtual plotting midpoint.
183	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
184	f_OUT = TP13 - TP14;	Calculates the differential output voltage; this is the meaningful transformer-secondary signal because the secondary is floating.
185	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
186	%% 15. Numerical results	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
187	fprintf('--- VERSION 2 RESULTS ---\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
188	fprintf('TP6 range : %.1f to %.1f mV\n', min(TP6)*1e3, max(TP6)*1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
189	fprintf('TP7 range : %.3f to %.3f V\n', min(TP7), max(TP7));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
190	fprintf('TP8 range : %.3f to %.3f V\n', min(TP8),	Prints formatted model parameters, results, or file-status information to the MATLAB



Line	MATLAB source line	English explanation
	max(TP8));	Command Window.
191	fprintf('Approx. Q6 threshold : %.3f V\n', VQ6_on);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
192	fprintf('Maximum Q6 drive : %.3f V\n', max(Q6_drive));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
193	fprintf('Frequency range : %.3f to %.3f kHz\n', min(f_inst)/1e3, max(f_inst)/1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
194	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
195	if max(TP8) > VQ6_on	Starts a conditional block; the following code executes only when this logical condition is true.
196	fprintf('\nRESULT: TP8 crosses the Q6 threshold. Frequency modulation is expected.\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
197	else	Selects the alternative branch of the current conditional block.
198	fprintf('\nRESULT: TP8 does NOT cross the Q6 threshold.\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
199	fprintf('The fast 0 -> 10 A step is still insufficient in this behavioral model.\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
200	fprintf('Next step: revisit LM358 gain, TP8 bias, D32/Q6 model, or coupling network.\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
201	end	Ends the current for-loop or if/else block.
202	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
203	%% 16. Figure 1 - Current and analog path	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
204	figure('Name','V2 Analog Control Path','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
205	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
206	subplot(4,1,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
207	plot(t*1e3,Iload,'LineWidth',1.3);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
208	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
209	ylabel('I_{LOAD} (A)');	Sets the y-axis label and therefore documents the plotted engineering unit.
210	title('Version 2 - Fast 0 to 10 A Current Step');	Sets the plot title so the test point or analysis purpose is explicit.
211	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
212	subplot(4,1,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
213	plot(t*1e3,TP6*1e3,'LineWidth',1.3);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
214	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
215	ylabel('TP6 (mV)');	Sets the y-axis label and therefore documents the plotted engineering unit.
216	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
217	subplot(4,1,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
218	plot(t*1e3,TP7,'LineWidth',1.3);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
219	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
220	ylabel('TP7 (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
221	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
222	subplot(4,1,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
223	plot(t*1e3,TP8,'LineWidth',1.3);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.



Line	MATLAB source line	English explanation
224	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
225	ylabel(VQ6_on,'--','Q6 threshold');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
226	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
227	ylabel('TP8 (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
228	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
229	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
230	%% 17. Figure 2 - TP8 and Q6 threshold	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
231	figure('Name','V2 Q6 Turn-On Test','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
232	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
233	plot(t*1e3,TP8,'LineWidth',1.5);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
234	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
235	ylabel(VQ6_on,'--','Q6 turn-on threshold');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
236	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
237	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
238	ylabel('Voltage (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
239	title('TP8 Versus Approximate BC558 Turn-On Threshold');	Sets the plot title so the test point or analysis purpose is explicit.
240	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
241	%% 18. Figure 3 - Q6 drive	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
242	figure('Name','V2 Q6 Drive','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
243	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
244	plot(t*1e3,Q6_drive,'LineWidth',1.5);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
245	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
246	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
247	ylabel('Q6 drive above threshold (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
248	title('Behavioral Q6 Drive');	Sets the plot title so the test point or analysis purpose is explicit.
249	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
250	%% 19. Figure 4 - Instantaneous frequency	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
251	figure('Name','V2 Instantaneous Frequency','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
252	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
253	plot(t*1e3,f_inst/1e3,'LineWidth',1.5);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
254	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
255	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
256	ylabel('Frequency (kHz)');	Sets the y-axis label and therefore documents the plotted engineering unit.
257	title('Version 2 - Modeled Instantaneous NE555 Frequency');	Sets the plot title so the test point or analysis purpose is explicit.
258	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
259	%% 20. Zoom around rising current edge	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
260	zoom_start = 0.90e-3;	Assigns or calculates the start time of the digital zoom window from the expression on the right-hand side.



Line	MATLAB source line	English explanation
261	zoom_stop = 1.60e-3;	Assigns or calculates the stop time of the digital zoom window from the expression on the right-hand side.
262	z = t >= zoom_start & t <= zoom_stop;	Builds a logical time mask so only the requested zoom interval is plotted.
263	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
264	figure('Name','V2 Rising Edge Detail','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
265	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
266	subplot(4,1,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
267	plot(t(z)*1e3,Iload(z),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
268	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
269	ylabel('A');	Sets the y-axis label and therefore documents the plotted engineering unit.
270	title('Rising-Edge Detail');	Sets the plot title so the test point or analysis purpose is explicit.
271	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
272	subplot(4,1,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
273	plot(t(z)*1e3,TP7(z),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
274	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
275	ylabel('TP7 (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
276	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
277	subplot(4,1,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
278	plot(t(z)*1e3,TP8(z),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
279	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
280	ylines(VQ6_on,'--');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
281	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
282	ylabel('TP8 (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
283	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
284	subplot(4,1,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
285	plot(t(z)*1e3,f_inst(z)/1e3,'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
286	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
287	ylabel('kHz');	Sets the y-axis label and therefore documents the plotted engineering unit.
288	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
289	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
290	%% 21. Zoom around falling current edge	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
291	zoom_start2 = 3.90e-3;	Assigns or calculates the MATLAB variable zoom_start2 from the expression on the right-hand side.
292	zoom_stop2 = 4.60e-3;	Assigns or calculates the MATLAB variable zoom_stop2 from the expression on the right-hand side.
293	z2 = t >= zoom_start2 & t <= zoom_stop2;	Builds a logical time mask so only the requested zoom interval is plotted.
294	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
295	figure('Name','V2 Falling Edge Detail','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.



Line	MATLAB source line	English explanation
296	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
297	subplot(4,1,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
298	plot(t(z2)*1e3,Iload(z2),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
299	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
300	ylabel('A');	Sets the y-axis label and therefore documents the plotted engineering unit.
301	title('Falling-Edge Detail');	Sets the plot title so the test point or analysis purpose is explicit.
302	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
303	subplot(4,1,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
304	plot(t(z2)*1e3,TP7(z2),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
305	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
306	ylabel('TP7 (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
307	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
308	subplot(4,1,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
309	plot(t(z2)*1e3,TP8(z2),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
310	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
311	ylene(VQ6_on,'--');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
312	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
313	ylabel('TP8 (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
314	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
315	subplot(4,1,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
316	plot(t(z2)*1e3,f_inst(z2)/1e3,'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
317	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
318	ylabel('kHz');	Sets the y-axis label and therefore documents the plotted engineering unit.
319	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
320	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
321	%% 22. Test-point oscilloscope view	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
322	% Use a short interval after the rising edge for fast digital signals.	Source-code comment: Use a short interval after the rising edge for fast digital signals.
323	osc_start = 1.2e-3;	Assigns or calculates the MATLAB variable osc_start from the expression on the right-hand side.
324	osc_stop = 1.6e-3;	Assigns or calculates the MATLAB variable osc_stop from the expression on the right-hand side.
325	zo = t >= osc_start & t <= osc_stop;	Builds a logical time mask so only the requested zoom interval is plotted.
326	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
327	figure('Name','V2 All Test Points','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
328	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
329	subplot(5,2,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
330	plot(t*1e3,TP6*1e3); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.



Line	MATLAB source line	English explanation
331	title('TP6'); ylabel('mV');	Sets the plot title so the test point or analysis purpose is explicit.
332	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
333	subplot(5,2,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
334	plot(t*1e3,TP7); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
335	title('TP7'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
336	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
337	subplot(5,2,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
338	plot(t*1e3,TP8); hold on; yline(VQ6_on,'--'); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
339	title('TP8'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
340	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
341	subplot(5,2,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
342	plot(t(zo)*1e3,TP9(zo)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
343	title('TP9'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
344	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
345	subplot(5,2,5);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
346	plot(t(zo)*1e3,TP10(zo)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
347	title('TP10'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
348	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
349	subplot(5,2,6);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
350	plot(t(zo)*1e3,TP11(zo)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
351	title('TP11'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
352	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
353	subplot(5,2,7);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
354	plot(t(zo)*1e3,TP12(zo)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
355	title('TP12'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
356	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
357	subplot(5,2,8);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
358	plot(t(zo)*1e3,TP13(zo)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
359	title('TP13'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
360	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
361	subplot(5,2,9);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
362	plot(t(zo)*1e3,TP14(zo)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
363	title('TP14'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.



Line	MATLAB source line	English explanation
364	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
365	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
366	subplot(5,2,10);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
367	plot(t(z0)*1e3,f_OUT(z0)); grid on;	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
368	title('TP13 - TP14'); ylabel('V');	Sets the plot title so the test point or analysis purpose is explicit.
369	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
370	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
371	%% 23. Optional export	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
372	% Uncomment after checking the figures:	Source-code comment: Uncomment after checking the figures:
373	%	Source-code comment explaining the model assumption or the purpose of the following lines.
374	% exportgraphics(figure(1),'V2_01_Analog_Path.png','Resolution',200);	Source-code comment: exportgraphics(figure(1),'V2_01_Analog_Path.png','Resolution',200);
375	% exportgraphics(figure(2),'V2_02_TP8_Threshold.png','Resolution',200);	Source-code comment: exportgraphics(figure(2),'V2_02_TP8_Threshold.png','Resolution',200);
376	% exportgraphics(figure(3),'V2_03_Q6_Drive.png','Resolution',200);	Source-code comment: exportgraphics(figure(3),'V2_03_Q6_Drive.png','Resolution',200);
377	% exportgraphics(figure(4),'V2_04_Frequency.png','Resolution',200);	Source-code comment: exportgraphics(figure(4),'V2_04_Frequency.png','Resolution',200);
378	% exportgraphics(figure(5),'V2_05_Rising_Edge.png','Resolution',200);	Source-code comment: exportgraphics(figure(5),'V2_05_Rising_Edge.png','Resolution',200);
379	% exportgraphics(figure(6),'V2_06_Falling_Edge.png','Resolution',200);	Source-code comment: exportgraphics(figure(6),'V2_06_Falling_Edge.png','Resolution',200);
380	% exportgraphics(figure(7),'V2_07_All_Test_Points.png','Resolution',200);	Source-code comment: exportgraphics(figure(7),'V2_07_All_Test_Points.png','Resolution',200);



Appendix C. Version 2.1 - Line-by-Line Annotated MATLAB Source

Every source line is reproduced exactly in the left column. The right column explains the purpose in plain English so a reader with little or no MATLAB experience can follow the program from top to bottom.

Line	MATLAB source line	English explanation
1	%% NE555 Current-Sense Model - VERSION 2.1	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
2	% Robust plotting version for MATLAB Online	Source-code comment: Robust plotting version for MATLAB Online
3	% Same electrical assumptions as Version 2.	Source-code comment: Same electrical assumptions as Version 2.
4	% Main goals:	Source-code comment: Main goals:
5	% 1) Force figures to render correctly in MATLAB Online.	Source-code comment: 1) Force figures to render correctly in MATLAB Online.
6	% 2) Keep the fast 0 -> 10 A current-step test.	Source-code comment: 2) Keep the fast 0 -> 10 A current-step test.
7	% 3) Show TP6...TP14 and the Q6 threshold clearly.	Source-code comment: 3) Show TP6...TP14 and the Q6 threshold clearly.
8	%	Source-code comment explaining the model assumption or the purpose of the following lines.
9	% NOTE:	Source-code comment: NOTE:
10	% Version 2 result already showed:	Source-code comment: Version 2 result already showed:
11	% TP8 max ~= 5.261 V	Source-code comment: TP8 max ~= 5.261 V
12	% Q6 threshold ~= 5.650 V	Source-code comment: Q6 threshold ~= 5.650 V
13	% Therefore Q6 does NOT turn on in this behavioral model and the	Source-code comment: Therefore Q6 does NOT turn on in this behavioral model and the
14	% instantaneous frequency remains at the nominal ~18.39 kHz.	Source-code comment: instantaneous frequency remains at the nominal ~18.39 kHz.
15	%	Source-code comment explaining the model assumption or the purpose of the following lines.
16	% This file DOES NOT artificially change gain, bias, or frequency.	Source-code comment: This file DOES NOT artificially change gain, bias, or frequency.
17	% It only makes the plotting more robust and explicit.	Source-code comment: It only makes the plotting more robust and explicit.
18	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
19	clear;	Removes variables from the MATLAB workspace so values from an earlier run cannot contaminate this simulation.
20	clc;	Clears the MATLAB Command Window so the new run starts with readable console output.
21	close all force;	Closes previously open figure windows so old plots are not confused with the current simulation results.
22	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
23	set(groot,'defaultFigureVisible','on');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
24	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
25	%% 1. Simulation settings	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
26	VCC = 5.0;	Assigns or calculates the 5 V supply used by the behavioral model from the expression on the right-hand side.
27	dt = 0.5e-6;	Assigns or calculates the simulation time step from the expression on the right-hand side.
28	Tsim = 6e-3;	Assigns or calculates the total simulated time from the expression on the right-hand side.
29	t = 0:dt:Tsim;	Assigns or calculates the simulation time vector from the expression on the right-hand side.
30	N = numel(t);	Calculates the number of time samples by counting the number of elements in the referenced vector.
31	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no



Line	MATLAB source line	English explanation
		executable effect.
32	%% 2. Circuit parameters	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
33	$R_{sense} = 5e-3;$	Assigns or calculates the current-shunt resistance from the expression on the right-hand side.
34	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
35	$R_{38} = 1e3;$	Assigns or calculates the 1 kOhm LM358 input-path resistor from the expression on the right-hand side.
36	$R_{37} = 10e3;$	Assigns or calculates the 10 kOhm LM358 input-path resistor from the expression on the right-hand side.
37	$R_{36} = 330e3;$	Assigns or calculates the LM358 gain-setting resistor from the expression on the right-hand side.
38	$C_{30} = 220e-9;$	Assigns or calculates the current-sense filtering capacitor from the expression on the right-hand side.
39	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
40	$R_{35} = 330e3;$	Assigns or calculates the main resistor in the transient-modulation path from the expression on the right-hand side.
41	$R_{34} = 24e3;$	Assigns or calculates the 24 kOhm modulation-path resistor from the expression on the right-hand side.
42	$C_{29} = 2e-9;$	Assigns or calculates the transient-modulation capacitor from the expression on the right-hand side.
43	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
44	$R_{32} = 10;$	Assigns or calculates the small series resistor in the NE555 charge path from the expression on the right-hand side.
45	$R_{33} = 5e3;$	Assigns or calculates the main NE555 charge resistor from the expression on the right-hand side.
46	$R_{29} = 1e3;$	Assigns or calculates the NE555 discharge resistor from the expression on the right-hand side.
47	$C_{27} = 10e-9;$	Assigns or calculates the NE555 timing capacitor from the expression on the right-hand side.
48	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
49	$V_{f_D31} = 0.60;$	Assigns or calculates the assumed D31 forward drop from the expression on the right-hand side.
50	$V_{CEsat} = 0.10;$	Assigns or calculates the assumed saturation voltage of the NE555 discharge transistor from the expression on the right-hand side.
51	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
52	$N_{pri} = 80;$	Assigns or calculates the transformer primary-turn count from the expression on the right-hand side.
53	$N_{sec} = 100;$	Assigns or calculates the transformer secondary-turn count from the expression on the right-hand side.
54	$turns_ratio = N_{sec}/N_{pri};$	Assigns or calculates the ideal transformer turns ratio from the expression on the right-hand side.
55	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
56	%% 3. Derived values	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
57	$A_{LM358} = R_{36}/(R_{38}+R_{37});$	Assigns or calculates the modeled LM358 closed-loop gain magnitude from the expression on the right-hand side.
58	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
59	$R_{th_C30} = 1/(1/R_{38} + 1/R_{37});$	Calculates the Thevenin resistance seen by C30 from the stated reciprocal relationship.
60	$\tau_{sense} = R_{th_C30}*C_{30};$	Assigns or calculates the current-sense first-order time constant from the



Line	MATLAB source line	English explanation
		expression on the right-hand side.
61	<code>fc_sense = 1/(2*pi*tau_sense);</code>	Calculates the corresponding current-sense cutoff frequency from the stated reciprocal relationship.
62	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
63	<code>tau_mod = (R35+R34)*C29;</code>	Assigns or calculates the transient-path time constant from the expression on the right-hand side.
64	<code>fc_mod = 1/(2*pi*tau_mod);</code>	Calculates the corresponding transient-path cutoff frequency from the stated reciprocal relationship.
65	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
66	<code>RA = R32 + R33;</code>	Assigns or calculates the NE555 effective charge resistance from the expression on the right-hand side.
67	<code>RB = R29;</code>	Assigns or calculates the NE555 discharge resistance from the expression on the right-hand side.
68	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
69	<code>VTL = VCC/3;</code>	Assigns or calculates the lower NE555 timing threshold (one-third VCC) from the expression on the right-hand side.
70	<code>VTH = 2*VCC/3;</code>	Assigns or calculates the upper NE555 timing threshold (two-thirds VCC) from the expression on the right-hand side.
71	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
72	<code>Vcharge_inf = VCC - Vf_D31;</code>	Assigns or calculates the asymptotic timing-capacitor charge voltage after the diode drop from the expression on the right-hand side.
73	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
74	<code>t_charge = RA*C27 * log((Vcharge_inf-VTL)/(Vcharge_inf-VTH));</code>	Calculates the calculated NE555 charge interval using the analytical exponential charge/discharge timing equation.
75	<code>t_discharge = RB*C27 * log((VTH-VCESat)/(VTL-VCESat));</code>	Calculates the calculated NE555 discharge interval using the analytical exponential charge/discharge timing equation.
76	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
77	<code>T555_nom = t_charge + t_discharge;</code>	Assigns or calculates the nominal NE555 oscillation period from the expression on the right-hand side.
78	<code>f555_nom = 1/T555_nom;</code>	Assigns or calculates the nominal NE555 frequency from the expression on the right-hand side.
79	<code>duty_nom = t_charge/T555_nom;</code>	Assigns or calculates the nominal high-state duty fraction from the expression on the right-hand side.
80	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
81	<code>% 4. Fast current-step profile</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
82	<code>Iload = zeros(size(t));</code>	Initializes the simulated load-current waveform with zeros, giving it the required array size before later samples are calculated.
83	<code>Iload(t >= 1e-3 & t < 4e-3) = 10;</code>	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
84	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
85	<code>% 5. TP6</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
86	<code>TP6 = -Iload*Rsense;</code>	Converts load current into the shunt voltage at TP6. The negative sign represents the polarity chosen for the inverting LM358 model.
87	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
88	<code>% 6. TP7</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
89	<code>TP7_target = -A_LM358*TP6;</code>	Calculates the ideal amplified current-sense voltage before the first-order



Line	MATLAB source line	English explanation
		output/filter response is applied.
90	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
91	TP7 = zeros(size(t));	Initializes the filtered behavioral LM358 output with zeros, giving it the required array size before later samples are calculated.
92	alpha_sense = dt/(tau_sense+dt);	Assigns or calculates the discrete first-order filter coefficient for TP7 from the expression on the right-hand side.
93	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
94	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
95	TP7(k) = TP7(k-1) + alpha_sense*(TP7_target(k)-TP7(k-1));	Updates TP7 with a first-order discrete low-pass equation, moving the previous output toward the instantaneous target.
96	end	Ends the current for-loop or if/else block.
97	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
98	TP7 = min(max(TP7,0),3.5);	Clamps the behavioral LM358 output to the assumed usable output range on a 5 V supply.
99	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
100	%% 7. TP8 behavioral transient node	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
101	Vbias_TP8 = 4.35;	Assigns or calculates the assumed TP8 bias/clamp level from the expression on the right-hand side.
102	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
103	TP7_lp_mod = zeros(size(t));	Initializes the low-pass state used to form the high-pass transient with zeros, giving it the required array size before later samples are calculated.
104	alpha_mod = dt/(tau_mod+dt);	Assigns or calculates the discrete transient-path filter coefficient from the expression on the right-hand side.
105	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
106	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
107	TP7_lp_mod(k) = TP7_lp_mod(k-1) + alpha_mod*(TP7(k)-TP7_lp_mod(k-1));	Updates the low-pass state used to subtract the slow component of TP7 and isolate the transient/high-pass component.
108	end	Ends the current for-loop or if/else block.
109	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
110	HP_mod = TP7 - TP7_lp_mod;	Subtracts the slow low-pass component from TP7 to obtain the transient/high-pass component that drives TP8.
111	TP8 = Vbias_TP8 + HP_mod;	Adds the transient component to the assumed TP8 DC bias, forming the behavioral control-node voltage.
112	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
113	TP8 = min(max(TP8,0),6.5);	Clamps the behavioral TP8 voltage to a bounded plotting range so extreme numerical values cannot dominate the visualization.
114	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
115	%% 8. Q6 threshold	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
116	VBE_PNP = 0.65;	Assigns or calculates the assumed BC558 base-emitter magnitude from the expression on the right-hand side.
117	VQ6_on = VCC + VBE_PNP;	Assigns or calculates the modeled Q6 turn-on threshold from the expression on the right-hand side.
118	Q6_drive = max(TP8 - VQ6_on,0);	Implements the behavioral Q6 threshold: values below threshold produce zero drive; only voltage above threshold contributes to modulation.
119	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
120	%% 9. Frequency	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
121	Kf = 5e3;	Assigns or calculates the behavioral frequency sensitivity coefficient from the expression on the right-hand side.
122	f_inst = f555_nom + Kf*Q6_drive;	Creates the behavioral frequency law by adding Kf times Q6 threshold overdrive to the nominal NE555 frequency.
123	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
124	%% 10. NE555 waveforms	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
125	phase = zeros(size(t));	Initializes the accumulated oscillator phase with zeros, giving it the required array size before later samples are calculated.
126	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
127	phase(k) = phase(k-1) + 2*pi*f_inst(k)*dt;	Numerically integrates instantaneous frequency into oscillator phase for this time sample.
128	end	Ends the current for-loop or if/else block.
129	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
130	phase01 = mod(phase,2*pi)/(2*pi);	Wraps accumulated phase into a normalized 0-to-1 cycle position used to synthesize the square-wave output.
131	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
132	TP10 = VCC*(phase01 < duty_nom);	Converts normalized oscillator phase and duty cycle into a 0 V / VCC square-wave approximation of the NE555 output.
133	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
134	TP9 = VCC*ones(size(t));	Initializes the modeled NE555 discharge-node voltage with ones of the required array size; later lines modify selected samples.
135	TP9(phase01 >= duty_nom) = VCEsat;	For time samples in the discharge phase, forces TP9 down to the assumed discharge-transistor saturation voltage.
136	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
137	%% 11. Primary and secondary	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
138	TP12 = TP10;	Uses TP10 directly as the second complementary primary-drive waveform.
139	TP11 = VCC - TP10;	Creates the complementary primary-drive waveform by logically inverting TP10 around the 5 V supply.
140	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
141	Vpri_diff = TP12 - TP11;	Calculates the differential voltage applied across the idealized transformer primary.
142	Vsec_diff = turns_ratio*Vpri_diff;	Applies the ideal turns ratio to the primary differential voltage to estimate the transformer secondary differential voltage.
143	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
144	TP13 = +0.5*Vsec_diff;	Assigns one secondary terminal to half of the differential secondary voltage about a virtual midpoint used only for plotting.
145	TP14 = -0.5*Vsec_diff;	Assigns the opposite secondary terminal to the negative half of the differential secondary voltage about the virtual plotting midpoint.
146	f_OUT = TP13 - TP14;	Calculates the differential output voltage; this is the meaningful transformer-secondary signal because the secondary is floating.
147	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
148	%% 12. Console summary	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
149	fprintf('\nVERSION 2.1 - ROBUST PLOTTING\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
150	fprintf('LM358 gain : %.2f V/V\n',A_LM358);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.



Line	MATLAB source line	English explanation
151	fprintf('Current-sense tau : %.1f us\n',tau_sense*1e6);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
152	fprintf('Current-sense fc : %.1f Hz\n',fc_sense);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
153	fprintf('Modulation-path tau : %.1f us\n',tau_mod*1e6);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
154	fprintf('Modulation-path fc : %.1f Hz\n',fc_mod);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
155	fprintf('Nominal NE555 frequency : %.3f kHz\n',f555_nom/1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
156	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
157	fprintf('\n--- RESULTS ---\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
158	fprintf('TP6 range : %.1f to %.1f mV\n',min(TP6)*1e3,max(TP6)*1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
159	fprintf('TP7 range : %.3f to %.3f V\n',min(TP7),max(TP7));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
160	fprintf('TP8 range : %.3f to %.3f V\n',min(TP8),max(TP8));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
161	fprintf('Q6 threshold : %.3f V\n',VQ6_on);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
162	fprintf('Maximum Q6 drive : %.3f V\n',max(Q6_drive));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
163	fprintf('Frequency range : %.3f to %.3f kHz\n',min(f_inst)/1e3,max(f_inst)/1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
164	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
165	% 13. FIGURE 1 - Slow analog/control signals	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
166	f1 = figure('Name','V2.1 Analog and Control Signals','NumberTitle','off',...	Assigns or calculates the MATLAB variable f1 from the expression on the right-hand side.
167	'Color','w','Visible','on');	Continuation line containing text, a table variable name, or another argument for the MATLAB statement started above.
168	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
169	ax1 = subplot(5,1,1);	Assigns or calculates the MATLAB variable ax1 from the expression on the right-hand side.
170	plot(t*1e3,Iload,'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
171	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
172	ylabel('A');	Sets the y-axis label and therefore documents the plotted engineering unit.
173	title('Load Current');	Sets the plot title so the test point or analysis purpose is explicit.
174	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
175	ax2 = subplot(5,1,2);	Assigns or calculates the MATLAB variable ax2 from the expression on the right-hand side.
176	plot(t*1e3,TP6*1e3,'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
177	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
178	ylabel('mV');	Sets the y-axis label and therefore documents the plotted engineering unit.
179	title('TP6 - Current Sense');	Sets the plot title so the test point or analysis purpose is explicit.
180	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
181	ax3 = subplot(5,1,3);	Assigns or calculates the MATLAB variable ax3 from the expression on the right-



Line	MATLAB source line	English explanation
		hand side.
182	plot(t*1e3,TP7,'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
183	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
184	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
185	title('TP7 - LM358 Output');	Sets the plot title so the test point or analysis purpose is explicit.
186	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
187	ax4 = subplot(5,1,4);	Assigns or calculates the MATLAB variable ax4 from the expression on the right-hand side.
188	plot(t*1e3,TP8,'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
189	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
190	plot(t*1e3,VQ6_on*ones(size(t)),'--','LineWidth',1.0);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
191	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
192	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
193	title('TP8 - Modulation Node / Q6 Threshold');	Sets the plot title so the test point or analysis purpose is explicit.
194	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
195	ax5 = subplot(5,1,5);	Assigns or calculates the MATLAB variable ax5 from the expression on the right-hand side.
196	plot(t*1e3,f_inst/1e3,'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
197	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
198	ylabel('kHz');	Sets the y-axis label and therefore documents the plotted engineering unit.
199	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
200	title('Instantaneous NE555 Frequency');	Sets the plot title so the test point or analysis purpose is explicit.
201	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
202	linkaxes([ax1 ax2 ax3 ax4 ax5],'x');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
203	xlim([0 Tsim*1e3]);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
204	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
205	drawnow;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
206	pause(0.1);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
207	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
208	% 14. FIGURE 2 - Zoom around the rising edge	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
209	z = t >= 0.90e-3 & t <= 1.80e-3;	Builds a logical time mask so only the requested zoom interval is plotted.
210	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
211	f2 = figure('Name','V2.1 Rising Edge Detail','NumberTitle','off',...	Assigns or calculates the MATLAB variable f2 from the expression on the right-hand side.
212	'Color','w','Visible','on');	Continuation line containing text, a table variable name, or another argument for the MATLAB statement started above.
213	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
214	<code>subplot(4,1,1);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
215	<code>plot(t(z)*1e3,Iload(z),'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
216	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
217	<code>ylabel('A');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
218	<code>title('0 A to 10 A Step');</code>	Sets the plot title so the test point or analysis purpose is explicit.
219	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
220	<code>subplot(4,1,2);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
221	<code>plot(t(z)*1e3,TP7(z),'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
222	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
223	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
224	<code>title('TP7');</code>	Sets the plot title so the test point or analysis purpose is explicit.
225	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
226	<code>subplot(4,1,3);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
227	<code>plot(t(z)*1e3,TP8(z),'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
228	<code>hold on;</code>	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
229	<code>plot(t(z)*1e3,VQ6_on*ones(1,sum(z)),'--','LineWidth',1.0);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
230	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
231	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
232	<code>title('TP8 and Q6 Threshold');</code>	Sets the plot title so the test point or analysis purpose is explicit.
233	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
234	<code>subplot(4,1,4);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
235	<code>plot(t(z)*1e3,f_inst(z)/1e3,'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
236	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
237	<code>ylabel('kHz');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
238	<code>xlabel('Time (ms)');</code>	Sets the x-axis label for the current plot.
239	<code>title('Instantaneous Frequency');</code>	Sets the plot title so the test point or analysis purpose is explicit.
240	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
241	<code>drawnow;</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
242	<code>pause(0.1);</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
243	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
244	<code>% 15. FIGURE 3 - Fast digital test points</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
245	<code>zo = t >= 1.20e-3 & t <= 1.60e-3;</code>	Builds a logical time mask so only the requested zoom interval is plotted.
246	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
247	<code>f3 = figure('Name','V2.1 Digital Test Points','NumberTitle','off',...</code>	Assigns or calculates the MATLAB variable f3 from the expression on the right-hand



Line	MATLAB source line	English explanation
		side.
248	'Color','w','Visible','on');	Continuation line containing text, a table variable name, or another argument for the MATLAB statement started above.
249	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
250	subplot(4,2,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
251	plot(t(z0)*1e3,TP9(z0),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
252	grid on; title('TP9'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
253	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
254	subplot(4,2,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
255	plot(t(z0)*1e3,TP10(z0),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
256	grid on; title('TP10'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
257	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
258	subplot(4,2,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
259	plot(t(z0)*1e3,TP11(z0),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
260	grid on; title('TP11'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
261	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
262	subplot(4,2,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
263	plot(t(z0)*1e3,TP12(z0),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
264	grid on; title('TP12'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
265	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
266	subplot(4,2,5);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
267	plot(t(z0)*1e3,TP13(z0),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
268	grid on; title('TP13'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
269	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
270	subplot(4,2,6);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
271	plot(t(z0)*1e3,TP14(z0),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
272	grid on; title('TP14'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
273	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
274	subplot(4,2,[7 8]);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
275	plot(t(z0)*1e3,f_OUT(z0),'LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
276	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to



Line	MATLAB source line	English explanation
		read.
277	title('Differential f-OUT = TP13 - TP14');	Sets the plot title so the test point or analysis purpose is explicit.
278	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
279	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
280	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
281	drawnow;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
282	pause(0.1);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
283	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
284	%% 16. Automatic diagnostic PNG export	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
285	% This is useful if MATLAB Online displays a figure window incorrectly.	Source-code comment: This is useful if MATLAB Online displays a figure window incorrectly.
286	% The PNG files will appear in the Current Folder.	Source-code comment: The PNG files will appear in the Current Folder.
287	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
288	exportgraphics(f1,'V2_1_Analog_Control.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
289	exportgraphics(f2,'V2_1_Rising_Edge.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
290	exportgraphics(f3,'V2_1_Digital_Test_Points.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
291	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
292	fprintf('\nThree PNG files were exported automatically:\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
293	fprintf(' V2_1_Analog_Control.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
294	fprintf(' V2_1_Rising_Edge.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
295	fprintf(' V2_1_Digital_Test_Points.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
296	fprintf('If MATLAB Online figure windows are blank, open these PNG files directly.\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
297	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
298	drawnow;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.



Appendix D. Version 2.2 - Line-by-Line Annotated MATLAB Source

Every source line is reproduced exactly in the left column. The right column explains the purpose in plain English so a reader with little or no MATLAB experience can follow the program from top to bottom.

Line	MATLAB source line	English explanation
1	%% NE555 Current-Sense Model - VERSION 2.2	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
2	% MATLAB Online "export-only" version.	Source-code comment: MATLAB Online "export-only" version.
3	% This version does NOT rely on interactive figure rendering.	Source-code comment: This version does NOT rely on interactive figure rendering.
4	% It calculates the same Version 2.1 signals and writes PNG files directly.	Source-code comment: It calculates the same Version 2.1 signals and writes PNG files directly.
5	%	Source-code comment explaining the model assumption or the purpose of the following lines.
6	% If the interactive MATLAB Online figure windows are blank but these PNG	Source-code comment: If the interactive MATLAB Online figure windows are blank but these PNG
7	% files contain the plots, the electrical model is fine and the problem is	Source-code comment: files contain the plots, the electrical model is fine and the problem is
8	% only the browser/figure renderer.	Source-code comment: only the browser/figure renderer.
9	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
10	clear;	Removes variables from the MATLAB workspace so values from an earlier run cannot contaminate this simulation.
11	clc;	Clears the MATLAB Command Window so the new run starts with readable console output.
12	close all force;	Closes previously open figure windows so old plots are not confused with the current simulation results.
13	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
14	%% 1. Simulation settings	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
15	VCC = 5.0;	Assigns or calculates the 5 V supply used by the behavioral model from the expression on the right-hand side.
16	dt = 0.5e-6;	Assigns or calculates the simulation time step from the expression on the right-hand side.
17	Tsim = 6e-3;	Assigns or calculates the total simulated time from the expression on the right-hand side.
18	t = 0:dt:Tsim;	Assigns or calculates the simulation time vector from the expression on the right-hand side.
19	N = numel(t);	Calculates the number of time samples by counting the number of elements in the referenced vector.
20	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
21	%% 2. Circuit parameters	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
22	Rsense = 5e-3;	Assigns or calculates the current-shunt resistance from the expression on the right-hand side.
23	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
24	R38 = 1e3;	Assigns or calculates the 1 kOhm LM358 input-path resistor from the expression on the right-hand side.
25	R37 = 10e3;	Assigns or calculates the 10 kOhm LM358 input-path resistor from the expression on the right-hand side.
26	R36 = 330e3;	Assigns or calculates the LM358 gain-setting resistor from the expression on the right-hand side.
27	C30 = 220e-9;	Assigns or calculates the current-sense filtering capacitor from the expression on the right-hand side.



Line	MATLAB source line	English explanation
28	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
29	R35 = 330e3;	Assigns or calculates the main resistor in the transient-modulation path from the expression on the right-hand side.
30	R34 = 24e3;	Assigns or calculates the 24 kOhm modulation-path resistor from the expression on the right-hand side.
31	C29 = 2e-9;	Assigns or calculates the transient-modulation capacitor from the expression on the right-hand side.
32	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
33	R32 = 10;	Assigns or calculates the small series resistor in the NE555 charge path from the expression on the right-hand side.
34	R33 = 5e3;	Assigns or calculates the main NE555 charge resistor from the expression on the right-hand side.
35	R29 = 1e3;	Assigns or calculates the NE555 discharge resistor from the expression on the right-hand side.
36	C27 = 10e-9;	Assigns or calculates the NE555 timing capacitor from the expression on the right-hand side.
37	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
38	Vf_D31 = 0.60;	Assigns or calculates the assumed D31 forward drop from the expression on the right-hand side.
39	VCEsat = 0.10;	Assigns or calculates the assumed saturation voltage of the NE555 discharge transistor from the expression on the right-hand side.
40	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
41	Npri = 80;	Assigns or calculates the transformer primary-turn count from the expression on the right-hand side.
42	Nsec = 100;	Assigns or calculates the transformer secondary-turn count from the expression on the right-hand side.
43	turns_ratio = Nsec/Npri;	Assigns or calculates the ideal transformer turns ratio from the expression on the right-hand side.
44	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
45	%% 3. Derived values	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
46	A_LM358 = R36/(R38+R37);	Assigns or calculates the modeled LM358 closed-loop gain magnitude from the expression on the right-hand side.
47	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
48	Rth_C30 = 1/(1/R38 + 1/R37);	Calculates the Thevenin resistance seen by C30 from the stated reciprocal relationship.
49	tau_sense = Rth_C30*C30;	Assigns or calculates the current-sense first-order time constant from the expression on the right-hand side.
50	fc_sense = 1/(2*pi*tau_sense);	Calculates the corresponding current-sense cutoff frequency from the stated reciprocal relationship.
51	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
52	tau_mod = (R35+R34)*C29;	Assigns or calculates the transient-path time constant from the expression on the right-hand side.
53	fc_mod = 1/(2*pi*tau_mod);	Calculates the corresponding transient-path cutoff frequency from the stated reciprocal relationship.
54	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
55	RA = R32 + R33;	Assigns or calculates the NE555 effective charge resistance from the expression on the right-hand side.
56	RB = R29;	Assigns or calculates the NE555 discharge resistance from the expression on the right-hand side.



Line	MATLAB source line	English explanation
57	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
58	VTL = VCC/3;	Assigns or calculates the lower NE555 timing threshold (one-third VCC) from the expression on the right-hand side.
59	VTH = 2*VCC/3;	Assigns or calculates the upper NE555 timing threshold (two-thirds VCC) from the expression on the right-hand side.
60	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
61	Vcharge_inf = VCC - Vf_D31;	Assigns or calculates the asymptotic timing-capacitor charge voltage after the diode drop from the expression on the right-hand side.
62	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
63	t_charge = RA*C27 * log((Vcharge_inf-VTL)/(Vcharge_inf-VTH));	Calculates the calculated NE555 charge interval using the analytical exponential charge/discharge timing equation.
64	t_discharge = RB*C27 * log((VTH-VCESat)/(VTL-VCESat));	Calculates the calculated NE555 discharge interval using the analytical exponential charge/discharge timing equation.
65	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
66	T555_nom = t_charge + t_discharge;	Assigns or calculates the nominal NE555 oscillation period from the expression on the right-hand side.
67	f555_nom = 1/T555_nom;	Assigns or calculates the nominal NE555 frequency from the expression on the right-hand side.
68	duty_nom = t_charge/T555_nom;	Assigns or calculates the nominal high-state duty fraction from the expression on the right-hand side.
69	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
70	%% 4. Fast 0 -> 10 A -> 0 A profile	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
71	Iload = zeros(size(t));	Initializes the simulated load-current waveform with zeros, giving it the required array size before later samples are calculated.
72	Iload(t >= 1e-3 & t < 4e-3) = 10;	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
73	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
74	%% 5. TP6	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
75	TP6 = -Iload*Rsense;	Converts load current into the shunt voltage at TP6. The negative sign represents the polarity chosen for the inverting LM358 model.
76	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
77	%% 6. TP7	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
78	TP7_target = -A_LM358*TP6;	Calculates the ideal amplified current-sense voltage before the first-order output/filter response is applied.
79	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
80	TP7 = zeros(size(t));	Initializes the filtered behavioral LM358 output with zeros, giving it the required array size before later samples are calculated.
81	alpha_sense = dt/(tau_sense+dt);	Assigns or calculates the discrete first-order filter coefficient for TP7 from the expression on the right-hand side.
82	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
83	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
84	TP7(k) = TP7(k-1) + alpha_sense*(TP7_target(k)-TP7(k-1));	Updates TP7 with a first-order discrete low-pass equation, moving the previous output toward the instantaneous target.
85	end	Ends the current for-loop or if/else block.
86	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
87	TP7 = min(max(TP7,0),3.5);	Clamps the behavioral LM358 output to the assumed usable output range on a 5 V supply.
88	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
89	%% 7. TP8	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
90	Vbias_TP8 = 4.35;	Assigns or calculates the assumed TP8 bias/clamp level from the expression on the right-hand side.
91	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
92	TP7_lp_mod = zeros(size(t));	Initializes the low-pass state used to form the high-pass transient with zeros, giving it the required array size before later samples are calculated.
93	alpha_mod = dt/(tau_mod*dt);	Assigns or calculates the discrete transient-path filter coefficient from the expression on the right-hand side.
94	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
95	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
96	TP7_lp_mod(k) = TP7_lp_mod(k-1) + alpha_mod*(TP7(k)-TP7_lp_mod(k-1));	Updates the low-pass state used to subtract the slow component of TP7 and isolate the transient/high-pass component.
97	end	Ends the current for-loop or if/else block.
98	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
99	HP_mod = TP7 - TP7_lp_mod;	Subtracts the slow low-pass component from TP7 to obtain the transient/high-pass component that drives TP8.
100	TP8 = Vbias_TP8 + HP_mod;	Adds the transient component to the assumed TP8 DC bias, forming the behavioral control-node voltage.
101	TP8 = min(max(TP8,0),6.5);	Clamps the behavioral TP8 voltage to a bounded plotting range so extreme numerical values cannot dominate the visualization.
102	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
103	%% 8. Q6 threshold	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
104	VBE_PNP = 0.65;	Assigns or calculates the assumed BC558 base-emitter magnitude from the expression on the right-hand side.
105	VQ6_on = VCC + VBE_PNP;	Assigns or calculates the modeled Q6 turn-on threshold from the expression on the right-hand side.
106	Q6_drive = max(TP8 - VQ6_on,0);	Implements the behavioral Q6 threshold: values below threshold produce zero drive; only voltage above threshold contributes to modulation.
107	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
108	%% 9. Frequency	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
109	Kf = 5e3;	Assigns or calculates the behavioral frequency sensitivity coefficient from the expression on the right-hand side.
110	f_inst = f555_nom + Kf*Q6_drive;	Creates the behavioral frequency law by adding Kf times Q6 threshold overdrive to the nominal NE555 frequency.
111	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
112	%% 10. NE555 signals	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
113	phase = zeros(size(t));	Initializes the accumulated oscillator phase with zeros, giving it the required array size before later samples are calculated.
114	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
115	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
116	phase(k) = phase(k-1) + 2*pi*f_inst(k)*dt;	Numerically integrates instantaneous frequency into oscillator phase for this time sample.



Line	MATLAB source line	English explanation
117	end	Ends the current for-loop or if/else block.
118	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
119	phase01 = mod(phase,2*pi)/(2*pi);	Wraps accumulated phase into a normalized 0-to-1 cycle position used to synthesize the square-wave output.
120	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
121	TP10 = VCC*(phase01 < duty_nom);	Converts normalized oscillator phase and duty cycle into a 0 V / VCC square-wave approximation of the NE555 output.
122	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
123	TP9 = VCC*ones(size(t));	Initializes the modeled NE555 discharge-node voltage with ones of the required array size; later lines modify selected samples.
124	TP9(phase01 >= duty_nom) = VCEsat;	For time samples in the discharge phase, forces TP9 down to the assumed discharge-transistor saturation voltage.
125	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
126	% 11. Transformer signals	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
127	TP12 = TP10;	Uses TP10 directly as the second complementary primary-drive waveform.
128	TP11 = VCC - TP10;	Creates the complementary primary-drive waveform by logically inverting TP10 around the 5 V supply.
129	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
130	Vpri_diff = TP12 - TP11;	Calculates the differential voltage applied across the idealized transformer primary.
131	Vsec_diff = turns_ratio*Vpri_diff;	Applies the ideal turns ratio to the primary differential voltage to estimate the transformer secondary differential voltage.
132	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
133	TP13 = +0.5*Vsec_diff;	Assigns one secondary terminal to half of the differential secondary voltage about a virtual midpoint used only for plotting.
134	TP14 = -0.5*Vsec_diff;	Assigns the opposite secondary terminal to the negative half of the differential secondary voltage about the virtual plotting midpoint.
135	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
136	f_OUT = TP13 - TP14;	Calculates the differential output voltage; this is the meaningful transformer-secondary signal because the secondary is floating.
137	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
138	% 12. Console summary	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
139	fprintf('\nVERSION 2.2 - EXPORT-ONLY PLOTTING\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
140	fprintf('LM358 gain : %.2f V/V\n', A_LM358);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
141	fprintf('TP8 max : %.3f V\n', max(TP8));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
142	fprintf('Q6 threshold : %.3f V\n', VQ6_on);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
143	fprintf('Max Q6 drive : %.3f V\n', max(Q6_drive));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
144	fprintf('Frequency : %.3f to %.3f kHz\n', ...	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
145	min(f_inst)/1e3,max(f_inst)/1e3);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
146	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
147	% 13. Simple graphics diagnostic image	MATLAB section header. It creates a collapsible/run-able code section and labels the



Line	MATLAB source line	English explanation
		next stage of the model.
148	<code>fig = figure('Visible','off','Color','w');</code>	Assigns or calculates the MATLAB variable fig from the expression on the right-hand side.
149	<code>set(fig,'Renderer','painters');</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
150	<code>plot([0 1 2],[0 1 0],'LineWidth',2);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
151	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
152	<code>xlabel('X');</code>	Sets the x-axis label for the current plot.
153	<code>ylabel('Y');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
154	<code>title('MATLAB Graphics Diagnostic');</code>	Sets the plot title so the test point or analysis purpose is explicit.
155	<code>print(fig,'Graphics_Diagnostic.png','-dpng','-r160');</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
156	<code>close(fig);</code>	Closes the specified figure after export to reduce graphics-resource use during automated execution.
157	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
158	<code>%% 14. Analog/control figure</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
159	<code>fig = figure('Visible','off','Color','w');</code>	Assigns or calculates the MATLAB variable fig from the expression on the right-hand side.
160	<code>set(fig,'Renderer','painters');</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
161	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
162	<code>subplot(5,1,1);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
163	<code>plot(t*1e3,Iload,'LineWidth',1.3);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
164	<code>grid on; ylabel('A'); title('Load Current');</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
165	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
166	<code>subplot(5,1,2);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
167	<code>plot(t*1e3,TP6*1e3,'LineWidth',1.3);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
168	<code>grid on; ylabel('mV'); title('TP6');</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
169	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
170	<code>subplot(5,1,3);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
171	<code>plot(t*1e3,TP7,'LineWidth',1.3);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
172	<code>grid on; ylabel('V'); title('TP7');</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
173	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
174	<code>subplot(5,1,4);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
175	<code>plot(t*1e3,TP8,'LineWidth',1.3);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
176	<code>hold on;</code>	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
177	<code>plot(t*1e3,VQ6_on*ones(size(t)),'--','LineWidth',1.0);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.



Line	MATLAB source line	English explanation
178	grid on; ylabel('V'); title('TP8 and Q6 Threshold');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
179	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
180	subplot(5,1,5);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
181	plot(t*1e3,f_inst/1e3,'LineWidth',1.3);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
182	grid on; ylabel('kHz'); xlabel('Time (ms)');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
183	title('Instantaneous Frequency');	Sets the plot title so the test point or analysis purpose is explicit.
184	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
185	print(fig,'V2_2_Analog_Control.png','-dpng','-r180');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
186	close(fig);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
187	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
188	%% 15. Rising edge figure	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
189	z = t >= 0.90e-3 & t <= 1.80e-3;	Builds a logical time mask so only the requested zoom interval is plotted.
190	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
191	fig = figure('Visible','off','Color','w');	Assigns or calculates the MATLAB variable fig from the expression on the right-hand side.
192	set(fig,'Renderer','painters');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
193	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
194	subplot(4,1,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
195	plot(t(z)*1e3,Iload(z),'LineWidth',1.3);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
196	grid on; ylabel('A'); title('0 A to 10 A Step');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
197	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
198	subplot(4,1,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
199	plot(t(z)*1e3,TP7(z),'LineWidth',1.3);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
200	grid on; ylabel('V'); title('TP7');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
201	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
202	subplot(4,1,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
203	plot(t(z)*1e3,TP8(z),'LineWidth',1.3);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
204	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
205	plot(t(z)*1e3,VQ6_on*ones(1,sum(z)),'--','LineWidth',1.0);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
206	grid on; ylabel('V'); title('TP8 and Q6 Threshold');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
207	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
208	subplot(4,1,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
209	plot(t(z)*1e3,f_inst(z)/1e3,'LineWidth',1.3);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
210	grid on; ylabel('kHz'); xlabel('Time (ms)');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
211	title('Instantaneous Frequency');	Sets the plot title so the test point or analysis purpose is explicit.
212	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
213	print(fig,'V2_2_Rising_Edge.png','-dpng','-r180');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
214	close(fig);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
215	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
216	%% 16. Digital test points	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
217	zo = t >= 1.20e-3 & t <= 1.60e-3;	Builds a logical time mask so only the requested zoom interval is plotted.
218	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
219	fig = figure('Visible','off','Color','w');	Assigns or calculates the MATLAB variable fig from the expression on the right-hand side.
220	set(fig,'Renderer','painters');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
221	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
222	subplot(4,2,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
223	plot(t(zo)*1e3,TP9(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
224	grid on; title('TP9'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
225	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
226	subplot(4,2,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
227	plot(t(zo)*1e3,TP10(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
228	grid on; title('TP10'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
229	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
230	subplot(4,2,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
231	plot(t(zo)*1e3,TP11(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
232	grid on; title('TP11'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
233	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
234	subplot(4,2,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
235	plot(t(zo)*1e3,TP12(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
236	grid on; title('TP12'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
237	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
238	subplot(4,2,5);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
239	plot(t(zo)*1e3,TP13(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
240	grid on; title('TP13'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
241	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
242	subplot(4,2,6);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
243	plot(t(zo)*1e3,TP14(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
244	grid on; title('TP14'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
245	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
246	subplot(4,2,[7 8]);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
247	plot(t(zo)*1e3,f_OUT(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
248	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
249	title('Differential f-OUT = TP13 - TP14');	Sets the plot title so the test point or analysis purpose is explicit.
250	ylabel('V'); xlabel('Time (ms)');	Sets the y-axis label and therefore documents the plotted engineering unit.
251	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
252	print(fig,'V2_2_Digital_Test_Points.png','-dpng','-r180');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
253	close(fig);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
254	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
255	%% 17. Save numeric data to CSV	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
256	T = table(t(:),Iload(:),TP6(:),TP7(:),TP8(:),TP9(:),TP10(:), ...	Starts construction of a MATLAB table that groups the calculated results for CSV export.
257	TP11(:),TP12(:),TP13(:),TP14(:),f_OUT(:),f_inst(:), ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
258	'VariableNames',{'Time_s','Iload_A','TP6_V','TP7_V','TP8_V',	Supplies human-readable column names for the MATLAB table being constructed on the surrounding continuation lines.
259	... 'TP9_V','TP10_V','TP11_V','TP12_V','TP13_V','TP14_V', ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
260	'fOUT_V','Frequency_Hz'});	Continuation line containing text, a table variable name, or another argument for the MATLAB statement started above.
261	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
262	writetable(T,'V2_2_Simulation_Data.csv');	Writes the MATLAB table to a CSV file so numeric results can be inspected independently or replotted later.
263	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
264	fprintf('\nFiles created in Current Folder:\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
265	fprintf(' Graphics_Diagnostic.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
266	fprintf(' V2_2_Analog_Control.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
267	fprintf(' V2_2_Rising_Edge.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.

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NE555 Current-Sense Controlled Frequency Generator | Study & Simulation Report

Email: ahds@engineer.com

Line	MATLAB source line	English explanation
268	<code>fprintf(' V2_2_Digital_Test_Points.png\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
269	<code>fprintf(' V2_2_Simulation_Data.csv\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
270	<code>fprintf('\nOpen the PNG files directly from the Current Folder.\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.



Appendix E. Version 2.3 - Line-by-Line Annotated MATLAB Source

Every source line is reproduced exactly in the left column. The right column explains the purpose in plain English so a reader with little or no MATLAB experience can follow the program from top to bottom.

Line	MATLAB source line	English explanation
1	%% NE555 Current-Sense Model - VERSION 2.3	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
2	% MATLAB Online exportgraphics version	Source-code comment: MATLAB Online exportgraphics version
3	%	Source-code comment explaining the model assumption or the purpose of the following lines.
4	% Purpose:	Source-code comment: Purpose:
5	% Re-run the Version 2 fast-current-step model and save all plots using	Source-code comment: Re-run the Version 2 fast-current-step model and save all plots using
6	% exportgraphics(), which is generally more reliable in MATLAB Online.	Source-code comment: exportgraphics(), which is generally more reliable in MATLAB Online.
7	%	Source-code comment explaining the model assumption or the purpose of the following lines.
8	% Electrical assumptions are intentionally unchanged from Versions 2.1/2.2:	Source-code comment: Electrical assumptions are intentionally unchanged from Versions 2.1/2.2:
9	% - VCC = 5 V	Source-code comment: - VCC = 5 V
10	% - R-sense = 5 mOhm	Source-code comment: - R-sense = 5 mOhm
11	% - Load current step = 0 A -> 10 A -> 0 A	Source-code comment: - Load current step = 0 A -> 10 A -> 0 A
12	% - LM358 gain ~= 30	Source-code comment: - LM358 gain ~= 30
13	% - TP8 behavioral bias ~= 4.35 V	Source-code comment: - TP8 behavioral bias ~= 4.35 V
14	% - BC558 turn-on threshold approximation ~= 5.65 V	Source-code comment: - BC558 turn-on threshold approximation ~= 5.65 V
15	% - Nominal NE555 frequency ~= 18.39 kHz	Source-code comment: - Nominal NE555 frequency ~= 18.39 kHz
16	%	Source-code comment explaining the model assumption or the purpose of the following lines.
17	% IMPORTANT:	Source-code comment: IMPORTANT:
18	% This remains a behavioral engineering model.	Source-code comment: This remains a behavioral engineering model.
19	% The purpose of Version 2.3 is mainly to solve the MATLAB Online	Source-code comment: The purpose of Version 2.3 is mainly to solve the MATLAB Online
20	% graphics/export problem while keeping the same electrical calculation.	Source-code comment: graphics/export problem while keeping the same electrical calculation.
21	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
22	clear;	Removes variables from the MATLAB workspace so values from an earlier run cannot contaminate this simulation.
23	clc;	Clears the MATLAB Command Window so the new run starts with readable console output.
24	close all force;	Closes previously open figure windows so old plots are not confused with the current simulation results.
25	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
26	%% 1. Simulation settings	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
27	VCC = 5.0;	Assigns or calculates the 5 V supply used by the behavioral model from the expression on the right-hand side.
28	dt = 0.5e-6; % 0.5 us	Assigns or calculates the simulation time step from the expression on the right-hand side.
29	Tsim = 6e-3; % 6 ms	Assigns or calculates the total simulated time from the expression on the right-hand side.
30	t = 0:dt:Tsim;	Assigns or calculates the simulation time vector from the expression on the right-hand side.
31	N = numel(t);	Calculates the number of time samples by counting the number of elements in the



Line	MATLAB source line	English explanation
		referenced vector.
32	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
33	%% 2. Circuit parameters	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
34	$R_{sense} = 5e-3;$ % 5 mOhm	Assigns or calculates the current-shunt resistance from the expression on the right-hand side.
35	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
36	$R38 = 1e3;$	Assigns or calculates the 1 kOhm LM358 input-path resistor from the expression on the right-hand side.
37	$R37 = 10e3;$	Assigns or calculates the 10 kOhm LM358 input-path resistor from the expression on the right-hand side.
38	$R36 = 330e3;$	Assigns or calculates the LM358 gain-setting resistor from the expression on the right-hand side.
39	$C30 = 220e-9;$	Assigns or calculates the current-sense filtering capacitor from the expression on the right-hand side.
40	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
41	$R35 = 330e3;$	Assigns or calculates the main resistor in the transient-modulation path from the expression on the right-hand side.
42	$R34 = 24e3;$	Assigns or calculates the 24 kOhm modulation-path resistor from the expression on the right-hand side.
43	$C29 = 2e-9;$	Assigns or calculates the transient-modulation capacitor from the expression on the right-hand side.
44	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
45	$R32 = 10;$	Assigns or calculates the small series resistor in the NE555 charge path from the expression on the right-hand side.
46	$R33 = 5e3;$	Assigns or calculates the main NE555 charge resistor from the expression on the right-hand side.
47	$R29 = 1e3;$	Assigns or calculates the NE555 discharge resistor from the expression on the right-hand side.
48	$C27 = 10e-9;$	Assigns or calculates the NE555 timing capacitor from the expression on the right-hand side.
49	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
50	$Vf_D31 = 0.60;$ % Approx. 1N914 forward drop	Assigns or calculates the assumed D31 forward drop from the expression on the right-hand side.
51	$VCEsat = 0.10;$ % Approx. 555 discharge-transistor saturation	Assigns or calculates the assumed saturation voltage of the NE555 discharge transistor from the expression on the right-hand side.
52	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
53	$Npri = 80;$	Assigns or calculates the transformer primary-turn count from the expression on the right-hand side.
54	$Nsec = 100;$	Assigns or calculates the transformer secondary-turn count from the expression on the right-hand side.
55	$turns_ratio = Nsec/Npri;$	Assigns or calculates the ideal transformer turns ratio from the expression on the right-hand side.
56	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
57	%% 3. Derived values	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
58	$A_{LM358} = R36/(R38+R37);$	Assigns or calculates the modeled LM358 closed-loop gain magnitude from the expression on the right-hand side.
59	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
60	$Rth_C30 = 1/(1/R38 + 1/R37);$	Calculates the Thevenin resistance seen by C30 from the stated reciprocal



Line	MATLAB source line	English explanation
		relationship.
61	<code>tau_sense = Rth_C30*C30;</code>	Assigns or calculates the current-sense first-order time constant from the expression on the right-hand side.
62	<code>fc_sense = 1/(2*pi*tau_sense);</code>	Calculates the corresponding current-sense cutoff frequency from the stated reciprocal relationship.
63	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
64	<code>tau_mod = (R35+R34)*C29;</code>	Assigns or calculates the transient-path time constant from the expression on the right-hand side.
65	<code>fc_mod = 1/(2*pi*tau_mod);</code>	Calculates the corresponding transient-path cutoff frequency from the stated reciprocal relationship.
66	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
67	<code>RA = R32 + R33;</code>	Assigns or calculates the NE555 effective charge resistance from the expression on the right-hand side.
68	<code>RB = R29;</code>	Assigns or calculates the NE555 discharge resistance from the expression on the right-hand side.
69	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
70	<code>VTL = VCC/3;</code>	Assigns or calculates the lower NE555 timing threshold (one-third VCC) from the expression on the right-hand side.
71	<code>VTH = 2*VCC/3;</code>	Assigns or calculates the upper NE555 timing threshold (two-thirds VCC) from the expression on the right-hand side.
72	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
73	<code>Vcharge_inf = VCC - Vf_D31;</code>	Assigns or calculates the asymptotic timing-capacitor charge voltage after the diode drop from the expression on the right-hand side.
74	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
75	<code>t_charge = RA*C27 * log((Vcharge_inf-VTL)/(Vcharge_inf-VTH));</code>	Calculates the calculated NE555 charge interval using the analytical exponential charge/discharge timing equation.
76	<code>t_discharge = RB*C27 * log((VTH-VCesat)/(VTL-VCesat));</code>	Calculates the calculated NE555 discharge interval using the analytical exponential charge/discharge timing equation.
77	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
78	<code>T555_nom = t_charge + t_discharge;</code>	Assigns or calculates the nominal NE555 oscillation period from the expression on the right-hand side.
79	<code>f555_nom = 1/T555_nom;</code>	Assigns or calculates the nominal NE555 frequency from the expression on the right-hand side.
80	<code>duty_nom = t_charge/T555_nom;</code>	Assigns or calculates the nominal high-state duty fraction from the expression on the right-hand side.
81	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
82	<code>%% 4. Fast current-step profile</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
83	<code>% 0 A from 0 to 1 ms</code>	Source-code comment: 0 A from 0 to 1 ms
84	<code>% 10 A from 1 to 4 ms</code>	Source-code comment: 10 A from 1 to 4 ms
85	<code>% 0 A after 4 ms</code>	Source-code comment: 0 A after 4 ms
86	<code>Iload = zeros(size(t));</code>	Initializes the simulated load-current waveform with zeros, giving it the required array size before later samples are calculated.
87	<code>Iload(t >= 1e-3 & t < 4e-3) = 10;</code>	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
88	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
89	<code>%% 5. TP6 - R-sense voltage</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
90	<code>% Polarity chosen for the inverting LM358 model.</code>	Source-code comment: Polarity chosen for the inverting LM358 model.



Line	MATLAB source line	English explanation
91	TP6 = -Iload*Rsense;	Converts load current into the shunt voltage at TP6. The negative sign represents the polarity chosen for the inverting LM358 model.
92	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
93	%% 6. TP7 - LM358 output	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
94	TP7_target = -A_LM358*TP6;	Calculates the ideal amplified current-sense voltage before the first-order output/filter response is applied.
95	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
96	TP7 = zeros(size(t));	Initializes the filtered behavioral LM358 output with zeros, giving it the required array size before later samples are calculated.
97	alpha_sense = dt/(tau_sense+dt);	Assigns or calculates the discrete first-order filter coefficient for TP7 from the expression on the right-hand side.
98	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
99	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
100	TP7(k) = TP7(k-1) + alpha_sense*(TP7_target(k)-TP7(k-1));	Updates TP7 with a first-order discrete low-pass equation, moving the previous output toward the instantaneous target.
101	end	Ends the current for-loop or if/else block.
102	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
103	% Approximate LM358 output range on 5-V supply	Source-code comment: Approximate LM358 output range on 5-V supply
104	TP7 = min(max(TP7,0),3.5);	Clamps the behavioral LM358 output to the assumed usable output range on a 5 V supply.
105	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
106	%% 7. TP8 - behavioral transient modulation node	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
107	Vbias_TP8 = 4.35;	Assigns or calculates the assumed TP8 bias/clamp level from the expression on the right-hand side.
108	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
109	TP7_lp_mod = zeros(size(t));	Initializes the low-pass state used to form the high-pass transient with zeros, giving it the required array size before later samples are calculated.
110	alpha_mod = dt/(tau_mod+dt);	Assigns or calculates the discrete transient-path filter coefficient from the expression on the right-hand side.
111	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
112	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
113	TP7_lp_mod(k) = TP7_lp_mod(k-1) + ...	Updates the low-pass state used to subtract the slow component of TP7 and isolate the transient/high-pass component.
114	alpha_mod*(TP7(k)-TP7_lp_mod(k-1));	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
115	end	Ends the current for-loop or if/else block.
116	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
117	HP_mod = TP7 - TP7_lp_mod;	Subtracts the slow low-pass component from TP7 to obtain the transient/high-pass component that drives TP8.
118	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
119	Kmod_node = 1.0; % no artificial gain increase	Assigns or calculates the behavioral TP8 shaping gain from the expression on the right-hand side.
120	TP8 = Vbias_TP8 + Kmod_node*HP_mod;	Adds the transient component to the assumed TP8 DC bias, forming the behavioral control-node voltage.
121	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
122	TP8 = min(max(TP8,0),6.5);	Clamps the behavioral TP8 voltage to a bounded plotting range so extreme numerical values cannot dominate the visualization.
123	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
124	%% 8. Q6 threshold and behavioral drive	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
125	VBE_PNP = 0.65;	Assigns or calculates the assumed BC558 base-emitter magnitude from the expression on the right-hand side.
126	VQ6_on = VCC + VBE_PNP; % approx. 5.65 V	Assigns or calculates the modeled Q6 turn-on threshold from the expression on the right-hand side.
127	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
128	Q6_drive = max(TP8 - VQ6_on,0);	Implements the behavioral Q6 threshold: values below threshold produce zero drive; only voltage above threshold contributes to modulation.
129	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
130	%% 9. Instantaneous frequency	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
131	Kf = 5e3; % Hz/V behavioral sensitivity	Assigns or calculates the behavioral frequency sensitivity coefficient from the expression on the right-hand side.
132	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
133	f_inst = f555_nom + Kf*Q6_drive;	Creates the behavioral frequency law by adding Kf times Q6 threshold overdrive to the nominal NE555 frequency.
134	f_inst = min(max(f_inst,5e3),40e3);	Clamps the behavioral frequency to a practical plotting range; this is a numerical safeguard rather than a measured circuit limit.
135	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
136	%% 10. Variable-frequency NE555 phase	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
137	phase = zeros(size(t));	Initializes the accumulated oscillator phase with zeros, giving it the required array size before later samples are calculated.
138	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
139	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
140	phase(k) = phase(k-1) + 2*pi*f_inst(k)*dt;	Numerically integrates instantaneous frequency into oscillator phase for this time sample.
141	end	Ends the current for-loop or if/else block.
142	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
143	phase01 = mod(phase,2*pi)/(2*pi);	Wraps accumulated phase into a normalized 0-to-1 cycle position used to synthesize the square-wave output.
144	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
145	%% 11. TP10 - NE555 output	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
146	TP10 = VCC*(phase01 < duty_nom);	Converts normalized oscillator phase and duty cycle into a 0 V / VCC square-wave approximation of the NE555 output.
147	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
148	%% 12. TP9 - NE555 discharge node	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
149	TP9 = VCC*ones(size(t));	Initializes the modeled NE555 discharge-node voltage with ones of the required array size; later lines modify selected samples.
150	TP9(phase01 >= duty_nom) = VCEsat;	For time samples in the discharge phase, forces TP9 down to the assumed discharge-transistor saturation voltage.
151	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
152	%% 13. TP11 / TP12 - primary-drive signals	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
153	TP12 = TP10;	Uses TP10 directly as the second complementary primary-drive waveform.
154	TP11 = VCC - TP10;	Creates the complementary primary-drive waveform by logically inverting TP10 around the 5 V supply.
155	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
156	%% 14. TP13 / TP14 - transformer secondary	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
157	Vpri_diff = TP12 - TP11;	Calculates the differential voltage applied across the idealized transformer primary.
158	Vsec_diff = turns_ratio*Vpri_diff;	Applies the ideal turns ratio to the primary differential voltage to estimate the transformer secondary differential voltage.
159	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
160	% Floating secondary represented with a virtual midpoint for visualization	Source-code comment: Floating secondary represented with a virtual midpoint for visualization
161	TP13 = +0.5*Vsec_diff;	Assigns one secondary terminal to half of the differential secondary voltage about a virtual midpoint used only for plotting.
162	TP14 = -0.5*Vsec_diff;	Assigns the opposite secondary terminal to the negative half of the differential secondary voltage about the virtual plotting midpoint.
163	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
164	f_OUT = TP13 - TP14;	Calculates the differential output voltage; this is the meaningful transformer-secondary signal because the secondary is floating.
165	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
166	%% 15. Console summary	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
167	fprintf('\nVERSION 2.3 - EXPORTGRAPHICS VERSION\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
168	fprintf('LM358 gain : %.2f V/V\n',A_LM358);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
169	fprintf('Current-sense tau : %.1f us\n',tau_sense*1e6);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
170	fprintf('Current-sense fc : %.1f Hz\n',fc_sense);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
171	fprintf('Modulation-path tau : %.1f us\n',tau_mod*1e6);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
172	fprintf('Modulation-path fc : %.1f Hz\n',fc_mod);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
173	fprintf('Nominal NE555 frequency : %.3f kHz\n',f555_nom/1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
174	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
175	fprintf('\n--- RESULTS ---\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
176	fprintf('TP6 range : %.1f to %.1f mV\n',min(TP6)*1e3,max(TP6)*1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
177	fprintf('TP7 range : %.3f to %.3f V\n',min(TP7),max(TP7));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
178	fprintf('TP8 range : %.3f to %.3f V\n',min(TP8),max(TP8));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
179	fprintf('Q6 threshold : %.3f V\n',VQ6_on);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
180	fprintf('Maximum Q6 drive : %.3f V\n',max(Q6_drive));	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
181	fprintf('Frequency range : %.3f to %.3f kHz\n', ...	Prints formatted model parameters, results, or file-status information to the



Line	MATLAB source line	English explanation
		MATLAB Command Window.
182	<code>min(f_inst)/1e3,max(f_inst)/1e3);</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
183	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
184	<code>if max(TP8) > VQ6_on</code>	Starts a conditional block; the following code executes only when this logical condition is true.
185	<code>fprintf('\nRESULT: TP8 crosses the Q6 threshold.\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
186	<code>fprintf('Frequency modulation is expected in this model.\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
187	<code>else</code>	Selects the alternative branch of the current conditional block.
188	<code>fprintf('\nRESULT: TP8 does NOT cross the Q6 threshold.\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
189	<code>fprintf('Q6 remains OFF in this behavioral model.\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
190	<code>fprintf('The NE555 frequency therefore remains nominal.\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
191	<code>end</code>	Ends the current for-loop or if/else block.
192	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
193	<code>%% 16. FIGURE 1 - Analog and control path</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
194	<code>fig1 = figure('Visible','off','Color','w');</code>	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
195	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
196	<code>subplot(5,1,1);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
197	<code>plot(t*1e3,Iload,'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
198	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
199	<code>ylabel('A');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
200	<code>title('Load Current');</code>	Sets the plot title so the test point or analysis purpose is explicit.
201	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
202	<code>subplot(5,1,2);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
203	<code>plot(t*1e3,TP6*1e3,'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
204	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
205	<code>ylabel('mV');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
206	<code>title('TP6 - Current Sense');</code>	Sets the plot title so the test point or analysis purpose is explicit.
207	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
208	<code>subplot(5,1,3);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
209	<code>plot(t*1e3,TP7,'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
210	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
211	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
212	<code>title('TP7 - LM358 Output');</code>	Sets the plot title so the test point or analysis purpose is explicit.
213	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
214	<code>subplot(5,1,4);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
215	<code>plot(t*1e3,TP8,'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
216	<code>hold on;</code>	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
217	<code>plot(t*1e3,VQ6_on*ones(size(t)),'--','LineWidth',1.0);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
218	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
219	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
220	<code>title('TP8 - Modulation Node and Q6 Threshold');</code>	Sets the plot title so the test point or analysis purpose is explicit.
221	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
222	<code>subplot(5,1,5);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
223	<code>plot(t*1e3,f_inst/1e3,'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
224	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
225	<code>ylabel('kHz');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
226	<code>xlabel('Time (ms)');</code>	Sets the x-axis label for the current plot.
227	<code>title('Instantaneous NE555 Frequency');</code>	Sets the plot title so the test point or analysis purpose is explicit.
228	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
229	<code>exportgraphics(fig1,'V2_3_01_Analog_Control.png','Resolution',180);</code>	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
230	<code>close(fig1);</code>	Closes the specified figure after export to reduce graphics-resource use during automated execution.
231	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
232	<code>%% 17. FIGURE 2 - Rising-edge detail</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
233	<code>z1 = t >= 0.90e-3 & t <= 1.80e-3;</code>	Builds a logical time mask so only the requested zoom interval is plotted.
234	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
235	<code>fig2 = figure('Visible','off','Color','w');</code>	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
236	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
237	<code>subplot(4,1,1);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
238	<code>plot(t(z1)*1e3,Iload(z1),'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
239	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
240	<code>ylabel('A');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
241	<code>title('Rising Edge: 0 A to 10 A');</code>	Sets the plot title so the test point or analysis purpose is explicit.
242	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
243	<code>subplot(4,1,2);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
244	<code>plot(t(z1)*1e3,TP7(z1),'LineWidth',1.4);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
245	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
246	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.



Line	MATLAB source line	English explanation
247	title('TP7');	Sets the plot title so the test point or analysis purpose is explicit.
248	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
249	subplot(4,1,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
250	plot(t(z1)*1e3,TP8(z1),'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
251	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
252	plot(t(z1)*1e3,VQ6_on*ones(1,sum(z1)),'--','LineWidth',1.0);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
253	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
254	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
255	title('TP8 and Q6 Threshold');	Sets the plot title so the test point or analysis purpose is explicit.
256	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
257	subplot(4,1,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
258	plot(t(z1)*1e3,f_inst(z1)/1e3,'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
259	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
260	ylabel('kHz');	Sets the y-axis label and therefore documents the plotted engineering unit.
261	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
262	title('Instantaneous Frequency');	Sets the plot title so the test point or analysis purpose is explicit.
263	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
264	exportgraphics(fig2,'V2_3_02_Rising_Edge.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
265	close(fig2);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
266	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
267	%% 18. FIGURE 3 - Falling-edge detail	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
268	z2 = t >= 3.90e-3 & t <= 4.80e-3;	Builds a logical time mask so only the requested zoom interval is plotted.
269	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
270	fig3 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
271	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
272	subplot(4,1,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
273	plot(t(z2)*1e3,Iload(z2),'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
274	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
275	ylabel('A');	Sets the y-axis label and therefore documents the plotted engineering unit.
276	title('Falling Edge: 10 A to 0 A');	Sets the plot title so the test point or analysis purpose is explicit.
277	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
278	subplot(4,1,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
279	plot(t(z2)*1e3,TP7(z2),'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.



Line	MATLAB source line	English explanation
280	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
281	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
282	title('TP7');	Sets the plot title so the test point or analysis purpose is explicit.
283	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
284	subplot(4,1,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
285	plot(t(z2)*1e3,TP8(z2),'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
286	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
287	plot(t(z2)*1e3,VQ6_on*ones(1,sum(z2)),'--','LineWidth',1.0);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
288	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
289	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
290	title('TP8 and Q6 Threshold');	Sets the plot title so the test point or analysis purpose is explicit.
291	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
292	subplot(4,1,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
293	plot(t(z2)*1e3,f_inst(z2)/1e3,'LineWidth',1.4);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
294	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
295	ylabel('kHz');	Sets the y-axis label and therefore documents the plotted engineering unit.
296	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
297	title('Instantaneous Frequency');	Sets the plot title so the test point or analysis purpose is explicit.
298	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
299	exportgraphics(fig3,'V2_3_03_Falling_Edge.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
300	close(fig3);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
301	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
302	%% 19. FIGURE 4 - Q6 drive	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
303	fig4 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
304	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
305	plot(t*1e3,Q6_drive,'LineWidth',1.5);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
306	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
307	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
308	ylabel('Voltage above threshold (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
309	title('Behavioral Q6 Drive');	Sets the plot title so the test point or analysis purpose is explicit.
310	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
311	exportgraphics(fig4,'V2_3_04_Q6_Drive.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
312	close(fig4);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
313	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no



Line	MATLAB source line	English explanation
		executable effect.
314	% 20. FIGURE 5 - Fast digital test points	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
315	zo = t >= 1.20e-3 & t <= 1.60e-3;	Builds a logical time mask so only the requested zoom interval is plotted.
316	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
317	fig5 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
318	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
319	subplot(4,2,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
320	plot(t(zo)*1e3,TP9(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
321	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
322	title('TP9');	Sets the plot title so the test point or analysis purpose is explicit.
323	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
324	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
325	subplot(4,2,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
326	plot(t(zo)*1e3,TP10(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
327	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
328	title('TP10');	Sets the plot title so the test point or analysis purpose is explicit.
329	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
330	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
331	subplot(4,2,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
332	plot(t(zo)*1e3,TP11(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
333	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
334	title('TP11');	Sets the plot title so the test point or analysis purpose is explicit.
335	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
336	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
337	subplot(4,2,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
338	plot(t(zo)*1e3,TP12(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
339	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
340	title('TP12');	Sets the plot title so the test point or analysis purpose is explicit.
341	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
342	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
343	subplot(4,2,5);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
344	plot(t(zo)*1e3,TP13(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
345	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
346	title('TP13');	Sets the plot title so the test point or analysis purpose is explicit.



Line	MATLAB source line	English explanation
347	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
348	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
349	subplot(4,2,6);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
350	plot(t(zo)*1e3,TP14(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
351	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
352	title('TP14');	Sets the plot title so the test point or analysis purpose is explicit.
353	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
354	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
355	subplot(4,2,[7 8]);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
356	plot(t(zo)*1e3,f_OUT(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
357	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
358	title('Differential f-OUT = TP13 - TP14');	Sets the plot title so the test point or analysis purpose is explicit.
359	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
360	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
361	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
362	exportgraphics(fig5,'V2_3_05_Digital_Test_Points.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
363	close(fig5);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
364	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
365	%% 21. FIGURE 6 - All test points overview	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
366	fig6 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
367	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
368	subplot(5,2,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
369	plot(t*1e3,TP6*1e3,'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
370	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
371	title('TP6');	Sets the plot title so the test point or analysis purpose is explicit.
372	ylabel('mV');	Sets the y-axis label and therefore documents the plotted engineering unit.
373	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
374	subplot(5,2,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
375	plot(t*1e3,TP7,'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
376	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
377	title('TP7');	Sets the plot title so the test point or analysis purpose is explicit.
378	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
379	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
380	subplot(5,2,3);	Selects a subplot position inside the current figure so several related waveforms



Line	MATLAB source line	English explanation
		can be shown together.
381	<code>plot(t*1e3,TP8,'LineWidth',1.1);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
382	<code>hold on;</code>	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
383	<code>plot(t*1e3,VQ6_on*ones(size(t)),'--','LineWidth',0.8);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
384	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
385	<code>title('TP8');</code>	Sets the plot title so the test point or analysis purpose is explicit.
386	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
387	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
388	<code>subplot(5,2,4);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
389	<code>plot(t(zo)*1e3,TP9(zo),'LineWidth',1.0);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
390	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
391	<code>title('TP9');</code>	Sets the plot title so the test point or analysis purpose is explicit.
392	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
393	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
394	<code>subplot(5,2,5);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
395	<code>plot(t(zo)*1e3,TP10(zo),'LineWidth',1.0);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
396	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
397	<code>title('TP10');</code>	Sets the plot title so the test point or analysis purpose is explicit.
398	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
399	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
400	<code>subplot(5,2,6);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
401	<code>plot(t(zo)*1e3,TP11(zo),'LineWidth',1.0);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
402	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
403	<code>title('TP11');</code>	Sets the plot title so the test point or analysis purpose is explicit.
404	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
405	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
406	<code>subplot(5,2,7);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
407	<code>plot(t(zo)*1e3,TP12(zo),'LineWidth',1.0);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
408	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
409	<code>title('TP12');</code>	Sets the plot title so the test point or analysis purpose is explicit.
410	<code>ylabel('V');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
411	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
412	<code>subplot(5,2,8);</code>	Selects a subplot position inside the current figure so several related waveforms can be shown together.
413	<code>plot(t(zo)*1e3,TP13(zo),'LineWidth',1.0);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.



Line	MATLAB source line	English explanation
414	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
415	title('TP13');	Sets the plot title so the test point or analysis purpose is explicit.
416	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
417	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
418	subplot(5,2,9);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
419	plot(t(zo)*1e3,TP14(zo),'LineWidth',1.0);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
420	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
421	title('TP14');	Sets the plot title so the test point or analysis purpose is explicit.
422	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
423	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
424	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
425	subplot(5,2,10);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
426	plot(t(zo)*1e3,f_OUT(zo),'LineWidth',1.0);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
427	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
428	title('TP13 - TP14');	Sets the plot title so the test point or analysis purpose is explicit.
429	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
430	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
431	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
432	exportgraphics(fig6,'V2_3_06_All_Test_Points.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
433	close(fig6);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
434	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
435	%% 22. Save numeric data to CSV	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
436	T = table(t(:),Iload(:),TP6(:),TP7(:),TP8(:),TP9(:),TP10(:), ...	Starts construction of a MATLAB table that groups the calculated results for CSV export.
437	TP11(:),TP12(:),TP13(:),TP14(:),f_OUT(:),f_inst(:), ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
438	'VariableNames',{'Time_s','Iload_A','TP6_V','TP7_V','TP8_V',	Supplies human-readable column names for the MATLAB table being constructed on the surrounding continuation lines.
439	'TP9_V','TP10_V','TP11_V','TP12_V','TP13_V','TP14_V', ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
440	'fOUT_V','Frequency_Hz'});	Continuation line containing text, a table variable name, or another argument for the MATLAB statement started above.
441	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
442	writetable(T,'V2_3_Simulation_Data.csv');	Writes the MATLAB table to a CSV file so numeric results can be inspected independently or replotted later.
443	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
444	%% 23. Final message	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
445	fprintf('\nFiles created in MATLAB Drive / Current Folder:\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
446	fprintf(' V2_3_01_Analog_Control.png\n');	Prints formatted model parameters, results, or file-status information to the



Line	MATLAB source line	English explanation
		MATLAB Command Window.
447	<code>fprintf(' V2_3_02_Rising_Edge.png\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
448	<code>fprintf(' V2_3_03_Falling_Edge.png\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
449	<code>fprintf(' V2_3_04_Q6_Drive.png\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
450	<code>fprintf(' V2_3_05_Digital_Test_Points.png\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
451	<code>fprintf(' V2_3_06_All_Test_Points.png\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
452	<code>fprintf(' V2_3_Simulation_Data.csv\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
453	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
454	<code>fprintf('\nOpen the PNG files directly from the Files panel.\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
455	<code>fprintf('The script intentionally creates no interactive figure windows.\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.



Appendix F. Version 3.0 - Line-by-Line Annotated MATLAB Source

Every source line is reproduced exactly in the left column. The right column explains the purpose in plain English so a reader with little or no MATLAB experience can follow the program from top to bottom.

Line	MATLAB source line	English explanation
1	%% NE555 Current-Sense Model - VERSION 3.0	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
2	% R36 / LM358 GAIN SWEEP	Source-code comment: R36 / LM358 GAIN SWEEP
3	%	Source-code comment explaining the model assumption or the purpose of the following lines.
4	% Purpose:	Source-code comment: Purpose:
5	% Keep the same circuit assumptions as Version 2.3, but sweep R36 to find	Source-code comment: Keep the same circuit assumptions as Version 2.3, but sweep R36 to find
6	% the approximate LM358 gain at which TP8 first crosses the BC558 (Q6)	Source-code comment: the approximate LM358 gain at which TP8 first crosses the BC558 (Q6)
7	% turn-on threshold.	Source-code comment: turn-on threshold.
8	%	Source-code comment explaining the model assumption or the purpose of the following lines.
9	% Tested R36 values:	Source-code comment: Tested R36 values:
10	% 330 kOhm, 390 kOhm, 430 kOhm, 470 kOhm, 510 kOhm, 560 kOhm	Source-code comment: 330 kOhm, 390 kOhm, 430 kOhm, 470 kOhm, 510 kOhm, 560 kOhm
11	%	Source-code comment explaining the model assumption or the purpose of the following lines.
12	% The script:	Source-code comment: The script:
13	% - Uses the same 0 A -> 10 A -> 0 A load-current step.	Source-code comment: - Uses the same 0 A -> 10 A -> 0 A load-current step.
14	% - Calculates TP6, TP7, TP8 for every R36 value.	Source-code comment: - Calculates TP6, TP7, TP8 for every R36 value.
15	% - Checks whether TP8 crosses the Q6 threshold.	Source-code comment: - Checks whether TP8 crosses the Q6 threshold.
16	% - Calculates the behavioral instantaneous frequency.	Source-code comment: - Calculates the behavioral instantaneous frequency.
17	% - Exports summary plots as PNG.	Source-code comment: - Exports summary plots as PNG.
18	% - Saves a CSV summary table.	Source-code comment: - Saves a CSV summary table.
19	%	Source-code comment explaining the model assumption or the purpose of the following lines.
20	% IMPORTANT:	Source-code comment: IMPORTANT:
21	% This remains a behavioral engineering model.	Source-code comment: This remains a behavioral engineering model.
22	% The purpose is to find the approximate gain threshold before changing	Source-code comment: The purpose is to find the approximate gain threshold before changing
23	% the real circuit.	Source-code comment: the real circuit.
24	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
25	clear;	Removes variables from the MATLAB workspace so values from an earlier run cannot contaminate this simulation.
26	clc;	Clears the MATLAB Command Window so the new run starts with readable console output.
27	close all force;	Closes previously open figure windows so old plots are not confused with the current simulation results.
28	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
29	%% 1. Simulation settings	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
30	VCC = 5.0;	Assigns or calculates the 5 V supply used by the behavioral model from the expression on the right-hand side.
31	dt = 0.5e-6;	Assigns or calculates the simulation time step from the expression on the right-hand side.
32	Tsim = 6e-3;	Assigns or calculates the total simulated time from the expression on the right-hand side.



Line	MATLAB source line	English explanation
33	<code>t = 0:dt:Tsim;</code>	Assigns or calculates the simulation time vector from the expression on the right-hand side.
34	<code>N = numel(t);</code>	Calculates the number of time samples by counting the number of elements in the referenced vector.
35	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
36	<code>%% 2. Fixed circuit parameters</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
37	<code>Rsense = 5e-3;</code>	Assigns or calculates the current-shunt resistance from the expression on the right-hand side.
38	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
39	<code>R38 = 1e3;</code>	Assigns or calculates the 1 kOhm LM358 input-path resistor from the expression on the right-hand side.
40	<code>R37 = 10e3;</code>	Assigns or calculates the 10 kOhm LM358 input-path resistor from the expression on the right-hand side.
41	<code>C30 = 220e-9;</code>	Assigns or calculates the current-sense filtering capacitor from the expression on the right-hand side.
42	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
43	<code>R35 = 330e3;</code>	Assigns or calculates the main resistor in the transient-modulation path from the expression on the right-hand side.
44	<code>R34 = 24e3;</code>	Assigns or calculates the 24 kOhm modulation-path resistor from the expression on the right-hand side.
45	<code>C29 = 2e-9;</code>	Assigns or calculates the transient-modulation capacitor from the expression on the right-hand side.
46	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
47	<code>R32 = 10;</code>	Assigns or calculates the small series resistor in the NE555 charge path from the expression on the right-hand side.
48	<code>R33 = 5e3;</code>	Assigns or calculates the main NE555 charge resistor from the expression on the right-hand side.
49	<code>R29 = 1e3;</code>	Assigns or calculates the NE555 discharge resistor from the expression on the right-hand side.
50	<code>C27 = 10e-9;</code>	Assigns or calculates the NE555 timing capacitor from the expression on the right-hand side.
51	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
52	<code>Vf_D31 = 0.60;</code>	Assigns or calculates the assumed D31 forward drop from the expression on the right-hand side.
53	<code>VCEsat = 0.10;</code>	Assigns or calculates the assumed saturation voltage of the NE555 discharge transistor from the expression on the right-hand side.
54	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
55	<code>Npri = 80;</code>	Assigns or calculates the transformer primary-turn count from the expression on the right-hand side.
56	<code>Nsec = 100;</code>	Assigns or calculates the transformer secondary-turn count from the expression on the right-hand side.
57	<code>turns_ratio = Nsec/Npri;</code>	Assigns or calculates the ideal transformer turns ratio from the expression on the right-hand side.
58	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
59	<code>%% 3. R36 sweep values</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
60	<code>R36_values = [330e3 390e3 430e3 470e3 510e3 560e3];</code>	Assigns or calculates the vector of R36 values tested in the gain sweep from the expression on the right-hand side.
61	<code>numCases = numel(R36_values);</code>	Calculates the number of R36 sweep cases by counting the number of elements in the referenced vector.



Line	MATLAB source line	English explanation
62	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
63	%% 4. Derived fixed values	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
64	Rin_eff = R38 + R37;	Assigns or calculates the effective LM358 input-path resistance from the expression on the right-hand side.
65	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
66	Rth_C30 = 1/(1/R38 + 1/R37);	Calculates the Thevenin resistance seen by C30 from the stated reciprocal relationship.
67	tau_sense = Rth_C30*C30;	Assigns or calculates the current-sense first-order time constant from the expression on the right-hand side.
68	fc_sense = 1/(2*pi*tau_sense);	Calculates the corresponding current-sense cutoff frequency from the stated reciprocal relationship.
69	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
70	tau_mod = (R35+R34)*C29;	Assigns or calculates the transient-path time constant from the expression on the right-hand side.
71	fc_mod = 1/(2*pi*tau_mod);	Calculates the corresponding transient-path cutoff frequency from the stated reciprocal relationship.
72	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
73	RA = R32 + R33;	Assigns or calculates the NE555 effective charge resistance from the expression on the right-hand side.
74	RB = R29;	Assigns or calculates the NE555 discharge resistance from the expression on the right-hand side.
75	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
76	VTL = VCC/3;	Assigns or calculates the lower NE555 timing threshold (one-third VCC) from the expression on the right-hand side.
77	VTH = 2*VCC/3;	Assigns or calculates the upper NE555 timing threshold (two-thirds VCC) from the expression on the right-hand side.
78	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
79	Vcharge_inf = VCC - Vf_D31;	Assigns or calculates the asymptotic timing-capacitor charge voltage after the diode drop from the expression on the right-hand side.
80	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
81	t_charge = RA*C27 * log((Vcharge_inf-VTL)/(Vcharge_inf-VTH));	Calculates the calculated NE555 charge interval using the analytical exponential charge/discharge timing equation.
82	t_discharge = RB*C27 * log((VTH-VCEsat)/(VTL-VCEsat));	Calculates the calculated NE555 discharge interval using the analytical exponential charge/discharge timing equation.
83	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
84	T555_nom = t_charge + t_discharge;	Assigns or calculates the nominal NE555 oscillation period from the expression on the right-hand side.
85	f555_nom = 1/T555_nom;	Assigns or calculates the nominal NE555 frequency from the expression on the right-hand side.
86	duty_nom = t_charge/T555_nom;	Assigns or calculates the nominal high-state duty fraction from the expression on the right-hand side.
87	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
88	%% 5. Q6 threshold	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
89	VBE_PNP = 0.65;	Assigns or calculates the assumed BC558 base-emitter magnitude from the expression on the right-hand side.
90	VQ6_on = VCC + VBE_PNP; % approx. 5.65 V	Assigns or calculates the modeled Q6 turn-on threshold from the expression on the right-hand side.



Line	MATLAB source line	English explanation
91	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
92	%% 6. Current profile	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
93	Iload = zeros(size(t));	Initializes the simulated load-current waveform with zeros, giving it the required array size before later samples are calculated.
94	Iload(t >= 1e-3 & t < 4e-3) = 10;	Defines the load current in the selected time interval, creating the stimulus used by the current-sense model.
95	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
96	%% 7. TP6	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
97	TP6 = -Iload*Rsense;	Converts load current into the shunt voltage at TP6. The negative sign represents the polarity chosen for the inverting LM358 model.
98	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
99	%% 8. Storage arrays	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
100	Gain_values = zeros(1,numCases);	Initializes storage for the LM358 gain calculated for each R36 value with zeros, giving it the required array size before later samples are calculated.
101	TP7_max_values = zeros(1,numCases);	Initializes storage for the maximum TP7 value of each case with zeros, giving it the required array size before later samples are calculated.
102	TP8_max_values = zeros(1,numCases);	Initializes storage for the maximum TP8 value of each case with zeros, giving it the required array size before later samples are calculated.
103	Q6_drive_max = zeros(1,numCases);	Initializes storage for the maximum threshold overdrive of each case with zeros, giving it the required array size before later samples are calculated.
104	f_max_values = zeros(1,numCases);	Initializes storage for the maximum modeled frequency of each case with zeros, giving it the required array size before later samples are calculated.
105	f_min_values = zeros(1,numCases);	Initializes storage for the minimum modeled frequency of each case with zeros, giving it the required array size before later samples are calculated.
106	CrossesThreshold = false(1,numCases);	Assigns or calculates Boolean results indicating whether TP8 crosses the Q6 threshold from the expression on the right-hand side.
107	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
108	TP7_all = zeros(numCases,N);	Initializes matrix storing every TP7 waveform with zeros, giving it the required array size before later samples are calculated.
109	TP8_all = zeros(numCases,N);	Initializes matrix storing every TP8 waveform with zeros, giving it the required array size before later samples are calculated.
110	f_all = zeros(numCases,N);	Initializes matrix storing every instantaneous-frequency waveform with zeros, giving it the required array size before later samples are calculated.
111	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
112	%% 9. Sweep loop	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
113	alpha_sense = dt/(tau_sense+dt);	Assigns or calculates the discrete first-order filter coefficient for TP7 from the expression on the right-hand side.
114	alpha_mod = dt/(tau_mod+dt);	Assigns or calculates the discrete transient-path filter coefficient from the expression on the right-hand side.
115	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
116	Vbias_TP8 = 4.35;	Assigns or calculates the assumed TP8 bias/clamp level from the expression on the right-hand side.
117	Kmod_node = 1.0;	Assigns or calculates the behavioral TP8 shaping gain from the expression on the right-hand side.
118	Kf = 5e3; % Hz/V	Assigns or calculates the behavioral frequency sensitivity coefficient from the expression on the right-hand side.
119	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.



Line	MATLAB source line	English explanation
120	for n = 1:numCases	Starts a loop. MATLAB repeats the following block for the stated index range.
121	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
122	R36 = R36_values(n);	Loads the current R36 value from the sweep vector for this loop iteration.
123	A_LM358 = R36/Rin_eff;	Assigns or calculates the modeled LM358 closed-loop gain magnitude from the expression on the right-hand side.
124	Gain_values(n) = A_LM358;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
125	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
126	% TP7 target	Source-code comment: TP7 target
127	TP7_target = -A_LM358*TP6;	Calculates the ideal amplified current-sense voltage before the first-order output/filter response is applied.
128	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
129	% TP7 low-pass response	Source-code comment: TP7 low-pass response
130	TP7 = zeros(size(t));	Initializes the filtered behavioral LM358 output with zeros, giving it the required array size before later samples are calculated.
131	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
132	TP7(k) = TP7(k-1) + alpha_sense*(TP7_target(k)-TP7(k-1));	Updates TP7 with a first-order discrete low-pass equation, moving the previous output toward the instantaneous target.
133	end	Ends the current for-loop or if/else block.
134	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
135	% LM358 output swing approximation	Source-code comment: LM358 output swing approximation
136	TP7 = min(max(TP7,0),3.5);	Clamps the behavioral LM358 output to the assumed usable output range on a 5 V supply.
137	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
138	% TP8 transient path	Source-code comment: TP8 transient path
139	TP7_lp_mod = zeros(size(t));	Initializes the low-pass state used to form the high-pass transient with zeros, giving it the required array size before later samples are calculated.
140	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
141	TP7_lp_mod(k) = TP7_lp_mod(k-1) + ...	Updates the low-pass state used to subtract the slow component of TP7 and isolate the transient/high-pass component.
142	alpha_mod*(TP7(k)-TP7_lp_mod(k-1));	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
143	end	Ends the current for-loop or if/else block.
144	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
145	HP_mod = TP7 - TP7_lp_mod;	Subtracts the slow low-pass component from TP7 to obtain the transient/high-pass component that drives TP8.
146	TP8 = Vbias_TP8 + Kmod_node*HP_mod;	Adds the transient component to the assumed TP8 DC bias, forming the behavioral control-node voltage.
147	TP8 = min(max(TP8,0),6.5);	Clamps the behavioral TP8 voltage to a bounded plotting range so extreme numerical values cannot dominate the visualization.
148	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
149	% Q6 behavioral drive	Source-code comment: Q6 behavioral drive
150	Q6_drive = max(TP8 - VQ6_on,0);	Implements the behavioral Q6 threshold: values below threshold produce zero drive; only voltage above threshold contributes to modulation.
151	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
152	% Frequency	Source-code comment: Frequency
153	f_inst = f555_nom + Kf*Q6_drive;	Creates the behavioral frequency law by adding Kf times Q6 threshold overdrive to the nominal NE555 frequency.
154	f_inst = min(max(f_inst,5e3),40e3);	Clamps the behavioral frequency to a practical plotting range; this is a numerical



Line	MATLAB source line	English explanation
		safeguard rather than a measured circuit limit.
155	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
156	% Save results	Source-code comment: Save results
157	TP7_all(n,:) = TP7;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
158	TP8_all(n,:) = TP8;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
159	f_all(n,:) = f_inst;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
160	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
161	TP7_max_values(n) = max(TP7);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
162	TP8_max_values(n) = max(TP8);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
163	Q6_drive_max(n) = max(Q6_drive);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
164	f_max_values(n) = max(f_inst);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
165	f_min_values(n) = min(f_inst);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
166	CrossesThreshold(n) = any(TP8 > VQ6_on);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
167	end	Ends the current for-loop or if/else block.
168	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
169	%% 10. Console summary	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
170	fprintf('\nVERSION 3.0 - R36 / GAIN SWEEP\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
171	fprintf('Nominal NE555 frequency : %.3f kHz\n',f555_nom/1e3);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
172	fprintf('Q6 threshold : %.3f V\n',VQ6_on);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
173	fprintf('Current-sense fc : %.1f Hz\n',fc_sense);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
174	fprintf('Modulation-path fc : %.1f Hz\n',fc_mod);	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
175	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
176	fprintf(' R36(k) Gain TP7max(V) TP8max(V) Q6drive(V) fmax(kHz) Cross?\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
177	fprintf('-----\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
178	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
179	for n = 1:numCases	Starts a loop. MATLAB repeats the following block for the stated index range.
180	if CrossesThreshold(n)	Starts a conditional block; the following code executes only when this logical condition is true.
181	crossText = 'YES';	Stores a short YES/NO text label used when printing whether the current case crosses the Q6 threshold.
182	else	Selects the alternative branch of the current conditional block.
183	crossText = 'NO ';	Stores a short YES/NO text label used when printing whether the current case crosses the Q6 threshold.
184	end	Ends the current for-loop or if/else block.
185	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no



Line	MATLAB source line	English explanation
		executable effect.
186	<code>fprintf(' %6.0f %5.2f %7.3f %7.3f %7.3f %7.3f %7.3f %s\n', ...</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
187	<code>R36_values(n)/1e3, ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
188	<code>Gain_values(n), ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
189	<code>TP7_max_values(n), ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
190	<code>TP8_max_values(n), ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
191	<code>Q6_drive_max(n), ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
192	<code>f_max_values(n)/1e3, ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
193	<code>crossText);</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
194	<code>end</code>	Ends the current for-loop or if/else block.
195	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
196	<code>firstCross = find(CrossesThreshold,1,'first');</code>	Searches the Boolean threshold-crossing vector and returns the index of the first successful R36 case.
197	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
198	<code>if isempty(firstCross)</code>	Starts a conditional block; the following code executes only when this logical condition is true.
199	<code>fprintf('\nRESULT: None of the tested R36 values crosses the Q6 threshold.\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
200	<code>else</code>	Selects the alternative branch of the current conditional block.
201	<code>fprintf('\nRESULT: First tested value crossing the Q6 threshold:\n');</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
202	<code>fprintf('R36 = %.0f kOhm\n',R36_values(firstCross)/1e3);</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
203	<code>fprintf('Gain = %.2f V/V\n',Gain_values(firstCross));</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
204	<code>fprintf('TP8 max = %.3f V\n',TP8_max_values(firstCross));</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
205	<code>fprintf('Q6 drive max = %.3f V\n',Q6_drive_max(firstCross));</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
206	<code>fprintf('f max = %.3f kHz\n',f_max_values(firstCross)/1e3);</code>	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
207	<code>end</code>	Ends the current for-loop or if/else block.
208	<code>(blank line)</code>	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
209	<code>%% 11. Export summary table</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
210	<code>Summary = table(...</code>	Starts construction of a MATLAB table that groups the calculated results for CSV export.
211	<code>R36_values(:)/1e3, ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
212	<code>Gain_values(:), ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
213	<code>TP7_max_values(:), ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
214	<code>TP8_max_values(:), ...</code>	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
215	<code>Q6_drive_max(:), ...</code>	Continuation line: this source line is part of a MATLAB statement that continues



Line	MATLAB source line	English explanation
		onto the next line.
216	f_min_values(:)/1e3, ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
217	f_max_values(:)/1e3, ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
218	CrossesThreshold(:), ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
219	'VariableNames', { ...	Supplies human-readable column names for the MATLAB table being constructed on the surrounding continuation lines.
220	'R36_kOhm', ...	Continues the list of table-column names used in the exported CSV file.
221	'LM358_Gain', ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
222	'TP7_Max_V', ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
223	'TP8_Max_V', ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
224	'Q6_Drive_Max_V', ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
225	'Frequency_Min_kHz', ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
226	'Frequency_Max_kHz', ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
227	'Crosses_Q6_Threshold'});	Continuation line containing text, a table variable name, or another argument for the MATLAB statement started above.
228	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
229	writetable(Summary,'V3_0_R36_Gain_Sweep_Summary.csv');	Writes the MATLAB table to a CSV file so numeric results can be inspected independently or replotted later.
230	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
231	%% 12. FIGURE 1 - TP8 max versus R36	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
232	fig1 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
233	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
234	plot(R36_values/1e3,TP8_max_values,'o-','LineWidth',1.5);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
235	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
236	plot(R36_values/1e3,VQ6_on*ones(size(R36_values)),'--','LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
237	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
238	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
239	xlabel('R36 (kOhm)');	Sets the x-axis label for the current plot.
240	ylabel('Maximum TP8 Voltage (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
241	title('Maximum TP8 Voltage vs R36');	Sets the plot title so the test point or analysis purpose is explicit.
242	legend('TP8 max','Q6 threshold','Location','best');	Creates a legend that identifies the plotted sweep cases or reference threshold.
243	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
244	exportgraphics(fig1,'V3_0_01_TP8_Max_vs_R36.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
245	close(fig1);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
246	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no



Line	MATLAB source line	English explanation
		executable effect.
247	% 13. FIGURE 2 - Maximum frequency versus R36	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
248	fig2 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
249	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
250	plot(R36_values/1e3,f_max_values/1e3,'o-','LineWidth',1.5);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
251	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
252	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
253	xlabel('R36 (kOhm)');	Sets the x-axis label for the current plot.
254	ylabel('Maximum Frequency (kHz)');	Sets the y-axis label and therefore documents the plotted engineering unit.
255	title('Maximum Modeled NE555 Frequency vs R36');	Sets the plot title so the test point or analysis purpose is explicit.
256	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
257	exportgraphics(fig2,'V3_0_02_Frequency_vs_R36.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
258	close(fig2);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
259	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
260	% 14. FIGURE 3 - TP8 waveforms for all R36 values	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
261	fig3 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
262	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
263	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
264	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
265	for n = 1:numCases	Starts a loop. MATLAB repeats the following block for the stated index range.
266	plot(t*1e3,TP8_all(n,:), 'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
267	end	Ends the current for-loop or if/else block.
268	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
269	plot(t*1e3,VQ6_on*ones(size(t)), 'k--','LineWidth',1.2);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
270	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
271	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
272	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
273	ylabel('TP8 (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
274	title('TP8 Waveforms for R36 Gain Sweep');	Sets the plot title so the test point or analysis purpose is explicit.
275	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
276	legend(...	Creates a legend that identifies the plotted sweep cases or reference threshold.
277	'330k','390k','430k','470k','510k','560k','Q6 threshold', ...	Continuation line: this source line is part of a MATLAB statement that continues onto the next line.
278	'Location','best');	Continuation line containing text, a table variable name, or another argument for the MATLAB statement started above.
279	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no



Line	MATLAB source line	English explanation
		executable effect.
280	exportgraphics(fig3,'V3_03_TP8_Waveforms.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
281	close(fig3);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
282	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
283	%% 15. FIGURE 4 - Frequency waveforms for all R36 values	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
284	fig4 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
285	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
286	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
287	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
288	for n = 1:numCases	Starts a loop. MATLAB repeats the following block for the stated index range.
289	plot(t*1e3,f_all(n,+)/1e3,'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
290	end	Ends the current for-loop or if/else block.
291	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
292	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
293	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
294	ylabel('Frequency (kHz)');	Sets the y-axis label and therefore documents the plotted engineering unit.
295	title('Instantaneous Frequency for R36 Gain Sweep');	Sets the plot title so the test point or analysis purpose is explicit.
296	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
297	legend('330k','390k','430k','470k','510k','560k','Location','best');	Creates a legend that identifies the plotted sweep cases or reference threshold.
298	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
299	exportgraphics(fig4,'V3_04_Frequency_Waveforms.png','Resolution',180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
300	close(fig4);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
301	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
302	%% 16. FIGURE 5 - TP7 max and gain	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
303	fig5 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
304	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
305	yyaxis left	Selects the left or right y-axis so two quantities with different numerical scales can share one graph.
306	plot(R36_values/1e3,TP7_max_values,'o-','LineWidth',1.5);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
307	ylabel('TP7 max (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
308	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
309	yyaxis right	Selects the left or right y-axis so two quantities with different numerical scales can share one graph.
310	plot(R36_values/1e3,Gain_values,'s-','LineWidth',1.5);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
311	ylabel('LM358 gain (V/V)');	Sets the y-axis label and therefore documents the plotted engineering unit.



Line	MATLAB source line	English explanation
312	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
313	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
314	xlabel('R36 (kOhm)');	Sets the x-axis label for the current plot.
315	title('TP7 Maximum and LM358 Gain vs R36');	Sets the plot title so the test point or analysis purpose is explicit.
316	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
317	exportgraphics(fig5, 'V3_0_05_TP7_and_Gain_vs_R36.png', 'Resolution', 180);	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
318	close(fig5);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
319	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
320	% 17. Optional detailed digital waveform for first crossing case	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
321	if ~isempty(firstCross)	Starts a conditional block; the following code executes only when this logical condition is true.
322	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
323	f_case = f_all(firstCross,:);	Assigns or calculates the frequency waveform for the selected threshold-crossing case from the expression on the right-hand side.
324	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
325	phase = zeros(size(t));	Initializes the accumulated oscillator phase with zeros, giving it the required array size before later samples are calculated.
326	for k = 2:N	Starts a loop. MATLAB repeats the following block for the stated index range.
327	phase(k) = phase(k-1) + 2*pi*f_case(k)*dt;	Numerically integrates instantaneous frequency into oscillator phase for this time sample.
328	end	Ends the current for-loop or if/else block.
329	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
330	phase01 = mod(phase, 2*pi)/(2*pi);	Wraps accumulated phase into a normalized 0-to-1 cycle position used to synthesize the square-wave output.
331	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
332	TP10 = VCC*(phase01 < duty_nom);	Converts normalized oscillator phase and duty cycle into a 0 V / VCC square-wave approximation of the NE555 output.
333	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
334	TP9 = VCC*ones(size(t));	Initializes the modeled NE555 discharge-node voltage with ones of the required array size; later lines modify selected samples.
335	TP9(phase01 >= duty_nom) = VCEsat;	For time samples in the discharge phase, forces TP9 down to the assumed discharge-transistor saturation voltage.
336	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
337	TP12 = TP10;	Uses TP10 directly as the second complementary primary-drive waveform.
338	TP11 = VCC - TP10;	Creates the complementary primary-drive waveform by logically inverting TP10 around the 5 V supply.
339	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
340	Vpri_diff = TP12 - TP11;	Calculates the differential voltage applied across the idealized transformer primary.
341	Vsec_diff = turns_ratio*Vpri_diff;	Applies the ideal turns ratio to the primary differential voltage to estimate the transformer secondary differential voltage.
342	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
343	TP13 = +0.5*Vsec_diff;	Assigns one secondary terminal to half of the differential secondary voltage about



Line	MATLAB source line	English explanation
		a virtual midpoint used only for plotting.
344	TP14 = -0.5*Vsec_diff;	Assigns the opposite secondary terminal to the negative half of the differential secondary voltage about the virtual plotting midpoint.
345	f_OUT = TP13 - TP14;	Calculates the differential output voltage; this is the meaningful transformer-secondary signal because the secondary is floating.
346	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
347	zo = t >= 1.20e-3 & t <= 1.60e-3;	Builds a logical time mask so only the requested zoom interval is plotted.
348	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
349	fig6 = figure('Visible','off','Color','w');	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
350	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
351	subplot(4,2,1);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
352	plot(t(zo)*1e3,TP9(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
353	grid on; title('TP9'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
354	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
355	subplot(4,2,2);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
356	plot(t(zo)*1e3,TP10(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
357	grid on; title('TP10'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
358	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
359	subplot(4,2,3);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
360	plot(t(zo)*1e3,TP11(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
361	grid on; title('TP11'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
362	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
363	subplot(4,2,4);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
364	plot(t(zo)*1e3,TP12(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
365	grid on; title('TP12'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
366	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
367	subplot(4,2,5);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
368	plot(t(zo)*1e3,TP13(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
369	grid on; title('TP13'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
370	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
371	subplot(4,2,6);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
372	plot(t(zo)*1e3,TP14(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.



Line	MATLAB source line	English explanation
373	grid on; title('TP14'); ylabel('V');	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
374	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
375	subplot(4,2,[7 8]);	Selects a subplot position inside the current figure so several related waveforms can be shown together.
376	plot(t(zo)*1e3,f_OUT(zo),'LineWidth',1.1);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
377	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
378	title(sprintf('Differential f-OUT, R36 = %.0f kOhm', ...	Sets the plot title so the test point or analysis purpose is explicit.
379	R36_values(firstCross)/1e3));	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
380	ylabel('V');	Sets the y-axis label and therefore documents the plotted engineering unit.
381	xlabel('Time (ms)');	Sets the x-axis label for the current plot.
382	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
383	exportgraphics(fig6,'V3_0_06_First_Threshold_Crossing_Digital.png', ...	Exports the selected MATLAB figure to a PNG file at the specified resolution for documentation and review.
384	'Resolution',180);	Continuation line containing text, a table variable name, or another argument for the MATLAB statement started above.
385	close(fig6);	Closes the specified figure after export to reduce graphics-resource use during automated execution.
386	end	Ends the current for-loop or if/else block.
387	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
388	%% 18. Final message	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
389	fprintf('\nFiles created:\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
390	fprintf(' V3_0_R36_Gain_Sweep_Summary.csv\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
391	fprintf(' V3_0_01_TP8_Max_vs_R36.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
392	fprintf(' V3_0_02_Frequency_vs_R36.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
393	fprintf(' V3_0_03_TP8_Waveforms.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
394	fprintf(' V3_0_04_Frequency_Waveforms.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
395	fprintf(' V3_0_05_TP7_and_Gain_vs_R36.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
396	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
397	if ~isempty(firstCross)	Starts a conditional block; the following code executes only when this logical condition is true.
398	fprintf(' V3_0_06_First_Threshold_Crossing_Digital.png\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
399	end	Ends the current for-loop or if/else block.
400	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
401	fprintf('\nOpen the PNG files directly from the MATLAB Files panel.\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.



Appendix G. Version 3.0 CSV Plot Test - Line-by-Line Annotated Source

Every source line is reproduced exactly in the left column. The right column explains the purpose in plain English so a reader with little or no MATLAB experience can follow the program from top to bottom.

Line	MATLAB source line	English explanation
1	%% V3.0 CSV Plot Test	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
2	% Minimal MATLAB plotting test using the already-created sweep CSV.	Source-code comment: Minimal MATLAB plotting test using the already-created sweep CSV.
3	% Purpose:	Source-code comment: Purpose:
4	% Verify whether MATLAB Online can display simple plots from numeric data.	Source-code comment: Verify whether MATLAB Online can display simple plots from numeric data.
5	%	Source-code comment explaining the model assumption or the purpose of the following lines.
6	% Required file in the Current Folder:	Source-code comment: Required file in the Current Folder:
7	% V3_0_R36_Gain_Sweep_Summary.csv	Source-code comment: V3_0_R36_Gain_Sweep_Summary.csv
8	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
9	clear;	Removes variables from the MATLAB workspace so values from an earlier run cannot contaminate this simulation.
10	clc;	Clears the MATLAB Command Window so the new run starts with readable console output.
11	close all force;	Closes previously open figure windows so old plots are not confused with the current simulation results.
12	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
13	%% 1. Read the CSV	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
14	filename = 'V3_0_R36_Gain_Sweep_Summary.csv';	Assigns or calculates the MATLAB variable filename from the expression on the right-hand side.
15	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
16	if ~isfile(filename)	Starts a conditional block; the following code executes only when this logical condition is true.
17	error('File not found: %s. Put the CSV in the Current Folder.', filename);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
18	end	Ends the current for-loop or if/else block.
19	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
20	T = readtable(filename);	Assigns or calculates the output table containing all time-domain samples from the expression on the right-hand side.
21	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
22	disp('CSV loaded successfully:');	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
23	disp(T);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
24	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
25	%% 2. Extract columns	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
26	R36_kOhm = T.R36_kOhm;	Assigns or calculates the MATLAB variable R36_kOhm from the expression on the right-hand side.
27	Gain = T.LM358_Gain;	Assigns or calculates the MATLAB variable Gain from the expression on the right-hand side.
28	TP8max = T.TP8_Max_V;	Assigns or calculates the MATLAB variable TP8max from the expression on the right-hand side.
29	Fmax_kHz = T.Frequency_Max_kHz;	Assigns or calculates the MATLAB variable Fmax_kHz from the expression on the right-hand side.



Line	MATLAB source line	English explanation
		side.
30	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
31	Q6_threshold = 5.650;	Assigns or calculates the MATLAB variable Q6_threshold from the expression on the right-hand side.
32	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
33	%% 3. Figure 1 - TP8 max vs R36	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
34	figure(1);	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
35	clf;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
36	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
37	plot(R36_kOhm, TP8max, 'o-', 'LineWidth', 1.8, 'MarkerSize', 7);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
38	hold on;	Keeps the current axes active so an additional waveform or threshold line can be overlaid.
39	plot(R36_kOhm, Q6_threshold*ones(size(R36_kOhm)), '--', 'LineWidth', 1.5);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
40	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
41	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
42	xlabel('R36 (kOhm)');	Sets the x-axis label for the current plot.
43	ylabel('Maximum TP8 Voltage (V)');	Sets the y-axis label and therefore documents the plotted engineering unit.
44	title('TP8 Maximum vs R36');	Sets the plot title so the test point or analysis purpose is explicit.
45	legend('TP8 max','Q6 threshold','Location','best');	Creates a legend that identifies the plotted sweep cases or reference threshold.
46	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
47	xlim([min(R36_kOhm)-10, max(R36_kOhm)+10]);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
48	ylim([min(TP8max)-0.1, max([TP8max; Q6_threshold])-0 + 0.15]);	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
49	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
50	drawnow;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
51	shg;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
52	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
53	%% 4. Figure 2 - Frequency vs R36	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
54	figure(2);	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
55	clf;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
56	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
57	plot(R36_kOhm, Fmax_kHz, 's-', 'LineWidth', 1.8, 'MarkerSize', 7);	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
58	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
59	grid on;	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
60	xlabel('R36 (kOhm)');	Sets the x-axis label for the current plot.
61	ylabel('Maximum Frequency (kHz)');	Sets the y-axis label and therefore documents the plotted engineering unit.
62	title('Maximum Modeled Frequency vs R36');	Sets the plot title so the test point or analysis purpose is explicit.



Line	MATLAB source line	English explanation
63	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
64	<code>xlim([min(R36_kOhm)-10, max(R36_kOhm)+10]);</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
65	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
66	<code>drawnow;</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
67	<code>shg;</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
68	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
69	<code>%% 5. Figure 3 - LM358 gain vs R36</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
70	<code>figure(3);</code>	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
71	<code>clf;</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
72	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
73	<code>plot(R36_kOhm, Gain, 'd-', 'LineWidth', 1.8, 'MarkerSize', 7);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
74	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
75	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
76	<code>xlabel('R36 (kOhm)');</code>	Sets the x-axis label for the current plot.
77	<code>ylabel('LM358 Gain (V/V)');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.
78	<code>title('LM358 Gain vs R36');</code>	Sets the plot title so the test point or analysis purpose is explicit.
79	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
80	<code>xlim([min(R36_kOhm)-10, max(R36_kOhm)+10]);</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
81	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
82	<code>drawnow;</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
83	<code>shg;</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
84	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
85	<code>%% 6. Simple graphics sanity check</code>	MATLAB section header. It creates a collapsible/run-able code section and labels the next stage of the model.
86	<code>figure(4);</code>	Creates a MATLAB figure. In later versions the figure is hidden during generation so it can be exported directly without interactive display.
87	<code>clf;</code>	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
88	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
89	<code>x = 0:0.1:10;</code>	Assigns or calculates the MATLAB variable x from the expression on the right-hand side.
90	<code>y = sin(x);</code>	Assigns or calculates the MATLAB variable y from the expression on the right-hand side.
91	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
92	<code>plot(x, y, 'LineWidth', 1.8);</code>	Plots the specified waveform against time or another x-axis variable. Scaling in the expression converts units when required.
93	<code>grid on;</code>	Turns on the plot grid to make time, voltage, current, or frequency values easier to read.
94	<code>xlabel('x');</code>	Sets the x-axis label for the current plot.
95	<code>ylabel('sin(x)');</code>	Sets the y-axis label and therefore documents the plotted engineering unit.



Line	MATLAB source line	English explanation
96	title('Simple MATLAB Graphics Test');	Sets the plot title so the test point or analysis purpose is explicit.
97	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
98	drawnow;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
99	shg;	Executes the MATLAB statement shown. Its role is part of the surrounding model section described by the nearby comments and equations.
100	(blank line)	Blank line used only as a visual separator between logical code blocks; it has no executable effect.
101	fprintf('\nFour simple figures were created.\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
102	fprintf('If these are still blank, the issue is with MATLAB Online graphics rendering,\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.
103	fprintf('not with the NE555 simulation calculations.\n');	Prints formatted model parameters, results, or file-status information to the MATLAB Command Window.



Appendix H. Data Dictionary and Source-File Map

The following compact reference identifies the principal signals and every MATLAB source file included in this complete learning edition.

Signal / file	Meaning / purpose
TP6	Current-sense/shunt voltage; 0 to about -50 mV for the selected 0-10 A polarity.
TP7	Behavioral LM358 amplified and filtered current-sense output.
TP8	Transient modulation/control node compared with the behavioral Q6 threshold.
TP9	Modeled NE555 discharge/timing node.
TP10	Modeled NE555 output.
TP11 / TP12	Complementary transformer-primary drive signals.
TP13 / TP14	Idealized floating transformer-secondary ends shown around a virtual midpoint.
f-OUT	Differential transformer output TP13 - TP14.
NE555_Current_Sense_All_Test_Points.m	Version 1 baseline with slow current ramps.
NE555_Current_Sense_All_Test_Points_V2.m	Version 2.0 hard current-step experiment.
NE555_Current_Sense_All_Test_Points_V2_1.m	Version 2.1 robust interactive plotting revision.
NE555_Current_Sense_All_Test_Points_V2_2.m	Version 2.2 export-only plotting revision.
NE555_Current_Sense_All_Test_Points_V2_3.m	Version 2.3 final 330 kOhm baseline with PNG and CSV export.
NE555_Current_Sense_R36_Gain_Sweep_V3_0.m	Version 3.0 R36/gain sweep that identified the first tested threshold crossing.
V3_0_Plot_Test_From_CSV.m	Minimal helper that reloads the Version 3.0 CSV and tests plotting independently of the main simulation.
V2_3_Simulation_Data.csv	High-resolution Version 2.3 time-domain dataset.
V3_0_R36_Gain_Sweep_Summary.csv	Six-case R36 sweep summary.



16. Final Engineering Conclusion and Closure of MATLAB Study Part 4

This fourth MATLAB study closes the NE555 current-sense frequency-modulation exercise at the behavioral-model level. The progression is technically useful because the unsuccessful cases were retained rather than hidden: Version 1 with $R36 = 330 \text{ k}\Omega$ and slow current ramps did not activate the desired modulation; Version 2.3 kept $R36 = 330 \text{ k}\Omega$ but applied a hard $0 \text{ A} \rightarrow 10 \text{ A} \rightarrow 0 \text{ A}$ step and still remained below the 5.65 V behavioral Q6 threshold; Version 3.0 then swept $R36$ and identified the first tested threshold-crossing region.

- $330 \text{ k}\Omega$: valid baseline, but TP8 remains below threshold; the desired frequency modulation is absent.
- $470 \text{ k}\Omega$: approximately 5.648 V at TP8, only about 2 mV below the behavioral threshold; this is too borderline to call robust.
- $510 \text{ k}\Omega$: approximately 5.758 V at TP8, about $+108 \text{ mV}$ of model margin, and approximately 18.928 kHz maximum modeled frequency. This is the selected optimum test point within the present sweep because it is the first clearly successful case without moving as aggressively as $560 \text{ k}\Omega$.
- $560 \text{ k}\Omega$: higher margin, but it is better treated as a comparison case until the real transistor/diode network and LM358 headroom are validated.
- The derived frequency equation is useful for understanding and predicting this MATLAB model, but $K_f = 5 \text{ kHz/V}$ and the 5.65 V threshold are still behavioral assumptions. They must be replaced or calibrated from real measurements before the law can be called a hardware transfer function.

Selected result of the MATLAB exercise

Use $R36 = 510 \text{ k}\Omega$ as the next bench/schematic validation point. In the present model it is the first tested value that produces a clear Q6 threshold crossing with moderate margin. The next step is not another arbitrary MATLAB gain change; it is correlation with the real circuit.

What comes next

The NE555 exercise is complete for this MATLAB series. Future work should move toward real schematics and measured behavior. A logical next project is a power-factor-correction (PFC) circuit study, for example a 120 VAC input design or a 347 VAC input design. That next study should begin from the actual topology, ratings, switching devices, magnetics, control IC, protection requirements, and measured waveforms rather than reusing assumptions from the NE555 exercise.

- Capture TP6, TP7, TP8, Q6 base/emitter/collector, TP9 and TP10 on real hardware at the selected $R36$ value.
- Measure the actual frequency response during controlled current transitions and fit a measured frequency law if the modulation is confirmed.
- Confirm D32 identity and use real device behavior for D32, D33, D34 and BC558 before finalizing the threshold model.
- When starting the PFC study, treat 120 VAC and 347 VAC as separate design cases because device stress, bus voltage, magnetics, sensing and safety constraints change substantially.